CS2 AI COACHING SYSTEM

Ultimate CS2 Coach

Part 2 — Services, Analysis and Database

Coaching Services, Knowledge & Retrieval, Analysis Engines, Database Schema, Training Pipeline

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Ultimate CS2 Coach — Part 2: Services, Analysis, Knowledge, Control, Progress, and Database

Topics: Coaching Services (4-mode pipeline: COPER/Hybrid/RAG/Base, Ollama LLM), Coaching Engines (11 game-theoretic and statistical analysis engines), Knowledge and Retrieval (RAG + COPER Experience Bank), Analysis Engines (Bayesian belief model, expectiminimax, momentum, roles, deception index, entropy, engagement range, utility economy, win probability), Processing and Feature Engineering (25-dim canonical vector, professional baselines), Control Module (lifecycle daemon, ingestion queue, ML control), Progress and Trends (longitudinal tracking), Database and Storage (three-tier SQLite WAL schema), Training and Orchestration Pipeline, Loss Functions.

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This document is the continuation of **Part 1B** — *The Senses and the Specialist*, which covers the RAP Coach model and all external data sources. Part 2 documents the coaching services that consume those neural outputs and turn them into actionable advice, together with the analysis engines, the tactical knowledge, and the storage and training infrastructure.

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5. Subsystem 3 — Coaching Services

Folder in the repo: `backend/services/` **Files:** `coaching_service.py` , `ollama_writer.py` , `analysis_orchestrator.py`

This subsystem is the **decision-making hub**: it takes everything the neural networks and knowledge systems produced and synthesizes it into actual coaching advice for the player.

-CoachingService (`coaching_service.py`)

The **central synthesis engine** that implements a **4-tier coaching fallback chain** with graceful degradation:

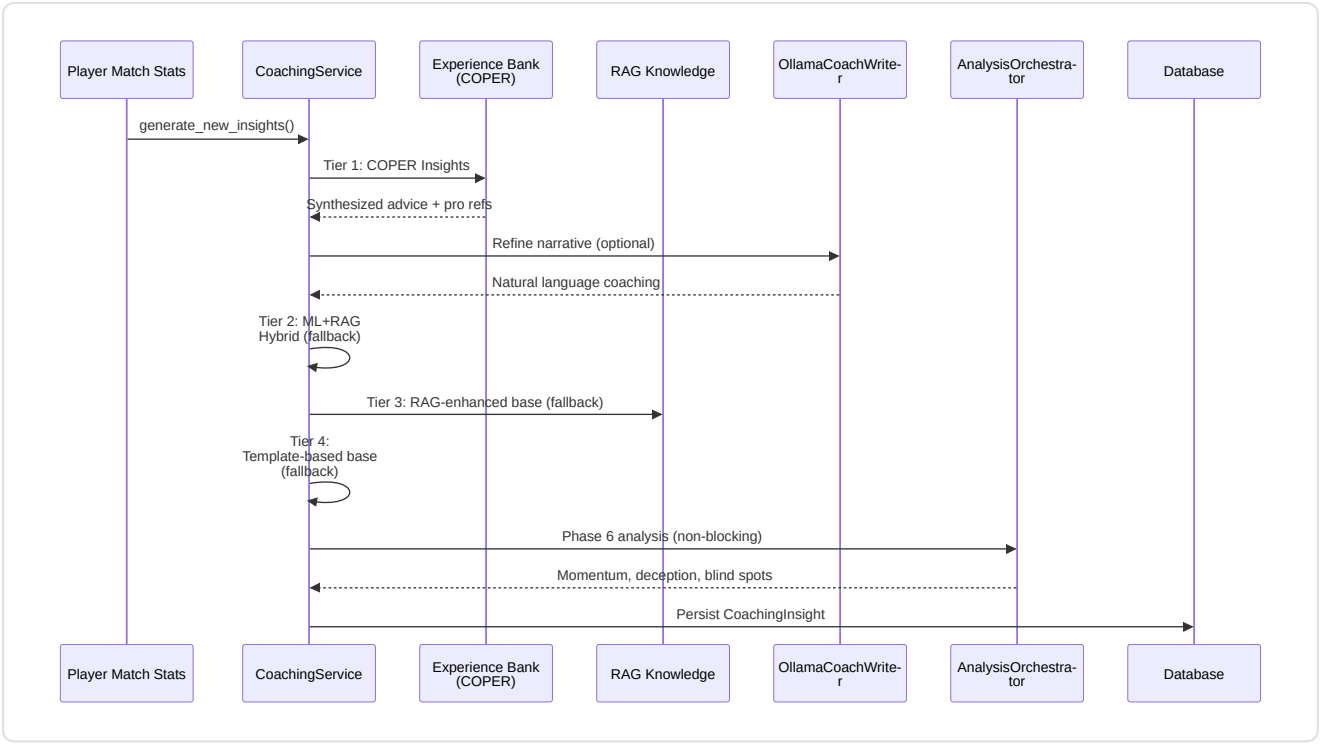
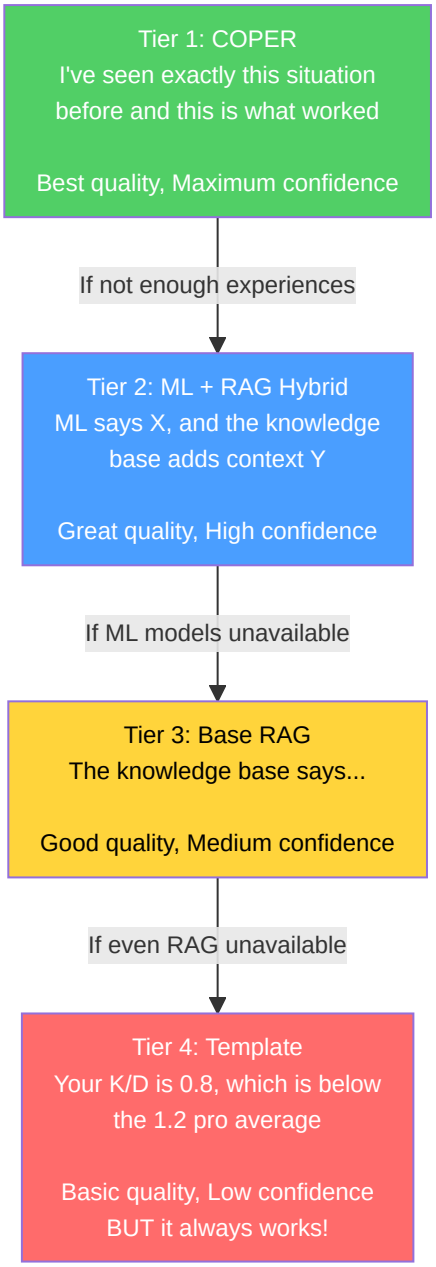


Diagram Explanation: Follow the arrows from top to bottom: this is the coach's "thought process" in order: (1) Match data arrives. (2) The coach first asks the Experience Bank: "Have we seen this situation before? What worked?". (3) If the answer sounds too robotic, it is optionally sent to Ollama (a local AI writer) to make it more natural. (4) If the Experience Bank has not enough data, it falls back to hybrid mode (ML predictions + RAG knowledge combined). (5) If even the ML models are unavailable, it falls back to RAG only (searching for relevant tips). (6) If even RAG fails, it uses simple statistical templates ("Your K/D ratio is 0.8, below average"). (7) Meanwhile, in the background, 7 analysis engines run special investigations across 5 pipelines (momentum, deception, entropy, strategy+blind spots, engagement range). (8) Everything is saved to the database for future reference.

4-tier fallback chain:

Tier	Method	Confidence	When used
1. **COPER**	`_generate_coper_insights()`	Maximum	Default — experience-based synthesis
2. **Hybrid**	`_generate_hybrid_insights()`	High	If COPER lacks enough experience
3. **Base RAG**	`_enhance_with_rag()`	Medium	If the ML models are unavailable
4. **Template**	Basic statistical template	Low	Last resort — always returns *something*



Key design: It never returns zero insights. The Phase 6 analysis is non-blocking (wrapped in try-catch, logged non-fatally).

> **Fix G-08 (Coaching Fallback):** Previously, the `_generate_coper_insights()` method did not receive the `deviations` and `rounds_played` parameters from the caller, causing a silent fallback to basic templates. After remediation, `generate_new_insights()` now explicitly passes both parameters (`deviations=deviations, rounds_played=rounds_played`) to the COPER handler, ensuring that Tier 1 COPER can generate contextual insights based on the player's real statistical deviations against the pro baseline. Without this fix, the system would always degrade to Tier 4 (template) even when COPER data was available.

Temporal baseline enrichment (Proposal 11): The coaching service now integrates time-weighted pro comparisons via two new methods:

- `_get_temporal_baseline(map_name)` — retrieves a pro baseline weighted according to `TemporalBaselineDecay` (half-life = 90 days) instead of static averages
- `_baseline_context_note(deviations, map_name)` — generates a natural-language description ("Based on recent pro data weighted by recency, your ADR is 12% below the current average meta on de_mirage") for COPER enrichment

This ensures coaching insights reflect the **current average meta** rather than outdated historical averages. If temporal data is insufficient (< 10 stat cards), the service transparently falls back to the legacy `get_pro_baseline()` function.

Round phase inference (`_infer_round_phase`): Equipment value → round phase classification:

Equipment value	Round phase
< $1,500	`pistol`
$1,500 – $2,999	`eco`
$3,000 – $3,999	`force`
≥ $4,000	`full_buy`

Health range classification (`_health_to_range`): Used for COPER context hashing: `"full"` (≥80), `"damaged"` (40–79), `"critical"` (<40).

-OllamaCoachWriter (`ollama_writer.py`)

Turns structured coaching information into natural language via a local LLM (Ollama).

- **Singleton** via the `get_ollama_writer()` factory
- **Feature-flagged:** `USE_OLLAMA_COACHING` setting (default: False)
- **Graceful degradation:** returns the original text if Ollama is unavailable
- **System prompt:** CS2 coaching expert tone, <100 words, actionable, encouraging

-AnalysisOrchestrator (`analysis_orchestrator.py`)

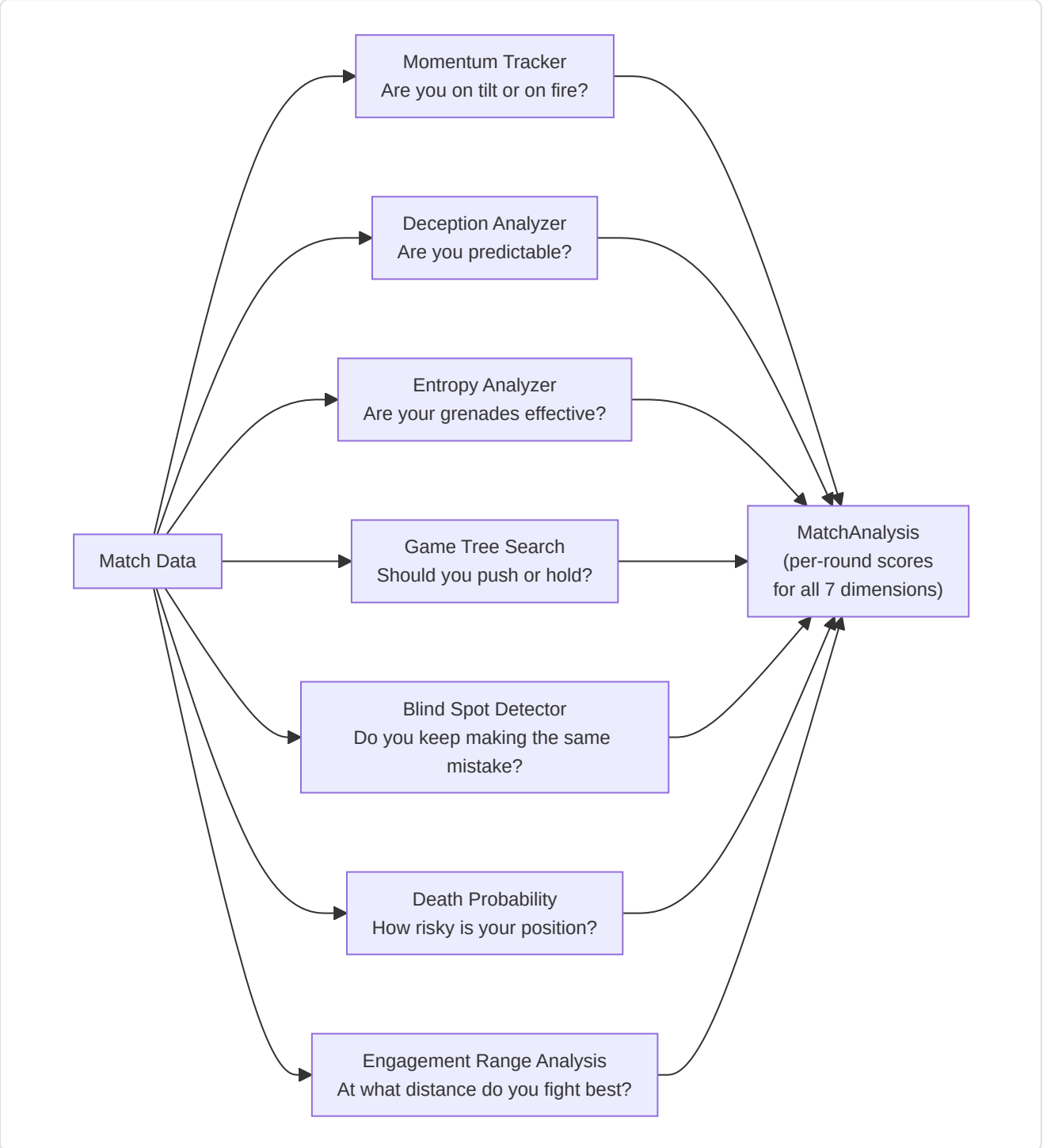
Synthesizes the advanced Phase 6 analysis by instantiating 7 engines and orchestrating 5 analysis pipelines:

Input: match tick data, events, player stats **Output:** `MatchAnalysis` with `RoundAnalysis` objects per round containing:

- `momentum_score` (tilt/win streak)
- `deception_score` (tactical sophistication)
- `utility_entropy` (effectiveness measurement)
- `blind_spots` (strategic gaps)

- `strategy_rec` (game tree recommendation)
- `engagement_range` (engagement distance analysis)

Engines instantiated: `belief_estimator`, `deception_analyzer`, `momentum_tracker`, `entropy_analyzer`, `game_tree`, `blind_spot_detector`, `engagement_analyzer` (7 engines). The `belief_estimator` is instantiated but is not currently invoked directly as a separate pipeline.



-Additional Services (Not previously documented)

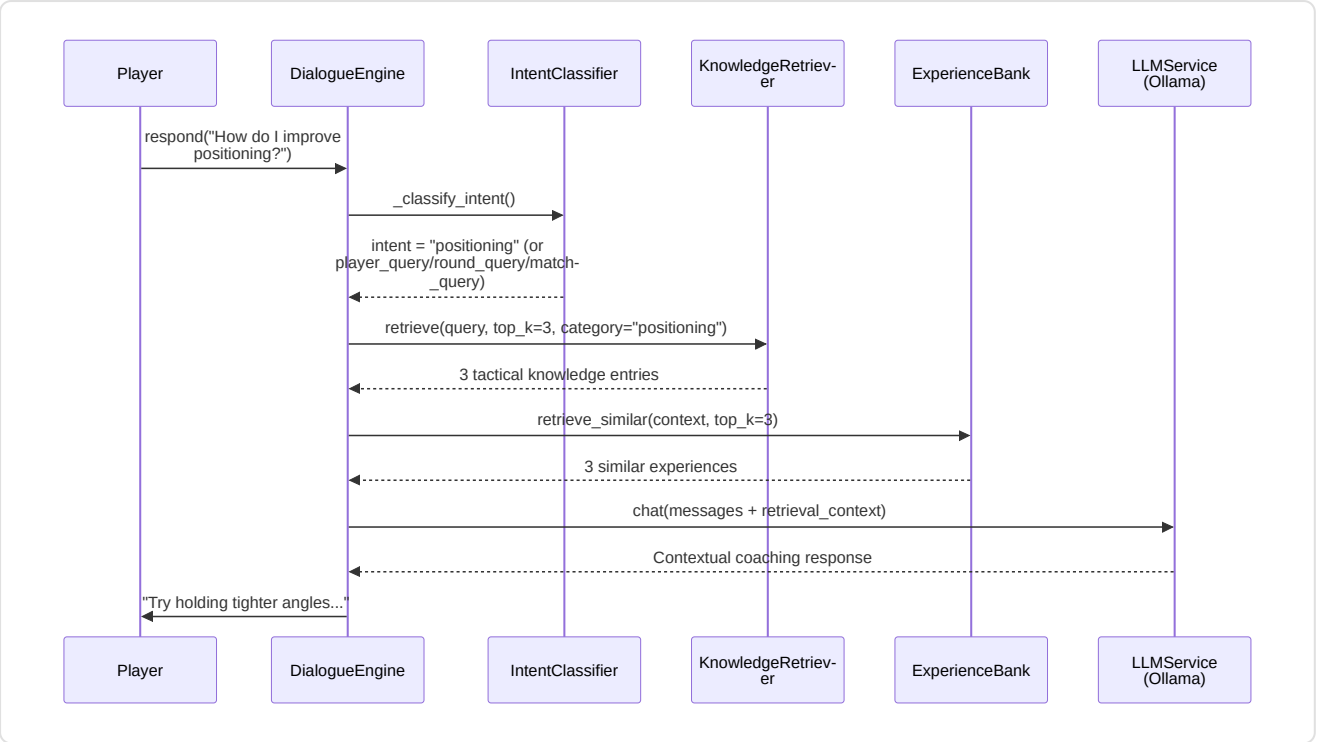
Beyond the three main services (CoachingService, OllamaCoachWriter, AnalysisOrchestrator), the `backend/services/` directory contains **7 additional services** that complete the coaching ecosystem:

CoachingDialogueEngine (`coaching_dialogue.py`)

Multi-turn dialogue engine with RAG and Experience Bank augmentation. Evolves the single-shot OllamaCoachWriter into an **interactive session** where players can ask follow-up questions about their performance.

Component	Detail
Intent classification	Keyword-based: 7 categories (positioning, utility, economy, aim, player_query, round_query, match_query) + general fallback
Sliding context window	`MAX_CONTEXT_TURNS = 6` (12 messages: 6 user + 6 assistant)
Retrieval augmentation	RAG top-3 + Experience Bank top-3, injected into the user message
Player entity detection	Integration with `PlayerLookupService` to detect mentions of professional players in the user message and inject "VERIFIED PLAYER DATA" blocks into the LLM context
Offline fallback	Template-based with RAG-only when Ollama is unavailable
Singleton	`get_dialogue_engine()` — module-level factory
Anti-hallucination (WR-78/79)	System prompt with rules: "use ONLY verified data", provenance markers ("pro data" vs "user data"), never fabricate tactical narratives

Response pipeline:



Session management: `start_session(player_name, demo_name)` → load context from DB (last 5 insights, primary focus area) → generate opening message via LLM → `respond(user_message)` for subsequent turns → `clear_session()` for reset.

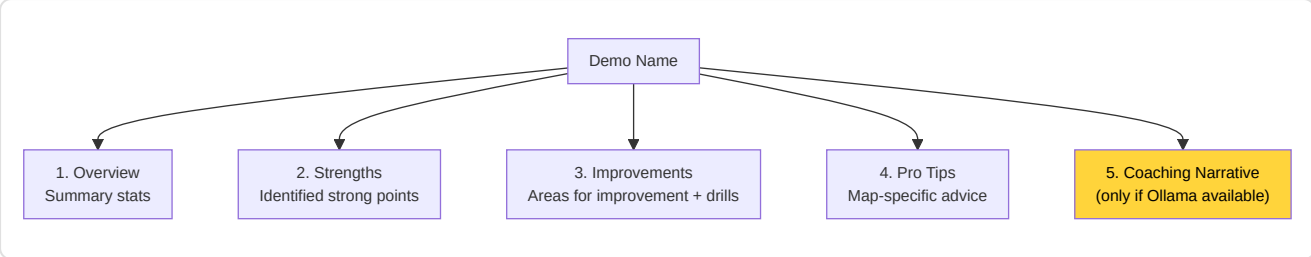
History safety (F5-06): The user message is added to history only *after* a valid response is obtained from the LLM, avoiding inconsistent states on exceptions.

LessonGenerator (`lesson_generator.py`)

Structured educational lesson generator starting from demo analysis:

Threshold	Constant	Value	Use
Strong ADR	`_ADR_STRONG_THRESHOLD`	75.0	Identifies strengths
Weak ADR	`_ADR_WEAK_THRESHOLD`	60.0	Identifies areas for improvement
Strong HS%	`_HS_STRONG_THRESHOLD`	0.40	Above-average precision
Weak HS%	`_HS_WEAK_THRESHOLD`	0.35	Below-average precision
Above-average rating	`_RATING_ABOVE_AVG`	1.0	Positive performance
Strong KAST	`_KAST_STRONG_THRESHOLD`	0.70	Consistent contribution
Death ratio	`_DEATH_RATIO_WARNING`	1.5×	deaths > kills × 1.5 = warning

Generated lesson structure:



Map-specific Pro Tips: Internal database with tips for mirage, inferno, dust2, ancient, nuke. Falls back to generic advice for unsupported maps.

LLMService (`llm_service.py`)

Ollama integration service for local LLM inference:

Parameter	Value	Description
`OLLAMA_URL`	`http://localhost:11434`	Ollama endpoint (env: `OLLAMA_URL`)
`DEFAULT_MODEL`	`llama3.1:8b`	8B general-purpose model (env: `OLLAMA_MODEL`)
`_AVAILABILITY_TTL`	60s	Availability cache
`temperature`	0.7	Response creativity
`top_p`	0.9	Nucleus sampling
`num_predict`	500	Response length limit

Supported APIs:

Method	Ollama endpoint	Use
`generate(prompt)`	`/api/generate`	Single-shot generation
`chat(messages)`	`/api/chat`	Multi-turn conversation
`generate_lesson(insights)`	`/api/generate`	Lessons from RAP insights
`explain_round_decision(round_data)`	`/api/generate`	Single-round explanation
`generate_pro_tip(context)`	`/api/generate`	Contextual tip

Auto-discovery model: If the configured model is not available, it automatically uses the first installed model. Singleton via `get_llm_service()`.

VisualizationService (`visualization_service.py`)

Generates Matplotlib radar charts for User vs Pro comparisons:

- `generate_performance_radar(user_stats, pro_stats, output_path)` → PNG file with radar overlay
- `plot_comparison_v2(p1_name, p2_name, p1_stats, p2_stats)` → `io.BytesIO` PNG buffer
- Features compared: avg_kills, avg_adr, avg_hs, avg_kast, accuracy

- Rendering wrapped in try/except (F5-19) to handle empty stats or missing matplotlib backend

ProfileService (`profile_service.py`)

External profile synchronization orchestrator:

- `fetch_steam_stats(steam_id)` → nickname, avatar, CS2 playtime (hours)
- `fetch_faceit_stats(nickname)` → faceit_elo, faceit_level, faceit_id
- `sync_all_external_data(steam_id, faceit_name)` → upsert to `PlayerProfile` in DB
- API keys loaded from environment/keyring (F5-22), never hardcoded

AnalysisService (`analysis_service.py`)

Analysis coordination service with drift detection:

- `analyze_latest_performance(player_name)` → latest stats from the DB
- `get_pro_comparison(player_name, pro_name)` → side-by-side comparison
- `check_for_drift(player_name)` → `detect_feature_drift()` on the last 100 matches

TelemetryClient (`telemetry_client.py`)

Async client for sending metrics to a central server:

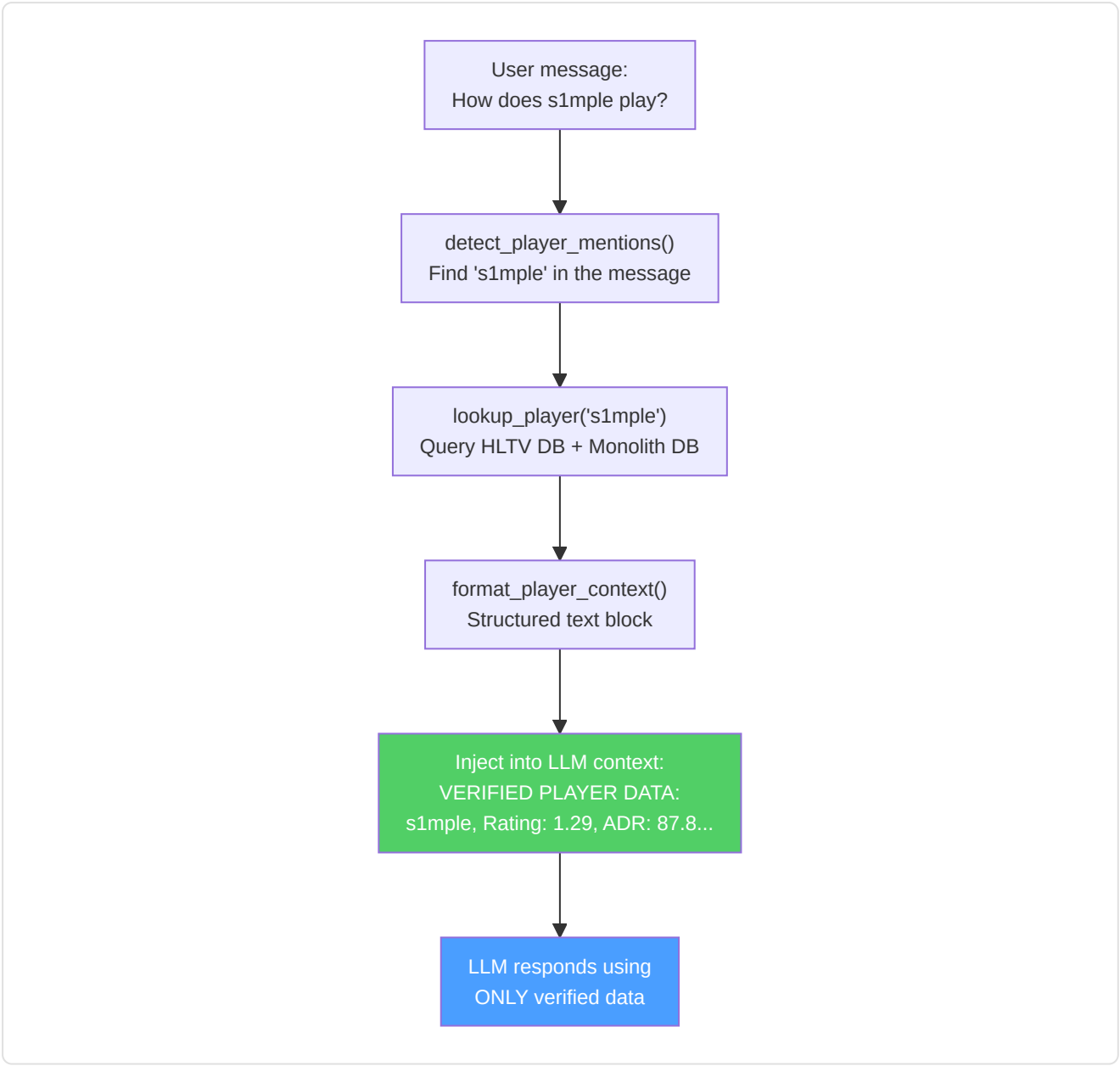
- Protocol: `httpx.Client` → `POST /api/ingest/telemetry`
- Configurable URL: `CS2_TELEMETRY_URL` (default: `http://127.0.0.1:8000`)
- Graceful fallback if httpx is not installed (optional feature)
- **Anti-fabrication compliant:** no synthetic data in the self-test

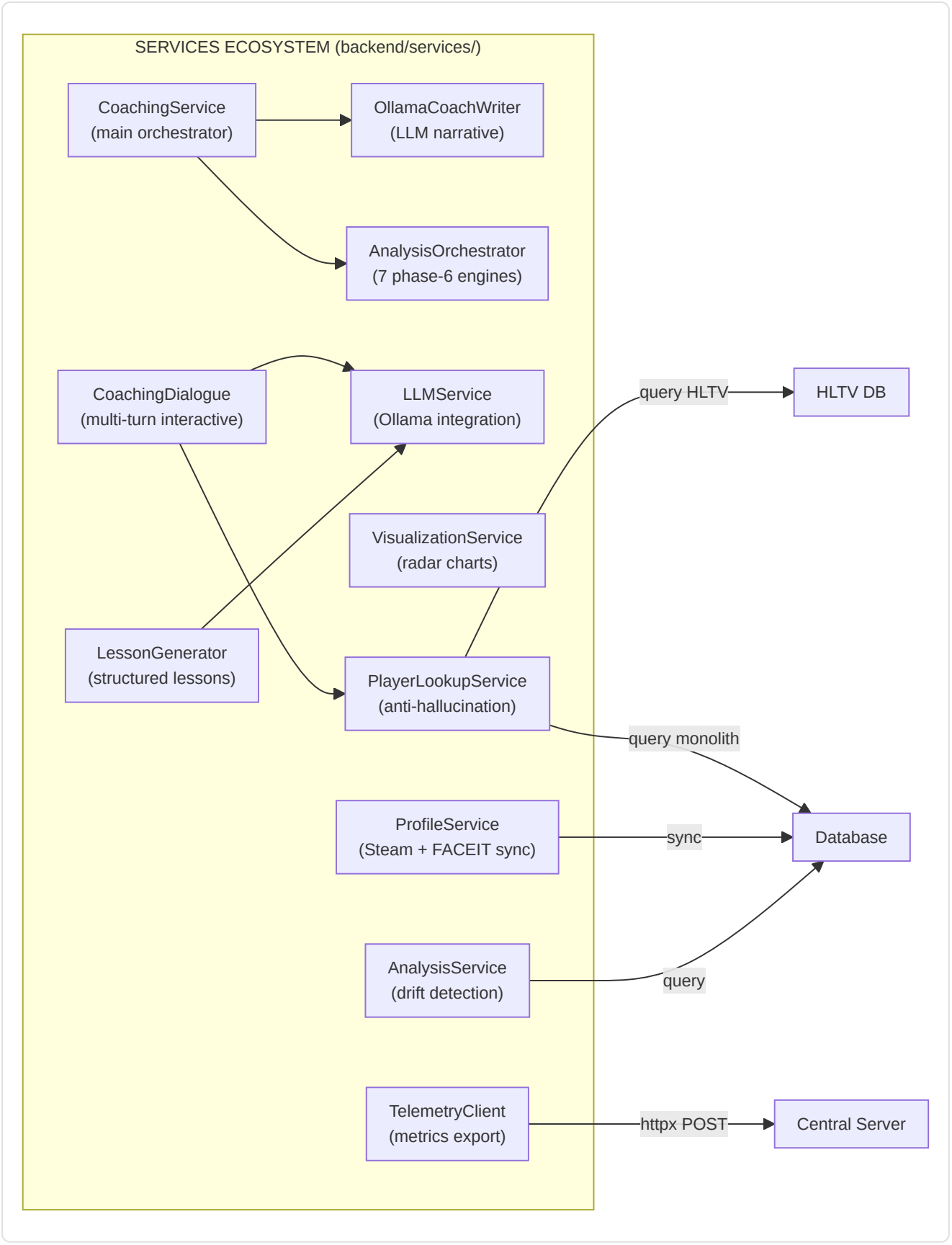
PlayerLookupService (`player_lookup.py`)

Professional player lookup service that prevents LLM hallucination by injecting verified data into the coaching dialogue context:

Component	Detail
3-tier matching	Exact (case-insensitive) → Fuzzy (SequenceMatcher ≥ 0.75) → Pattern (nickname regex)
Nickname cache	TTL 60s, loads all `ProPlayer.nickname` from the HLTV DB at startup
Stop-word filter	89 common English words filtered to avoid false positives
Output	`ProPlayerProfile` dataclass: nickname, hltv_id, real_name, country, team, HLTV statistics (rating, KPR, ADR, KAST), performance from local demos
Integration	`CoachingDialogueEngine` → `detect_player_mentions()` → `lookup_player()` → `format_player_context()` → "VERIFIED PLAYER DATA" block injected into the LLM context
Singleton	`get_player_lookup_service()`

Anti-hallucination pipeline:




SERVICES ECOSYSTEM (backend/services/)
CoachingService (main orchestrator)
OllamaCoachWriter (LLM narrative)
AnalysisOrchestrator (7 phase-6 engines)
CoachingDialogue (multi-turn interactive)
LLMService (Ollama integration)
query HLTV
HLTV DB
VisualizationService (radar charts)
LessonGenerator (structured lessons)
PlayerLookupService (anti-hallucination)
query monolith
ProfileService (Steam + FACEIT sync)
sync
Database
AnalysisService (drift detection)
query
TelemetryClient (metrics export)
httpx POST
Central Server

-FastAPI Server (`server.py`)

Standalone REST server that exposes endpoints for telemetry, insights, health, and remote service control. **It is not integrated into the main application flow** — it is a separate utility service, launchable independently.

Rate Limiting: `RateLimiter` class with sliding window — maximum 10 requests per minute per IP address. The internal `_requests` structure maps each IP to a list of `float` timestamps, purging expired timestamps on each check.

Lifecycle: The server uses FastAPI's asynchronous lifespan handler to initialize the database at startup via `init_database()`.

Pydantic Models:

Model	Type	Main Fields
`InsightRead`	Response	`id`, `player_name`, `title`, `message`, `focus_area`, `severity`, `created_at`
`MatchTelemetry`	Request	`player_id`, `match_id`, `stats` (dict), `timestamp` (float)

Endpoint Table:

Method	Path	Rate Limited	Description
`POST`	`/api/ingest/telemetry`	Yes	Match telemetry data ingestion. Background write to disk via `_write_telemetry_to_disk()`
`GET`	`/api/insights`	No	Returns coaching insights for a player (query param: `player_name`)
`GET`	`/api/status`	No	Application status: version, uptime, database state
`GET`	`/api/console/health`	No	Health check for service monitoring
`POST`	`/api/training/start`	No	Starts ML training via Console backend
`POST`	`/api/training/stop`	No	Stops ML training
`POST`	`/api/training/pause`	No	Pauses ML training
`POST`	`/api/training/resume`	No	Resumes ML training from pause
`POST`	`/api/services/hunter/start`	No	Starts HLTV Hunter service
`POST`	`/api/services/hunter/stop`	No	Stops HLTV Hunter service
`GET`	`/api/services/status`	No	Status of all managed services

Telemetry security: `_write_telemetry_to_disk(data)` sanitizes filenames derived from `player_id` and `match_id` to prevent path traversal attacks — dangerous characters are removed before constructing the destination path.

-Onboarding (`new_user_flow.py`)

Manages the new user experience through a **3-stage progressive onboarding** process that guides the player from installation to full coaching.

Onboarding Stages:

Constant	Value	Meaning	Condition
`AWAITING_FIRST_DEMO`	`"awaiting_first_demo"`	No demos uploaded	0 demos
`BUILDING_BASELINE`	`"building_baseline"`	Demos uploaded but insufficient for stable baseline	1-2 demos
`COACH_READY`	`"coach_ready"`	Full coaching active	3+ demos

`UserOnboardingManager` class:

- `MIN_INITIAL_DEMOS = 1` — minimum to start basic coaching
- `RECOMMENDED_DEMOS = 3` — minimum for stable statistical baseline
- `_CACHE_TTL_SECONDS = 60` — cache TTL for demo count (avoids repeated queries during rapid UI refreshes)

Methods:

- `get_status(user_id) → OnboardingStatus` — returns the full onboarding status with demo count, stage, messages, and `coach_ready` / `baseline_stable` flags
- `_count_user_demos(user_id)` — counts processed (non-pro) demos with TTL-based cache. Filter: `player_name == user_id AND is_pro == False`
- `invalidate_cache(user_id=None)` — invalidates cache after new demo upload (for specific user or entire cache)
- `_determine_stage(count)` — maps demo count to stage
- `_get_stage_message(stage, count)` — human-readable message for each stage

Dataclass `OnboardingStatus`: `stage`, `demos_uploaded`, `demos_required` (1), `demos_recommended` (3), `coach_ready`, `baseline_stable`, `message`.

Singleton: `get_onboarding_manager()` — module-level factory (not thread-safe with double-checked locking — acceptable for this low-contention use case).

5B. Subsystem 3B — Coaching Engines

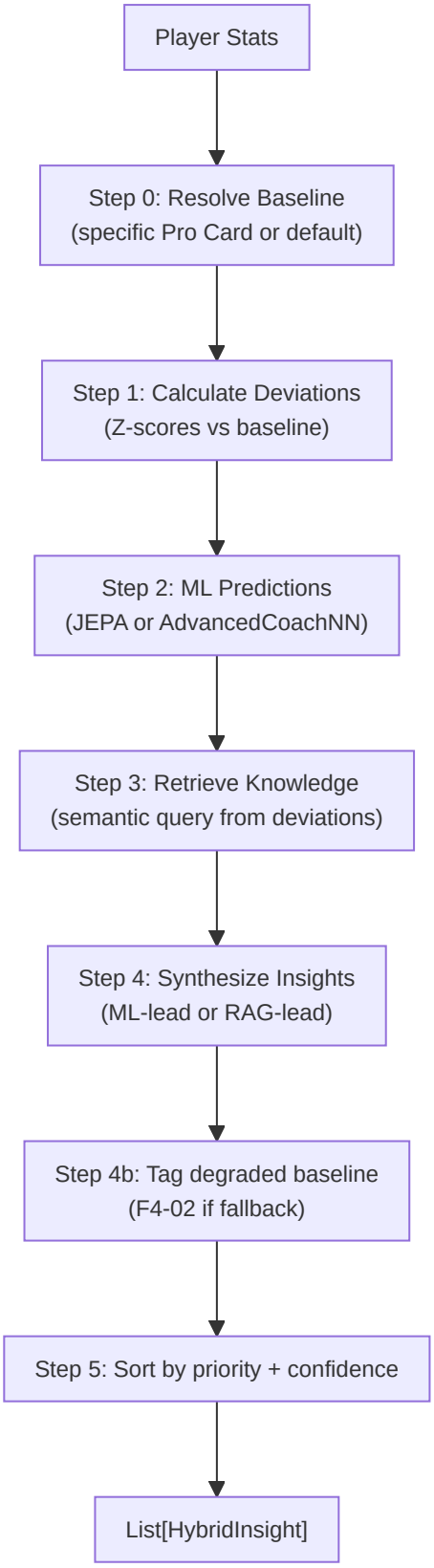
Folder in the repo: `backend/coaching/` **Files:** 8 modules

This subsystem contains the **coaching decision engines** that transform raw statistical deviations into prioritized and contextualized advice. Unlike the Services (Section 5), which orchestrate and present, the Coaching Engines contain the coach's **reasoning logic**.

-HybridCoachingEngine (`hybrid_engine.py`)

The **decision-making heart** of coaching: synthesizes ML predictions with RAG knowledge to generate unified, deduplicated insights.

5-stage pipeline:



Classes and dataclasses:

Type	Name	Description
`Enum`	`InsightPriority`	CRITICAL (|Z|>2.5, conf>0.8), HIGH (|Z|>2.0, conf>0.6), MEDIUM (|Z|>1.0, conf>0.4), LOW
`dataclass`	`HybridInsight`	title, message, priority, confidence, feature, ml_z_score, knowledge_refs, pro_examples, tick_range, demo_name

Confidence formula:

```
base_confidence = |z_score|/3.0 × 0.6 + knowledge_effectiveness × 0.4
confidence = base_confidence × MetaDriftEngine.get_meta_confidence_adjustment()
```

Where `knowledge_effectiveness = min(1.0, mean(usage_count) / 100)`.

Synthesis strategy:

Condition	Strategy
|Z| > 2 (high ML confidence)	Lead with ML, support with RAG
|Z| < 1 (low ML confidence)	Lead with RAG
1 ≤ |Z| ≤ 2	Balanced approach
No significant deviation	Knowledge-only insight (LOW priority)

TASK 2.7.1 — Reference Clip: Each `HybridInsight` can include `tick_range: (start, end)` and `demo_name` to allow the UI to jump directly to the evidence in the demo file.

Baseline fallback (F4-02): If `get_pro_baseline()` fails, uses hardcoded static values and marks all insights with `baseline_quality=degraded` in the message.

-CorrectionEngine (`correction_engine.py`)

Generates the **top-3 weighted corrections** from Z-score deviations:

Constant	Value	Use
`CONFIDENCE_ROUNDS_CEILING`	300	Confidence scaling: `min(1.0, rounds / 300)`

Importance weights (DEFAULT_IMPORTANCE):

Feature	Weight
`avg_kast`	1.5
`avg_adr`	1.5
`accuracy`	1.4
`impact_rounds`	1.3
`avg_hs`	1.2
`econ_rating`	1.1
`positional_aggression_score`	1.0

Pipeline: `deviations` → apply confidence scaling × rounds_played → optional `apply_nn_refinement()` → sort by `|weighted_z| × importance` → return top 3.

User override: Weights are overridable via `get_setting("COACH_WEIGHT_OVERRIDES")`.

-ExplanationGenerator (`explainability.py`)

Translates RL latent signals into **understandable narratives** organized along the 5 skill axes (`SkillAxes`):

Axis	Negative Template	Action Template
MECHANICS	"Your {feature} is {delta}% below professional standards..."	"Focus on crosshair height when clearing corners..."
POSITIONING	"Positioning at {location} was suboptimal..."	"Try holding a tighter angle at {location}..."
UTILITY	"Utility timing with {weapon} was suboptimal..."	"Wait for a clear sound cue before deploying..."
TIMING	"Engagement timing is lagging behind..."	"Coordinate with teammates to trade-frag..."
DECISION	"Decision efficiency is {delta}% lower..."	"In clutch scenarios, prioritize the round objective..."

"Silence is a Valid Action" principle:

Threshold	Constant	Value	Behavior
Silence	`SILENCE_THRESHOLD`	0.2	|delta| < 0.2 → no feedback (silence)
High severity	`SEVERITY_HIGH_BOUNDARY`	1.5	|delta| > 1.5 → "High"
Medium severity	`SEVERITY_MEDIUM_BOUNDARY`	0.8	|delta| > 0.8 → "Medium"

Complexity filter: For `skill_level < 3` (beginners), negative narratives are simplified to the suggested action alone, avoiding cognitive overload.

-NNRefinement (`nn_refinement.py`)

Applies neural-network weight adjustments to pre-computed corrections:

```
refined_z = weighted_z × (1 + nn_adjustments["{feature}_weight"])
```

Lightweight module that scales corrections from the `CorrectionEngine` using weights learned by the ML model, allowing the neural system to influence advice prioritization.

-ProBridge (`pro_bridge.py`)

Translation layer that assimilates HLTV Player Cards into the coach's cognitive model:

Class	Responsibility
`PlayerCardAssimilator`	Converts `ProPlayerStatCard` → coach-compatible baseline
`get_pro_baseline_for_coach()`	Direct factory function

Legacy constant: `ESTIMATED_ROUNDS_PER_MATCH = 24.0` — present in the code but **no longer used** for KPR/DPR conversion after the P3-02 fix.

Metric mapping:

HLTV metric	Coach metric	Transformation
`card.kpr`	`avg_kills`	**direct** (P3-02: NOT multiplied by 24)
`card.dpr`	`avg_deaths`	**direct** (P3-02: NOT multiplied by 24)
`card.adr`	`avg_adr`	direct
`card.kast`	`avg_kast`	V-2: defensive normalization (`/100` if `kast > 1.0` , otherwise direct)
`card.impact`	`impact_rounds`	direct
`card.rating_2_0`	`rating`	direct
`detailed_stats.headshot_pct`	`avg_hs`	V-2: defensive normalization (`/100` if `> 1.0` , default 0.45)
`detailed_stats.total_opening_kills`	`entry_rate`	`/100` (heuristic)
`detailed_stats.utility_damage_per_round`	`utility_damage`	direct (default 45.0)

> **Fix P3-02 — KPR/DPR scale:** The previous code multiplied `kpr × 24` and `dpr × 24`, producing total kills/deaths per match (values 15-20) instead of per-round values (0.6-0.8). This made all comparison z-scores invalid since the user stats extracted by `base_features.py` and `pro_baseline.py` are already on a per-round scale. After remediation, `get_coach_baseline()` uses the per-round rates directly.
>
> **Fix V-2 — Legacy defensive normalization:** Older HLTV records in the database may contain percentage values (e.g. `kast=72.0` instead of `kast=0.72`). Conditional normalization (`/100 if > 1.0`) transparently handles both formats.

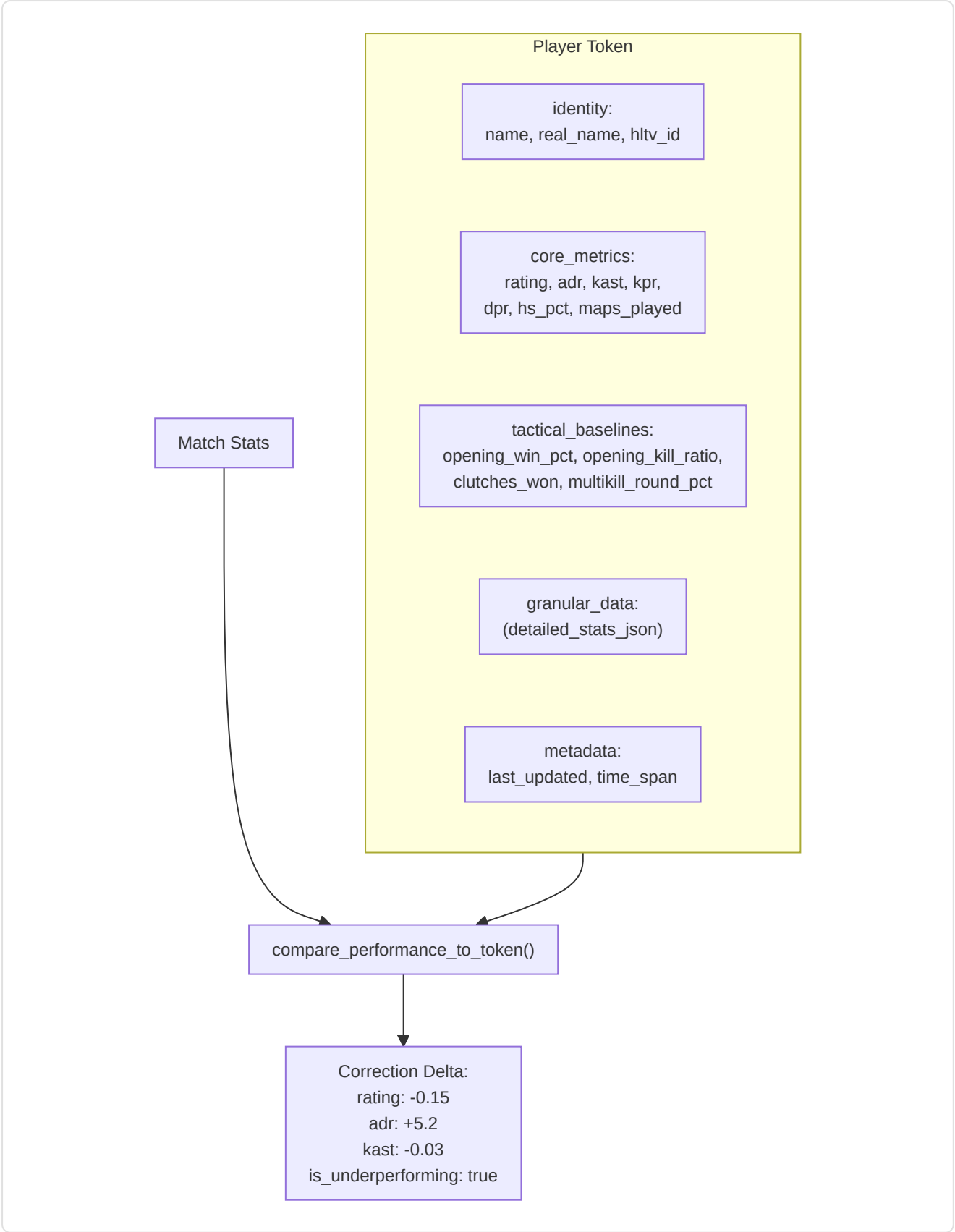
Archetype classification: `get_player_archetype()` → Star Fragger (impact>1.3), Support Anchor (kast>0.75), Sniper Specialist (AWP kills>40%), All-Rounder (default).

-PlayerTokenResolver (`token_resolver.py`)

Resolves **static tokens** (Card) for dynamic comparison:

- `get_player_token(player_name)` → token dict with identity, core_metrics, tactical_baselines, granular_data, metadata
- `compare_performance_to_token(match_stats, token)` → `Correction Delta` with deltas for rating, adr, kast, accuracy_vs_hs, and an `is_underperforming` flag (rating < 85% of the pro)

Token structure:

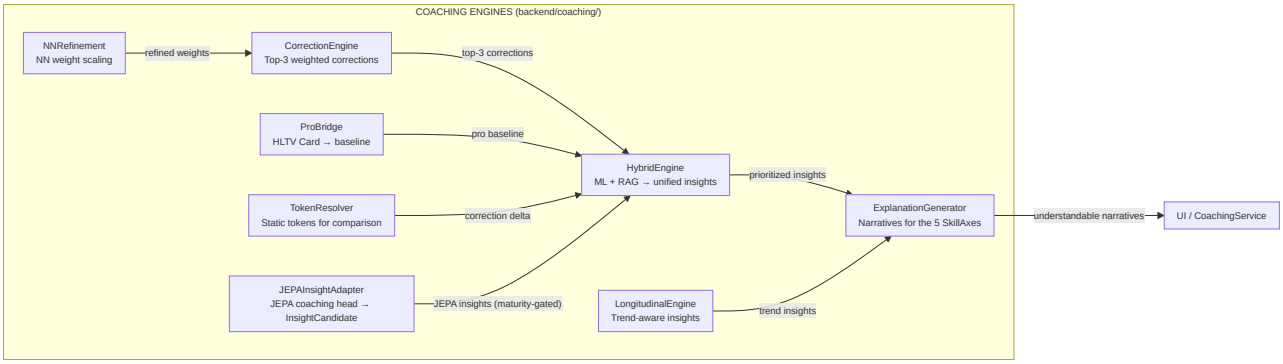


-LongitudinalEngine (`longitudinal_engine.py`)

Generates trend-aware insights by comparing performance trajectories over time with NN stability signals:

- **Input:** `List[FeatureTrend]` (from the progress module) + `nn_signals` (from the training pipeline)

- **Confidence filter:** Only trends with `confidence ≥ 0.6`
- **Output:** Max 3 insights, categorized as:
  - **Regression:** slope < 0 → severity "Medium" (or "High" if `nn_signals.stability_warning` is active)
  - **Improvement:** slope > 0 → severity "Positive", focus "Reinforcement"



-JEPAInsightAdapter (`jepa_insight_adapter.py`)

Converts raw output vectors from the **JEPA Coaching Head** into `InsightCandidate` objects ready for the coaching pipeline (F1.1 / W5.1).

Dimensional contract: Operates on the first 10 elements of the 25-dim contract (target features: health, armor, has_helmet, has_defuser, equipment_value, is_crouching, is_scoped, is_blinded, enemies_visible, pos_x) mapped to five SkillAxes (`survival` , `economy` , `movement` , `utility` , `positioning`).

Maturity ladder:

State	Confidence multiplier	Max candidates	Mode
`doubt` / `crisis`	0.5×	1	Hedged: "Observation: …"
`learning`	0.8×	2	Direct
`conviction` / `mature`	1.0×	3	Direct

Activation flag: `USE_JEPA_MODEL` (default `False`). When off, `generate_jepa_insights()` returns an empty list — no change to existing behavior (F1.2 byte-identical). Respects the **NO-WALLHACK** constraint (F1.3): consumes exclusively the observed player's tick window (25-dim POV contract, P-X-01).

---

6. Subsystem 4 — Knowledge and Retrieval

Folder in the repo: `backend/knowledge/` **Files:** `rag_knowledge.py` , `experience_bank.py`

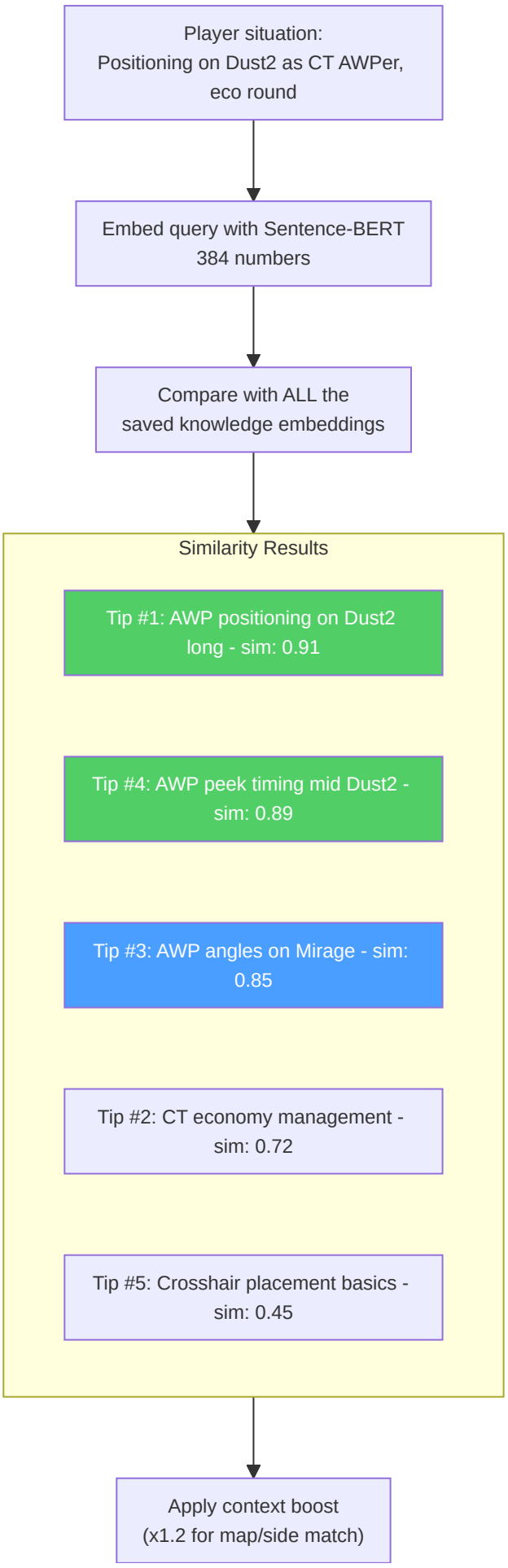
This subsystem manages two complementary data types: **tactical knowledge** (strategies, positioning, utility usage documented and indexed for semantic search) and **coaching experiences** (records of every coaching session with outcome, effectiveness, and match context, used for case-based learning).

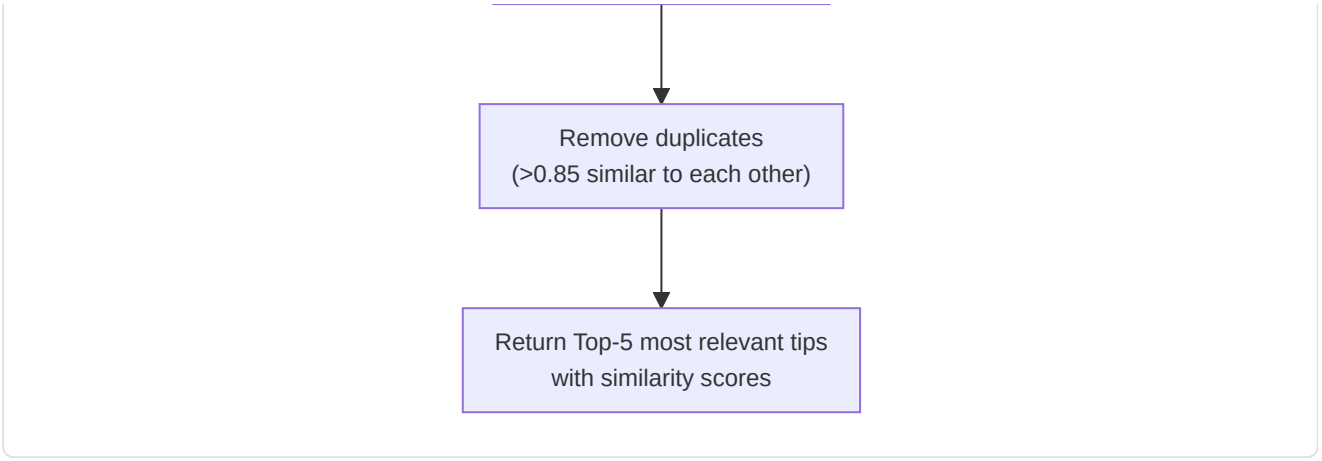
-RAG Knowledge Base (`rag_knowledge.py`)

Implements a **retrieval-augmented generation** pipeline using dense vector similarity search:

Component	Detail
Embedding model	`sentence-transformers/all-MiniLM-L6-v2` (384-dimensional vectors)
Fallback	Hash-based embeddings if Sentence-BERT is unavailable
Storage	`TacticalKnowledge` SQLite table (embedding stored as a JSON-encoded float array)
Retrieval	Cosine similarity via `scipy.spatial.distance.cosine`
Top-k	Configurable, default k=5
Versioning	`CURRENT_VERSION = "v3"` (2026-04, Coach Book refactor, Premier S4 active duty alignment); stale v2 embeddings automatically recomputed via `trigger_reembedding()`
Categories	14: aim, positioning, utility, movement, economy, strategy, crosshair placement, communication, mental, game sense, trading, **mid_round**, **retakes_post_plant**, **aim_and_duels**

> **Fix M-07 — Zero-norm vector rejection:** `VectorIndex.search()` validates the query vector norm before the search. If the norm is zero (typically due to an empty fallback embedding or corrupted input), the method returns `None` with a log warning instead of propagating a division-by-zero error in cosine similarity. This protects the RAG pipeline from degenerate queries without interrupting the coaching flow.

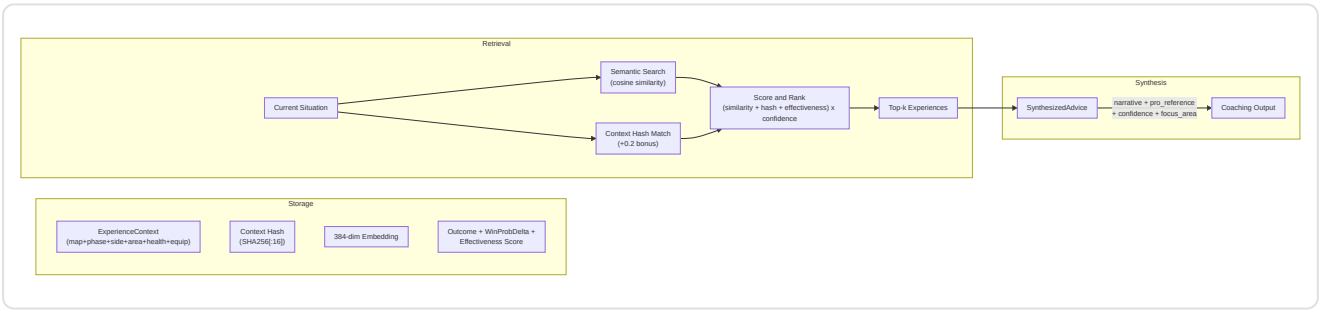

Player situation:
Positioning on Dust2 as CT AWPer,
eco round
Embed query with Sentence-BERT
384 numbers
Compare with ALL the
saved knowledge embeddings
Similarity Results
Tip #1: AWP positioning on Dust2
long - sim: 0.91
Tip #4: AWP peek timing mid Dust2 -
sim: 0.89
Tip #3: AWP angles on Mirage - sim:
0.85
Tip #2: CT economy management -
sim: 0.72
Tip #5: Crosshair placement basics -
sim: 0.45
Apply context boost
(x1.2 for map/side match)



Query construction: natural-language dynamic queries from player stats, map, side, role. Elements matching the context get a 1.2x relevance multiplier. Deduplication filters elements with > 0.85 similarity against already-selected results.

-Experience Bank (`experience_bank.py`) — COPER Framework (KT-01 Enhanced)

Implements the **Contextual Observation–Prediction–Experience–Retrieval (COPER)** framework with CRUD semantics, prioritized replay, and TrueSkill integration:

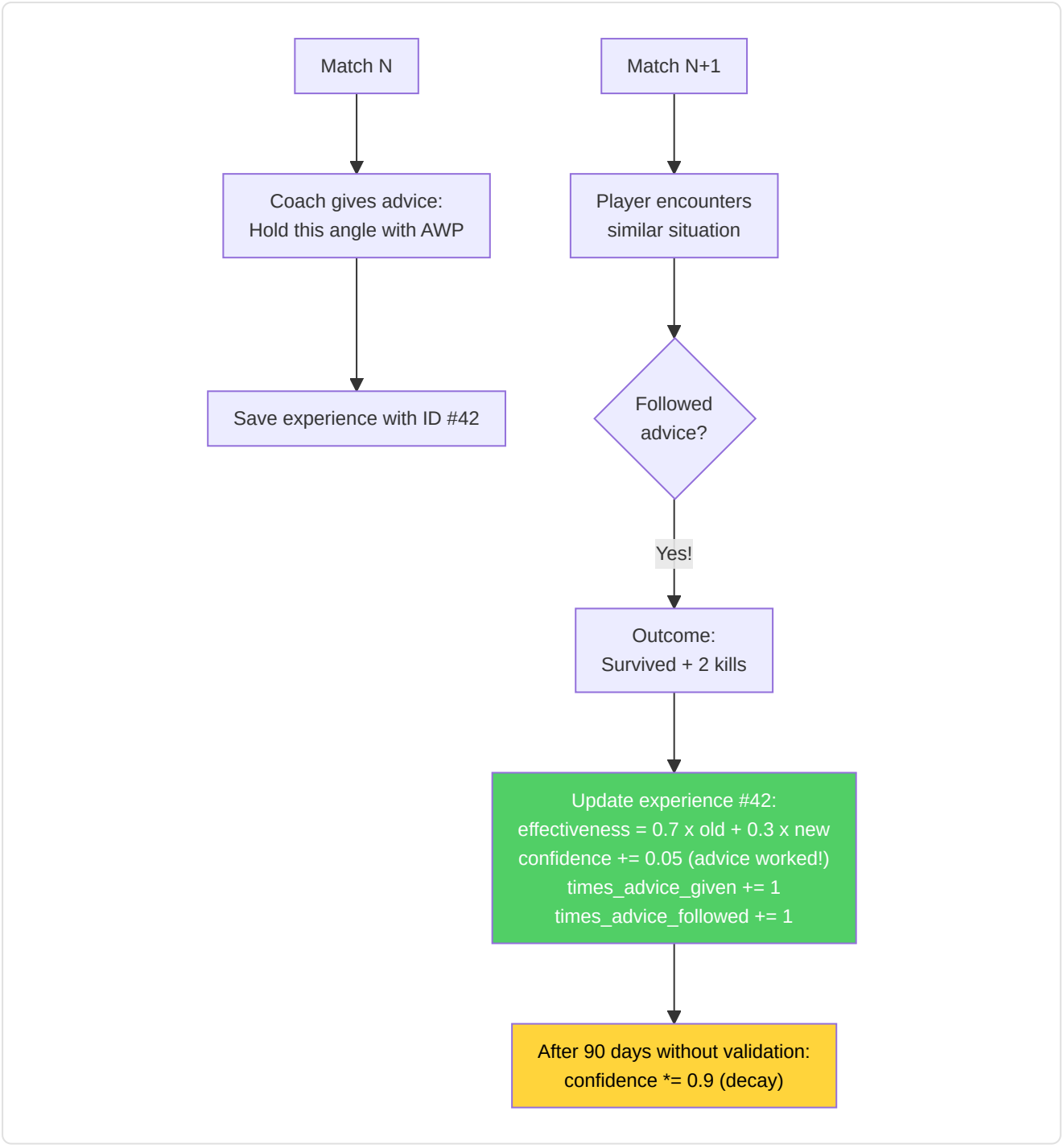


Dual retrieval strategy:

1. **User experiences:** Past situations drawn from the user's game history.
2. **Pro experiences:** How the pros handled analogous situations.
3. **Pattern analysis:** Identifies recurring weaknesses, improvement trends, contextual correlations.

Feedback loop (EMA-based):

- Each experience tracks `outcome_validated`, `effectiveness_score`, `times_advice_given`, `times_advice_followed`
- Follow-up matches update effectiveness: `new_score = 0.7 × old_score + 0.3 × outcome_value`
- Stale experiences (>90 days without validation): confidence decreases by 10%
- Usage tracking increments `usage_count` on every retrieval

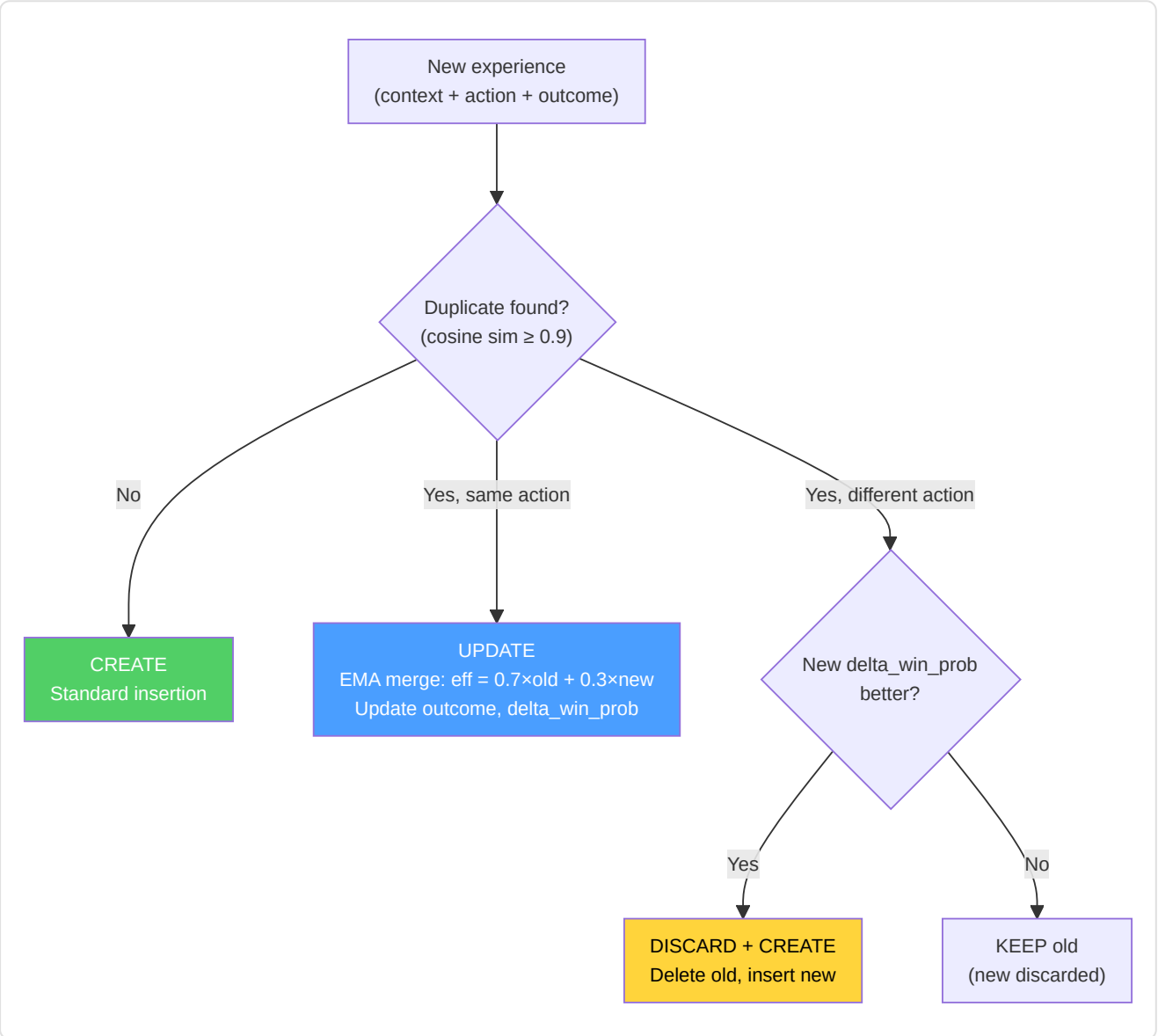


Experience extraction from demos: Groups events by tick, identifies player kills/deaths, creates context from a tick snapshot, infers the action (scoped_hold, crouch_peek, pushed, held_angle), determines the outcome.

KT-01 enhancements — CRUD semantics and prioritized replay:

Constant	Value	Purpose
`DUPLICATE_SIMILARITY_THRESHOLD`	0.9	Cosine similarity for duplicate detection
`CRUD_EMA_FACTOR`	0.3	EMA weight for effectiveness merge on UPDATE
`REPLAY_ALPHA`	0.6	Replay priority exponent (lower = more uniform)
`REPLAY_GATE`	0.4	Minimum confidence to be eligible for replay
`_MIN_EFFECTIVENESS_TRIALS`	5	Minimum trials before effectiveness affects retrieval

CRUD decision on insertion:



Prioritized replay (KT-01): Experiences are sampled for replay with probability proportional to `priority^REPLAY_ALPHA`, where `priority = effectiveness_score × confidence_score`. Only experiences with `confidence_score ≥ REPLAY_GATE` are eligible. This balances exploitation (effective experiences) with exploration (less tested experiences).

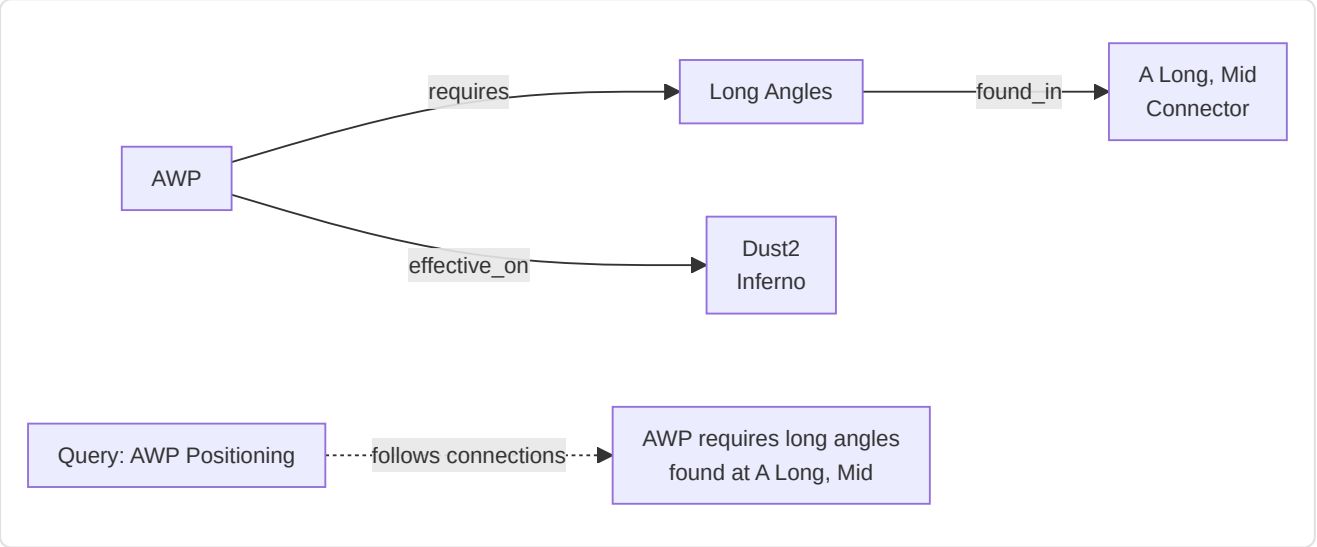
TrueSkill integration (KT-01): `mu_skill` and `sigma_skill` fields for Bayesian tracking of player competence in the specific situation. TrueSkill priors influence the weight of the experience in retrieval: experiences with high uncertainty (`sigma` high) are penalized relative to those with a stable signal.

Embedding compression: 384-dim embeddings are now encoded as `base64(float32)` instead of a JSON array, achieving 4× compression on database storage space without loss of precision.

Pro reference linking: Each experience can include `pro_player_name`, `pro_match_id`, `source_demo` to link directly to how a specific professional handled an analogous situation.

-Knowledge Graph

A lightweight **entity-relation graph** stored in SQLite (`kg_entities`, `kg_relations` tables). Supports `query_subgraph(entity_name)` at 1 hop for multi-hop reasoning to complement semantic similarity.



-Knowledge Base Initialization (`init_knowledge_base.py`)

Orchestration script that populates the RAG database with tactical knowledge from two sources:

1. **Manual knowledge:** Loading from JSON files (`data/tactical_knowledge.json`) with manually curated tips per category and map
2. **Automatic mining:** Invocation of `ProDemoMiner.mine_all_pro_demos()` to extract tactical patterns from professional demos

After loading, it generates a report per category and map using aggregate COUNT queries (without loading every record into memory).

-ProDemoMiner (`pro_demo_miner.py`)

Extracts tactical knowledge from professional demos using pattern detection:

Class	Lines	Description
`ProDemoMiner`	~180	Pattern extraction from match metadata (map, team, success rate)
`AdvancedProDemoMiner(ABC)`	~60	Abstract base (F5-05) for demo parsing with demoparser2

Quality thresholds:

Constant	Value	Meaning
`MIN_SUCCESS_RATE`	0.70	Pattern accepted only if success ≥70%
`MIN_SAMPLE_SIZE`	5	At least 5 occurrences for a valid pattern

Supported maps: mirage, dust2, inferno, nuke, overpass, vertigo, ancient, anubis.

Types of knowledge generated:

- Map-specific knowledge (positioning, utility, rotations)
- Team-specific knowledge (strategies, tendencies, styles)
- Successful-execution knowledge (patterns with ≥70% rate)

-Round Utils (`round_utils.py`)

Shared utility for round economic phase classification, extracted to eliminate duplication between the knowledge and services layers:

Equipment Value	Phase	Constant
< $1,500	`pistol`	`_PISTOL_MAX_EQUIP`
$1,500 – $2,999	`eco`	`_ECO_MAX_EQUIP`
$3,000 – $3,999	`force`	`_FORCE_MAX_EQUIP`
≥ $4,000	`full_buy`	—

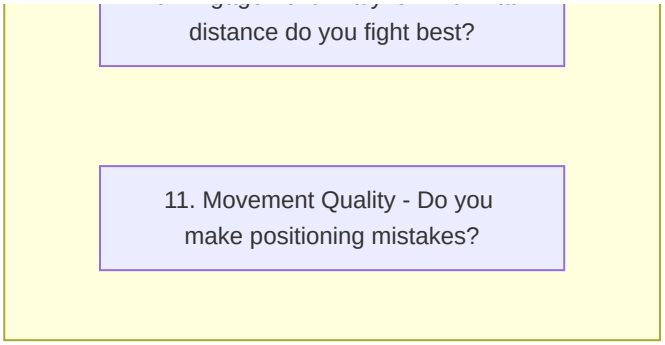
---

7. Subsystem 5 — Analysis Engines

Folder in the repo: `backend/analysis/` **11 files, ~2,600 lines of production code**

This subsystem contains **11 specialized analysis engines**, each designed to investigate a different dimension of gameplay. They run as Phase 6 analyses, providing insights that go beyond what neural networks alone can offer.


THE 11 ANALYSIS ENGINES — YOUR SPECIALIST TEAM
1. Role Classifier - What position do you play?
2. Win Probability - What are the odds right now?
3. Game Tree - What is the best move?
4. Bayesian Death - How likely are you to die?
5. Deception Index - How unpredictable are you?
6. Momentum Tracker - Are you on fire or on tilt?
7. Entropy Analyzer - Are your grenades effective?
8. Blind Spot Detector - What mistake do you repeat?
9. Utility & Economy - Buy optimizer
10. Engagement Analyzer - At what



-Role Classifier (`role_classifier.py`)

Assigns one of 6 roles using **learned statistical thresholds**:

Role	Primary signal	Secondary signal
AWPer	AWP kill ratio vs threshold	—
Entry Fragger	Entry rate + first-death bonus (0.3×)	—
Support	Assist rate + utility damage bonus (max 0.3×)	—
IGL	Survival rate + KD balance bonus	—
Lurker	Solo kill ratio vs threshold	—
Flex	Fallback when confidence is low	—

Cold-start protection: `RoleThresholdStore` requires ≥10 samples and ≥3 valid thresholds to exit cold-start. Returns `(FLEX, 0.0)` if in cold-start. Thresholds are **persisted in the database** via `persist_to_db()` and `load_from_db()` — fully implemented, not stubs.

Team balance audit (`audit_team_balance()`): detects multiple AWPers (HIGH), missing Entry (HIGH), missing Support (MEDIUM), no diversity (CRITICAL), multiple Lurkers (MEDIUM).

-Win Probability Predictor (`win_probability.py`)

12-feature neural network that estimates P(round_win | game_state):

Architecture: `Linear(12, 64) → ReLU → Dropout(0.2) → Linear(64, 32) → ReLU → Dropout(0.1) → Linear(32, 1) → Sigmoid`.

12 Features:

#	Feature	Normalization
1	team_economy	/16000
2	enemy_economy	/16000
3	economy differential	(team−enemy)/16000
4	players_alive	/5
5	enemies_alive	/5
6	player count differential	(alive−enemy)/5
7	utility_remaining	/5
8	map_control_pct	[0, 1]
9	time_remaining	/115
10	bomb_planted	binary
11	is_ct	binary
12	equipment value ratio	min(team/enemy, 2)/2

Heuristic overrides: 3+ advantage → floor at 85%, 3+ disadvantage → ceiling at 15%, 0 alive → 0%, planted-bomb adjustments (T: +0.10, CT: −0.10) — additive on base probability, economic caps of ± $8000.

> **Note A-12 — Cross-load guard:** This 12-feature predictor (`WinProbabilityNN`) is a *separate and incompatible* model from the 9-feature `WinProbabilityTrainerNN` described in Section 12. The checkpoints are not interchangeable: at load time, the `state_dict` dimensionality is validated and, on mismatch, the model is re-initialized from scratch with a log warning.

Elo-Augmented Predictor (KT-07):

The `EloAugmentedPredictor` wraps the base `WinProbabilityNN` with an optional Elo system to leverage match history:

Constant	Value	Description
`_ELO_INITIAL`	1500.0	Initial Elo for players with no history
`_ELO_K_FACTOR`	32.0	Base K-factor for update magnitude
`_ELO_RECENCY_HALF_LIFE`	20 matches	A match's weight halves every 20 matches

Elo update formula with recency weight:

```
new_elo = old_elo + K × w × (S - E)

where:
  S = actual score (1 for win, 0 for loss)
  E = expected score = 1 / (1 + 10^((opp_elo - elo) / 400))
  w = recency weight = 2^((match_index - N + 1) / half_life)
```

The recency weight (adapted from Glickman, 1999) ensures that recent matches contribute more to the final rating. A match from 20 matches ago contributes half of the K-factor compared to the latest match.

NN + Elo blending:

```
final_prob = (1 - α) × nn_prob + α × elo_prob    (α = 0.15 default)
```

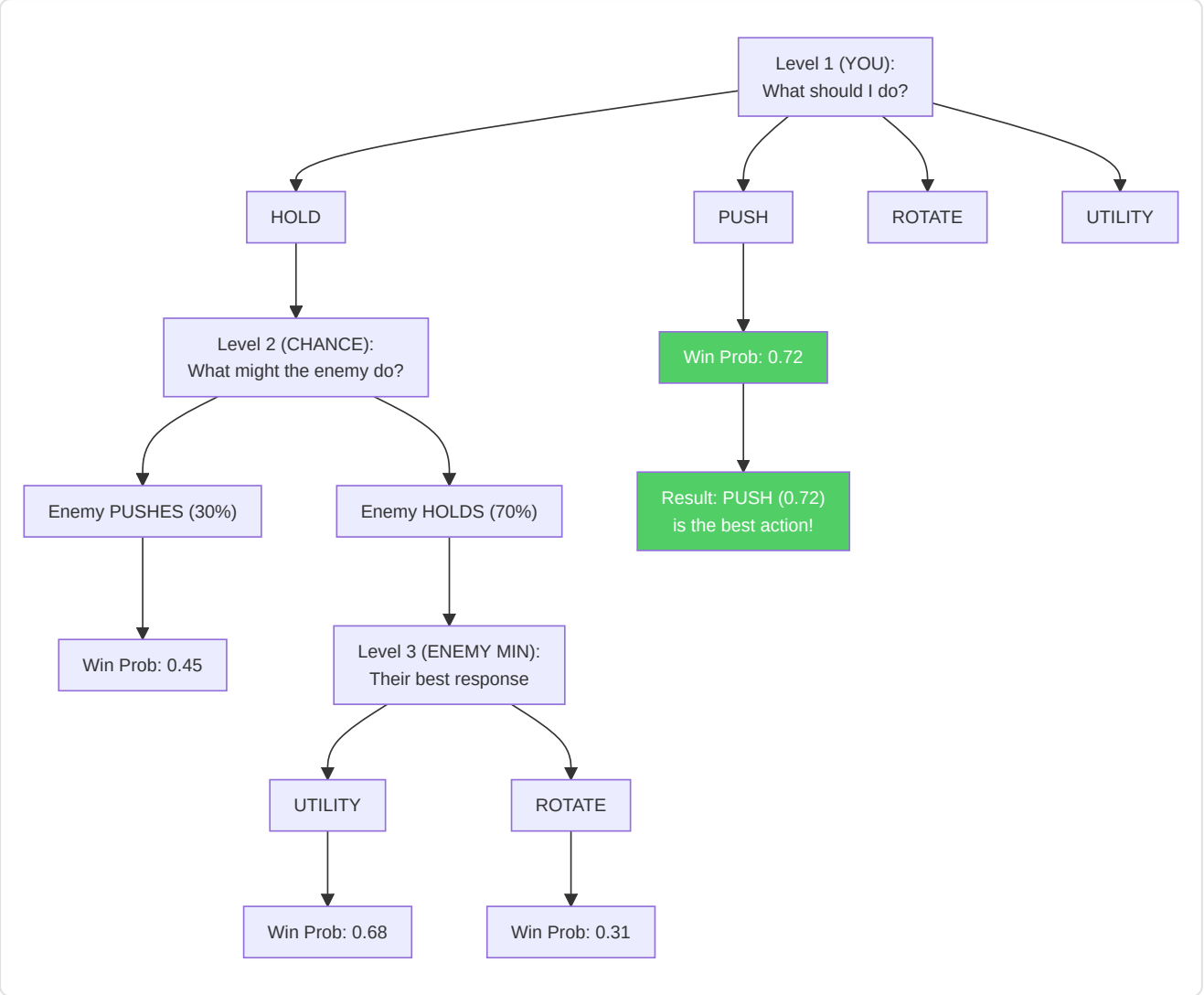
`compute_elo_differential(team_histories, enemy_histories)` computes the average Elo differential between the two teams, normalized by 400 (one Elo "class"), and converts it to probability via the standard logistic formula. The blend is conservative (α = 0.15) because Elo only captures historical information, while the NN sees the current round state.

Note: Elo is an **optional augmentation** — the base 12-feature architecture of `WinProbabilityNN` remains unchanged. If history is unavailable, the predictor falls back to pure NN probability.

-Expectiminimax Game Tree (`game_tree.py`)

Implements **expectiminimax search** with adaptive opponent modeling:

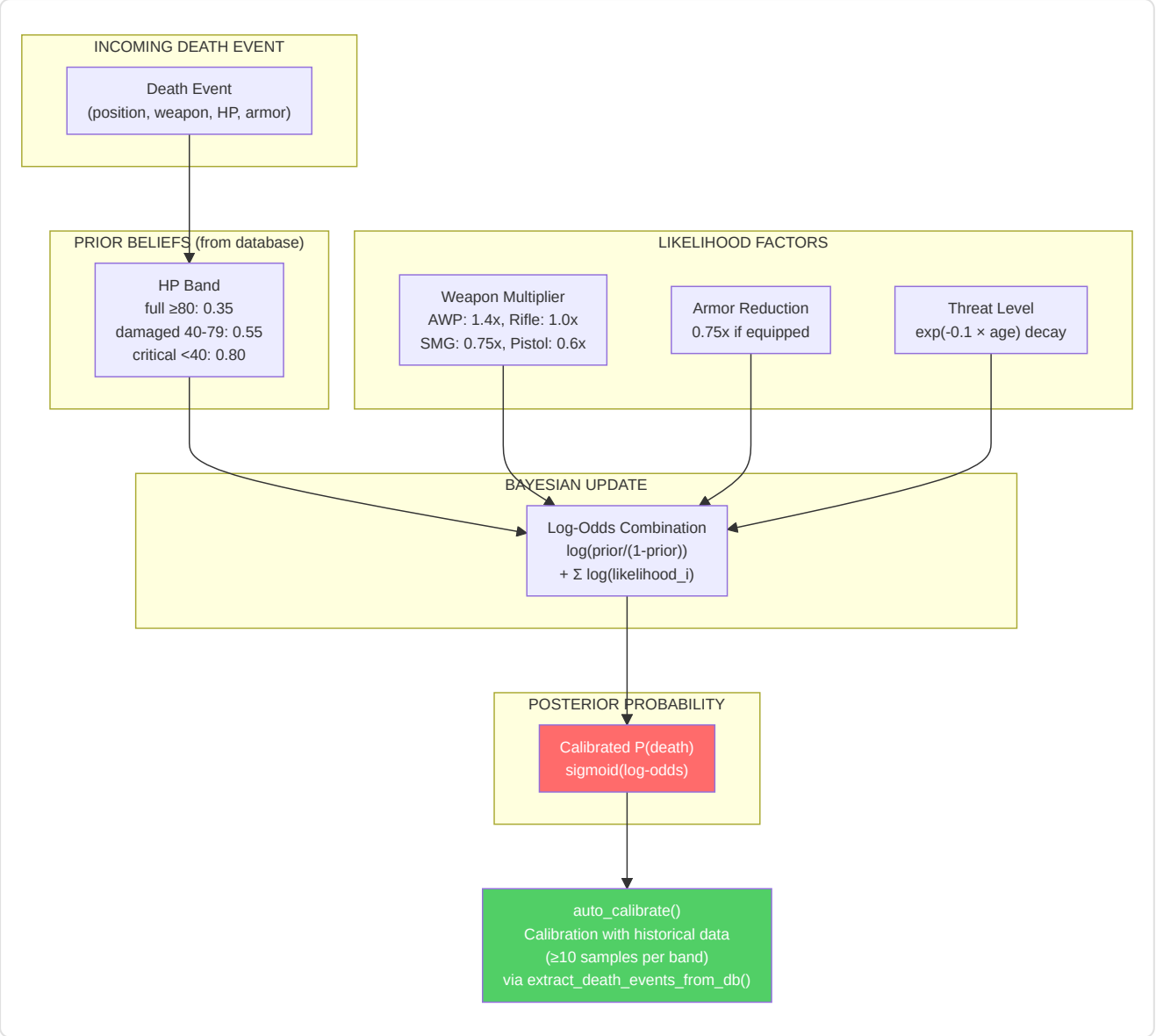
- **Actions:** push, hold, rotate, use_utility
- **Opponent model:** Economic priorities (eco/force/full buy), side adjustments, advantage adjustments, time pressure
- **Depth:** 3 levels (max → chance → min)
- **Node budget:** 1000 (prevents explosion)
- **Leaf evaluation:** `WinProbabilityPredictor` (lazy load)
- **Opponent learning:** Incremental EMA update (α clipped at 0.5)



-Bayesian Death Estimator (`belief_model.py`)

Models P(death | belief, HP, armor, weapon_class):

- **Prior:** Death rates per HP band (full ≥80: 0.35, damaged 40-79: 0.55, critical <40: 0.80)
- **Likelihood factors:** Threat level (with exponential decay exp(−0.1 × age)), armor reduction (0.75×), weapon multipliers (AWP: 1.4×, Rifle: 1.0×, SMG: 0.75×, Pistol: 0.6×, Knife: 0.3×)
- **Posterior:** Logistic combination in log-odds space
- **Calibration:** `calibrate(historical_rounds)` learns empirical priors (≥10 samples per band)



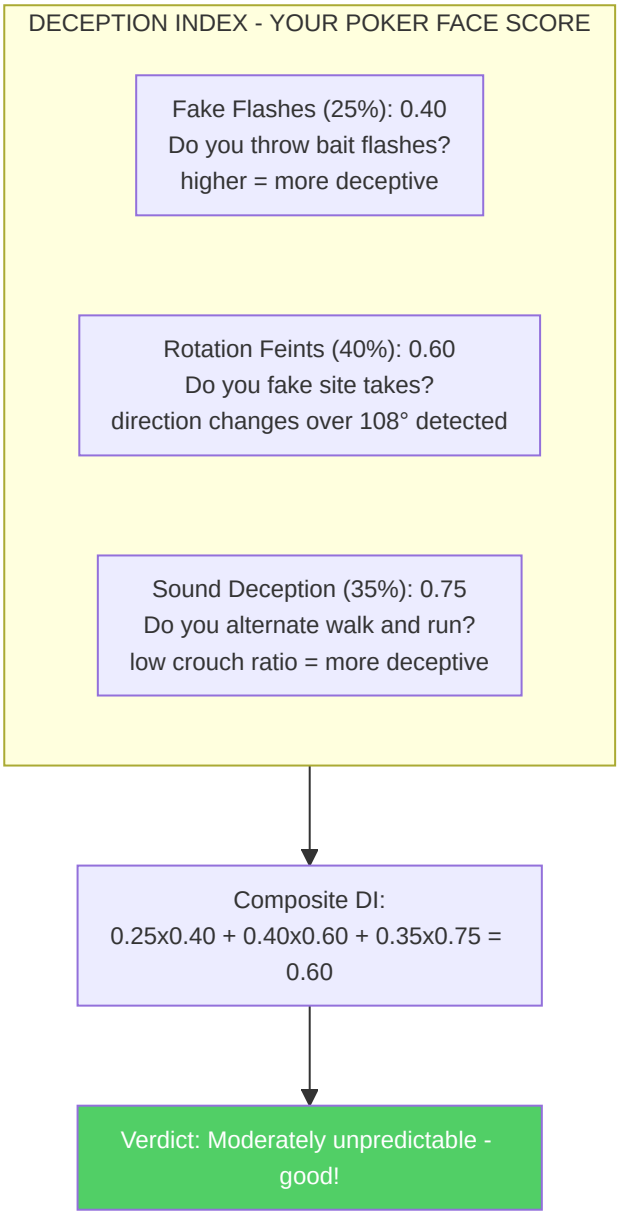
> **Calibration note (G-07):** The Bayesian Death Estimator is now a **live component** thanks to the wiring completed during remediation. The `extract_death_events_from_db()` function in the Session Engine automatically extracts death events from the database and passes them to `auto_calibrate()`, allowing the model to refine its priors based on real accumulated data. This feedback loop turns the estimator from a static model into a self-calibrating system driven by experience.

-Deception Index (`deception_index.py`)

Quantifies tactical deception via three sub-metrics:

Sub-metric	Weight	Detection method
Fake-flash frequency	0.25	Flashes that do not blind enemies — `bait_rate = 1 - effective/total`
Rotation-feint frequency	0.40	Direction changes >108° detected via angular velocity sampling (20 position intervals)
Sound-deception score	0.35	Inverse of crouch ratio — `1.0 - crouch_ratio × 2.0`

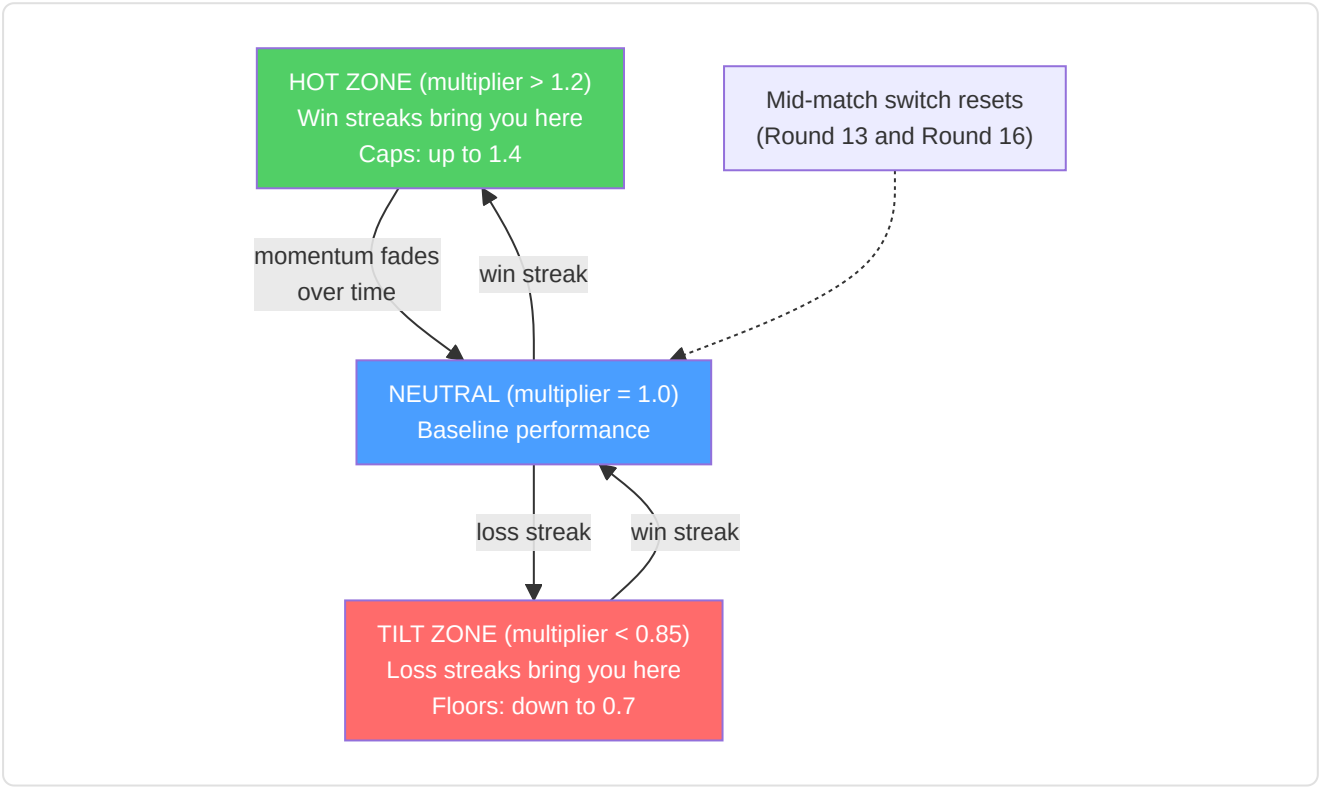
Composite: `DI = 0.25·fake_flash + 0.40·rotation_feint + 0.35·sound_deception`, clamped to [0, 1].



-Momentum Tracker (`momentum.py`)

Models psychological momentum as a performance multiplier that decays over time:

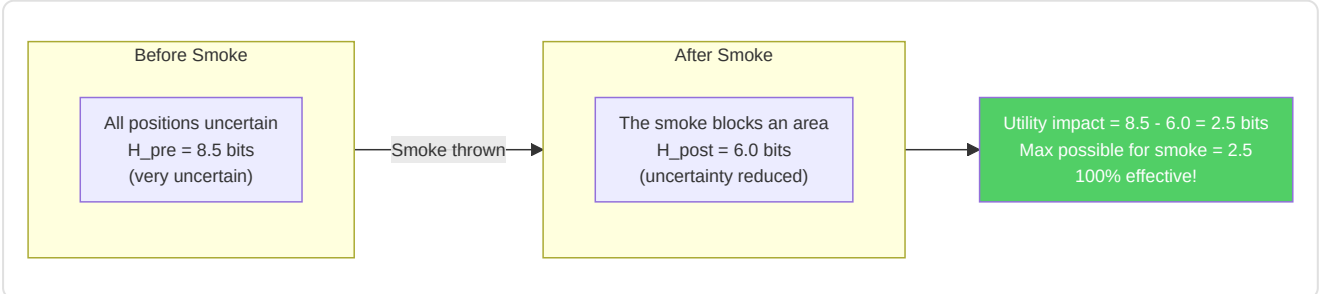
- Winning streak: multiplier = 1.0 + 0.05 × streak length × decay
- Losing streak: multiplier = 1.0 − 0.04 × streak length × decay
- Decay: exp(−0.15 × gap_rounds)
- Bounds: [0.7, 1.4]
- Tilt detection: multiplier < 0.85
- Hot detection: multiplier > 1.2
- Mid-switch reset: Round 13 (MR12) and 16 (MR13)



-Entropy Analyzer (`entropy_analysis.py`)

Measures utility effectiveness via **Shannon entropy reduction** of enemy positions:

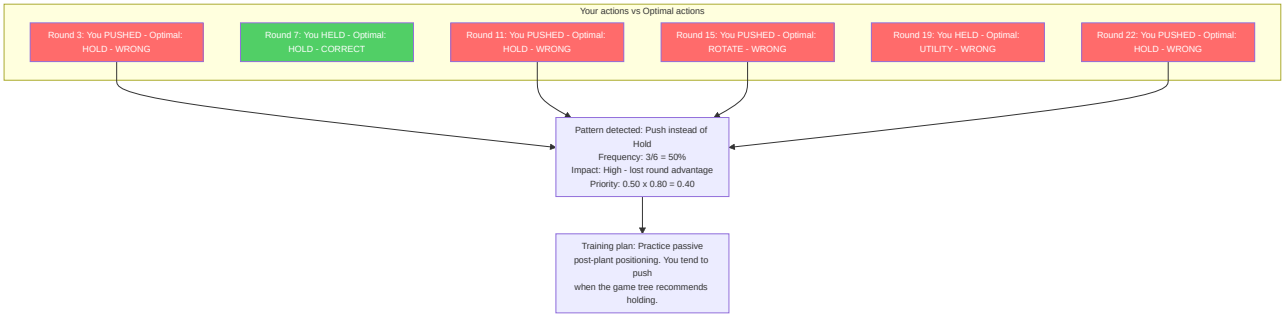
- Discretizes positions into a 32×32 grid
- Computes `H = −Σ p(cell) × log₂(p(cell))`
- Utility impact = `H_pre − H_post` (positive = information gained)
- Maximum entropy reductions: Smoke 2.5 bits, Molotov 2.0, Flash 1.8, HE 1.5



-Blind Spot Detector (`blind_spots.py`)

Identifies recurring suboptimal decisions relative to the game tree's recommendations:

- Compares real player actions with the optimal actions of `ExpectiminimaxSearch`
- Classifies situations (post-plant, clutch, eco, late round, numerical advantage)
- Priority = `frequency × impact_rating`
- Generates natural-language training plans for the top blind spots



-Engagement Range Analyzer (`engagement_range.py`)

Analyzes kill distances to build role- and position-specific **engagement profiles**:

Core components:

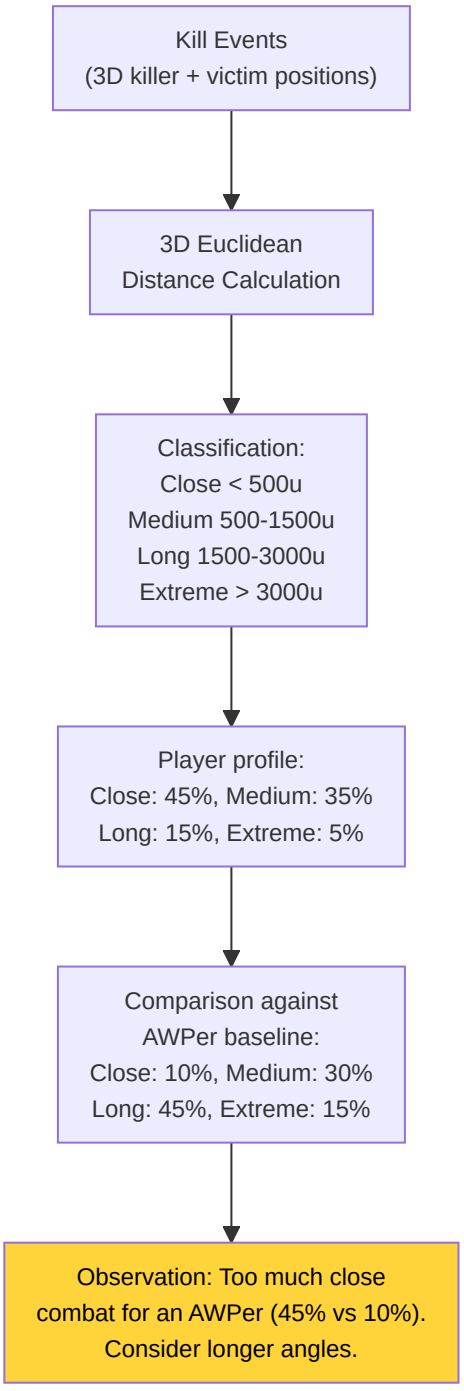
Component	Purpose
`NamedPositionRegistry`	Registry of per-map callouts (e.g. "A Site", "Window", "Banana") with 3D coordinates and radius
`EngagementRangeAnalyzer`	Euclidean killer-victim distance calculation, classification, and comparison against pro baselines
`EngagementProfile`	Distribution % per band: close (<500u), medium (500-1500u), long (1500-3000u), extreme (>3000u)

Pro baselines by role:

Role	Close	Medium	Long	Extreme
AWPer	10%	30%	45%	15%
Entry Fragger	40%	40%	15%	5%
Support	25%	45%	25%	5%
Lurker	35%	35%	20%	10%
IGL/Flex	25%	40%	25%	10%

Deviation threshold: A >15% difference from the role baseline generates a coaching observation (e.g. "More close kills than typical for an AWPer — consider longer angles").

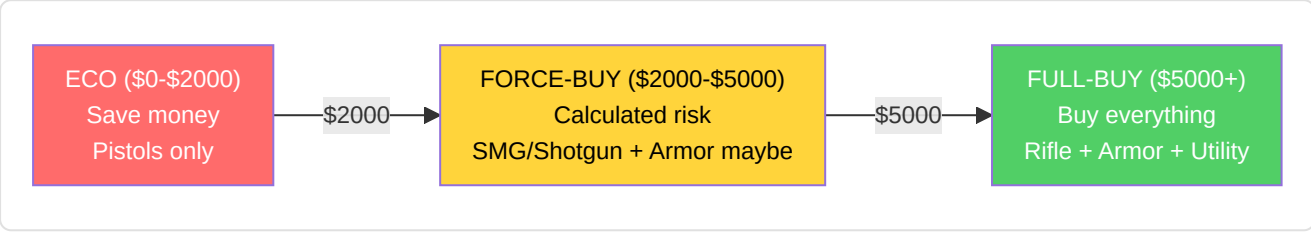
Supported maps: de_mirage, de_inferno, de_dust2, de_anubis, de_nuke, de_ancient, de_overpass, de_vertigo, de_train (not in the current Active Duty pool, supported for historical/workshop demos) — extendable via JSON.



-Utility and Economy Analyzer (`utility_economy.py`)

Utility analyzer: Effectiveness score per type vs pro baselines (Molotov: 35 damage/throw, Flash: 1.2 enemies/flash, etc.)

Economy optimizer: Buy advice based on economic thresholds ($5000 full buy, $2000 force, <$2000 eco), round context, score differential, and loss bonus.



-Movement Quality Analyzer (`movement_quality.py`)

Detects 4 common positioning mistakes based on the MLMove paper (SIGGRAPH 2024, Stanford/Activision/NVIDIA):

4 types of mistakes detected:

#	Type	Detection condition	Description
1	`high_ground_abandoned`	Descent ≥100 units without combat context	High ground abandoned without need
2	`position_abandoned`	Position held ≥3s left without new enemy info	Consolidated position abandoned
3	`over_aggressive_trade`	Solo push after teammate death with <2 teammates left	Too aggressive trading
4	`over_passive_support`	Idle when a teammate creates an opening + numerical advantage	Too passive support

Key thresholds:

Constant	Value	Meaning
`_ESTABLISHED_HOLD_TICKS`	384 (3s)	Minimum time for "established position"
`_HIGH_GROUND_DROP`	100.0 units	Minimum descent to flag high ground
`_TRADE_WINDOW_TICKS`	640 (5s)	Time window for trade analysis
`_AUDIO_RANGE_DISTANCE`	1500.0 units	Maximum distance for "within audio range"
`_MOVEMENT_THRESHOLD`	300.0 units	Minimum displacement to count as "moved"
`_COMBAT_PROXIMITY_TICKS`	64 (~0.5s)	Ticks around a kill/death event for "in combat" context

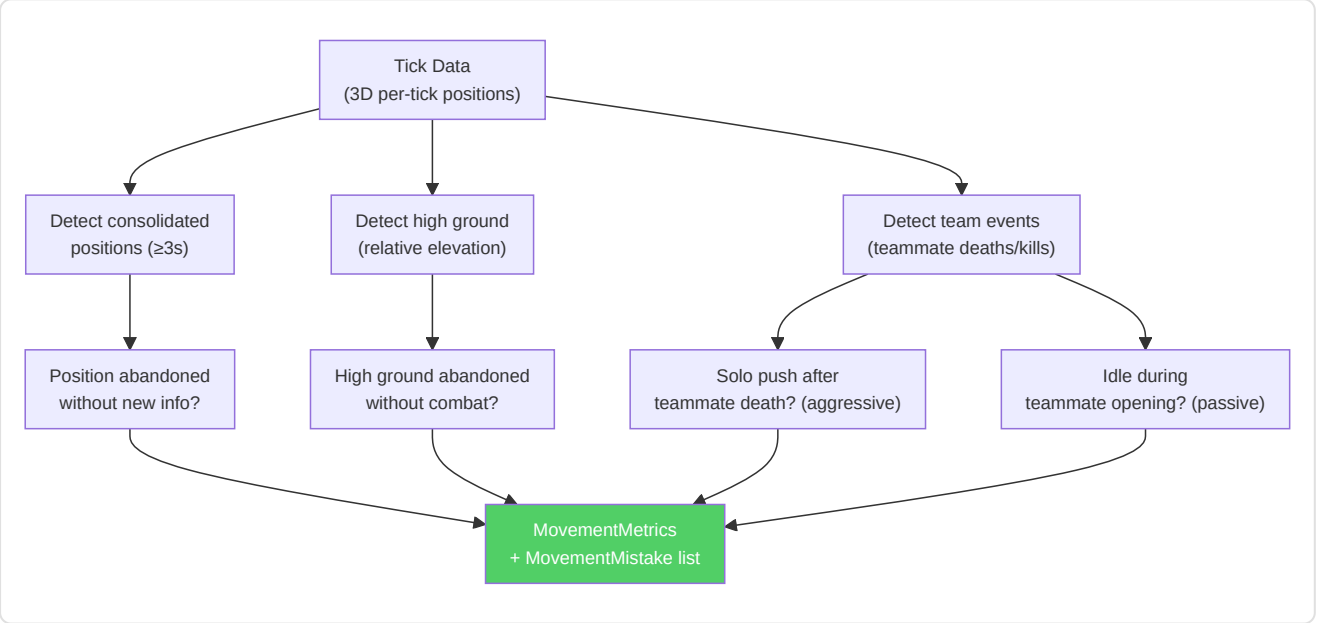
Output dataclasses:

- `MovementMistake`: type, round, tick, time in round, description, callout (map position), severity [0-1]
- `MovementMetrics`: map_coverage_score, high_ground_utilization, position_stability, total_rounds_analyzed, mistakes list + `mistakes_per_round` property

Public API:

Method	Input	Output
`analyze_round_ticks(ticks, map_name, player, round)`	Ticks for a round	`List[MovementMistake]`
`analyze_match_ticks(all_ticks, map_name, player)`	All match ticks	`MovementMetrics` (aggregated)
`get_movement_quality_analyzer()`	—	`MovementQualityAnalyzer` singleton

ADDITIVE: Does NOT modify METADATA_DIM=25. Movement metrics are computed as features derived from existing tick data, stored in analysis results.



Summary of the 11 Analysis Engines

#	Engine	File	Input	Output	Complexity
1	Role Classifier	`role_classifier.py`	PlayerMatchStats	Role (6 classes) + confidence	O(n) features
2	Win Probability	`win_probability.py`	12 round state features	P(CT win) ∈ [0,1]	O(1) forward pass
3	Game Tree	`game_tree.py`	Round state + actions	Optimal node (minimax)	O(b^d) branching
4	Bayesian Death	`belief_model.py`	Position + time + round	P(death) + risk factors	O(n) prior update
5	Deception Index	`deception_index.py`	Round history + positions	Unpredictability score [0,1]	O(n×m) pattern match
6	Momentum Tracker	`momentum.py`	Round sequence	hot/cold/neutral state	O(n) sliding window
7	Entropy Analyzer	`entropy_analysis.py`	Utility damage per type	Effectiveness score vs pro	O(k) per utility type
8	Blind Spot Detector	`blind_spots.py`	Death positions + angles	Repeated patterns	O(n²) clustering
9	Utility & Economy	`utility_economy.py`	Round economy + utility	Buy advice + rating	O(1) threshold check
10	Engagement Analyzer	`engagement_range.py`	3D kill events	Distance profile (4 bands)	O(n) Euclidean dist
11	Movement Quality	`movement_quality.py`	3D per-round tick data	MovementMetrics (4 mistake types)	O(n) per-tick scan

---

8. Subsystem 6 — Processing and Feature Engineering

Directory: `backend/processing/`

This subsystem handles all **data preparation**, transforming raw game recordings into the precise numerical formats the neural networks need for training and inference.

-Unified Feature Extractor (`vectorizer.py`)

The **single source of truth** for tick-level feature vectors. Both training (`RAPStateReconstructor`) and inference (`GhostEngine`) MUST use this class.

25-dimensional feature vector contract:

Index	Feature	Normalization	Range
0	`health`	`/100`	[0, 1]
1	`armor`	`/100`	[0, 1]
2	`has_helmet`	binary	{0, 1}
3	`has_defuser`	binary	{0, 1}
4	`equipment_value`	`/10000`	[0, 1]
5	`is_crouching`	binary	{0, 1}
6	`is_scoped`	binary	{0, 1}
7	`is_blinded`	binary	{0, 1}
8	`enemies_visible`	`/5` (clipped)	[0, 1]
9	`pos_x`	`/4096`	[−1, 1]
10	`pos_y`	`/4096`	[−1, 1]
11	`pos_z`	`/1024`	[−1, 1]
12	`view_yaw_sin`	`sin(yaw_rad)`	[−1, 1]
13	`view_yaw_cos`	`cos(yaw_rad)`	[−1, 1]
14	`view_pitch`	`/90`	[−1, 1]
15	`z_penalty`	`compute_z_penalty()`	[0, 1]
16	`kast_estimate`	KAST from stats or 0.70 default	[0, 1]
17	`map_id`	Deterministic hash-based encoding	[0, 1]
18	`round_phase`	0=pistol, 0.33=eco, 0.66=force, 1=full	[0, 1]
19	`weapon_class`	Weapon class mapping (0-1)	[0, 1]
20	`time_in_round`	`/115` (seconds in round)	[0, 1]
21	`bomb_planted`	binary	{0, 1}
22	`teammates_alive`	`/4` (alive teammates)	[0, 1]
23	`enemies_alive`	`/5` (alive enemies)	[0, 1]
24	`team_economy`	`/16000` (average team money)	[0, 1]

Design decisions:

- **Cyclical yaw encoding** (sin/cos at indices 12-13) eliminates the ±180° discontinuity
- **Z penalty** (index 15) quantifies wrong-level risk for multilevel maps
- **Tactical context integration** (indices 19-24) gives the model game-situation awareness
- **KAST estimate fallback chain:** explicit value → stats-based computation → 0.70 default
- **Map identity encoding:** deterministic hash enables map-specific learning

- **HeuristicConfig** (`base_features.py`) allows overriding all normalization bounds via JSON

> **Design decisions:** The **cyclical yaw encoding** (sin/cos) solves angular discontinuity: an angle of -179° and +179° differ by only 2° in reality but by 358° in linear representation. The (sin θ, cos θ) pair maps the angle onto a unit circle where Euclidean distance reflects the actual angular distance. The **Z penalty** (`z_penalty` in the 25-dim vector) explicitly flags the risk of vertical floor error — critical on multilevel maps (Nuke, Vertigo) where confusing upper/lower leads to completely wrong tactical decisions. **Tactical integration** (economy, alive counts, time) provides the model with the context needed to evaluate whether an aggressive play is correct given the numerical advantage, economic budget, and remaining time in the round.

-HLTV 2.0 Rating (`rating.py`)

The **unified rating module** that prevents training-inference skew:

```
R = (R_kill + R_survival + R_kast + R_impact + R_damage) / 5

where:
R_kill = KPR / 0.679
R_survival = (1 − DPR) / 0.317
R_kast = KAST / 0.70
R_impact = (2.13·KPR + 0.42·ADR/100) / 1.0
R_damage = ADR / 73.3
```

Used by: demo_parser.py (analysis), base_features.py (aggregation), coaching_service.py (insights).

-PlusMinus and Role-Adjusted Rating Metrics (`rating.py`) — KT-06

Complementary module to the HLTV 2.0 rating that provides two additional metrics designed to capture aspects Rating 2.0 overlooks:

PlusMinus:

```
PlusMinus = (kills - deaths) / max(rounds_played, 1) + team_contribution_bonus
```

Component	Formula	Typical range
Net frag differential	`(kills - deaths) / rounds`	[-1.0, +1.0]
Team contribution bonus	`_TEAM_CONTRIBUTION_SCALE × (team_win_rate - 0.5)`	[-0.05, +0.05]
`_TEAM_CONTRIBUTION_SCALE`	0.10	—

The team-contribution bonus rewards players on winning teams and penalizes those on losing teams, analogous to +/- in basketball/hockey.

Role-Adjusted Rating (Bayesian):

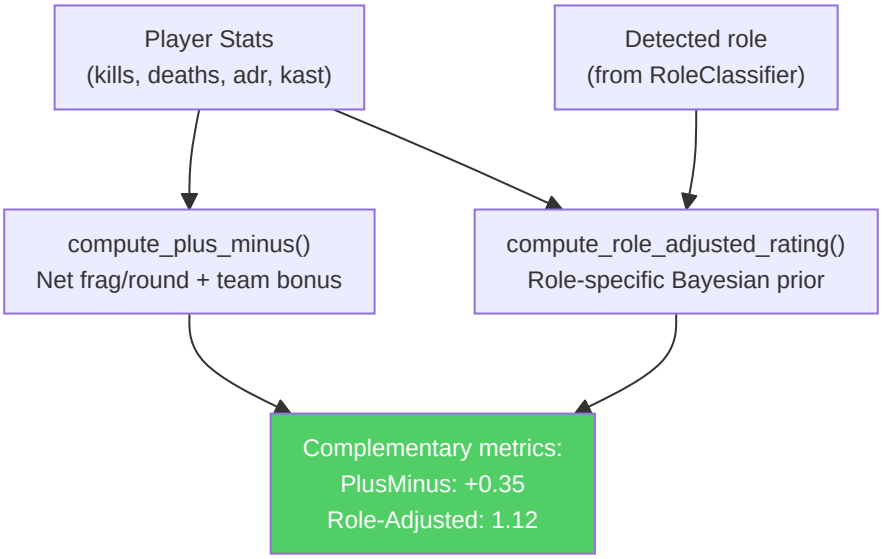
Applies role-specific Bayesian priors so that AWPers are not penalized for lower KAST and supports are not penalized for lower K/D:

Role	K/D Prior	KAST Prior	ADR Prior	Weight
AWPer	1.15	0.68	75.0	5.0
Entry	0.95	0.72	80.0	5.0
Support	0.90	0.78	65.0	5.0
Lurker	1.05	0.70	72.0	5.0
IGL	0.88	0.74	68.0	5.0

Priors calibrated from the average data of HLTV top-30 teams (2024-2025 season). Composite formula:

```
adj_metric = (n × observed + weight × prior) / (n + weight)
role_adjusted_rating = 0.40 × adj_kd + 0.35 × adj_kast + 0.25 × adj_adr_norm
```

Where `adj_adr_norm = adj_adr / 120` normalizes ADR to [0, 1]. The framework is inspired by TrueSkill (Herbrich et al., NeurIPS 2006) — when the sample is small (low `n`), the prior dominates; when the sample is large, the observed data dominates.

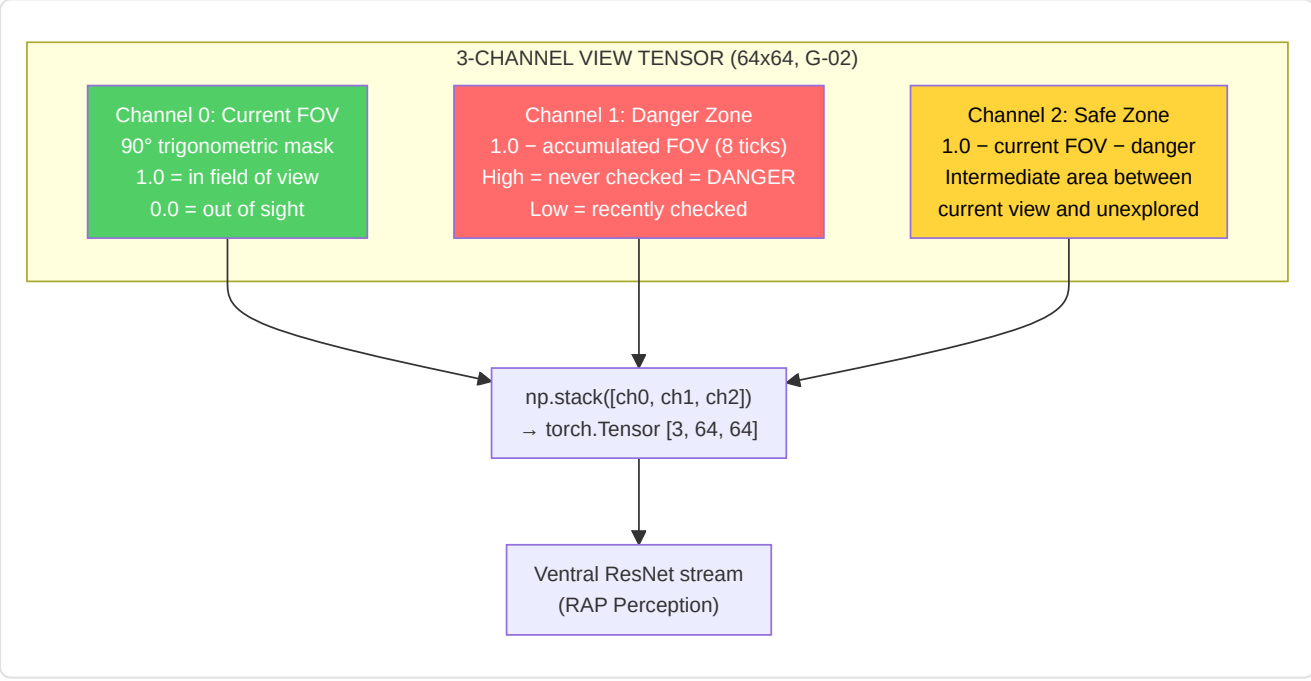


-Tensor Factory (`tensor_factory.py`)

Converts raw tick data into 64x64 image tensors for the RAP perception layer:

Tensor	Channels	Content
Map	3 (R/G/B)	R: player position, G: teammates (α-blended), B: enemies (α-blended)
View	3	**Ch0:** current FOV mask (90° default, trigonometric masking); **Ch1:** danger zone (= 1 − FOV accumulated over the last 8 ticks), areas never checked are potential enemy positions; **Ch2:** safe zone (= 1 − current FOV − danger zone), the area between current view and unexplored zones
Motion	2 or 3	Velocity/acceleration heatmaps (Gaussian-blurred 2D blobs)

> **Fix G-02 (Danger Zone):** Previously the view tensor had 3 identical channels (placeholder). After remediation, the 3 channels encode distinct tactical information: the **current FOV** shows what the player sees now, the **danger zone** accumulates the FOV history of the last 8 ticks (~125ms at 64 Hz) and inverts the result to identify areas never checked — where enemies could hide — and the **safe zone** represents the intermediate area between current view and unexplored zones. The danger zone calculation uses `np.maximum(accumulated_fov, tick_fov)` for each historical tick, then `danger_zone = 1.0 - accumulated_fov`. This gives the RAP model spatial awareness of visual coverage, turning the view tensor from a simple static input into a **temporal tactical indicator**.



-Heatmap Engine (`heatmap_engine.py`)

High-performance Gaussian occupancy maps for tactical visualization:

- Thread-safe data generation (`generate_heatmap_data()`)
- Texture creation on the main thread only (OpenGL)
- Differential heatmaps with hotspot detection for positional training

-Data Pipeline (`data_pipeline.py`)

`ProDataPipeline` manages ML-ready data preparation:

1. **Fetches** all `PlayerMatchStats` from the database
2. **Cleans** outliers (`avg_adr < 400`, `avg_kills < 3.0`)
3. **Scales** via `StandardScaler`
4. **Splits** temporally (70/15/15) with chronological ordering per group (pro/user)
5. **Keeps** the `dataset_split` column in place

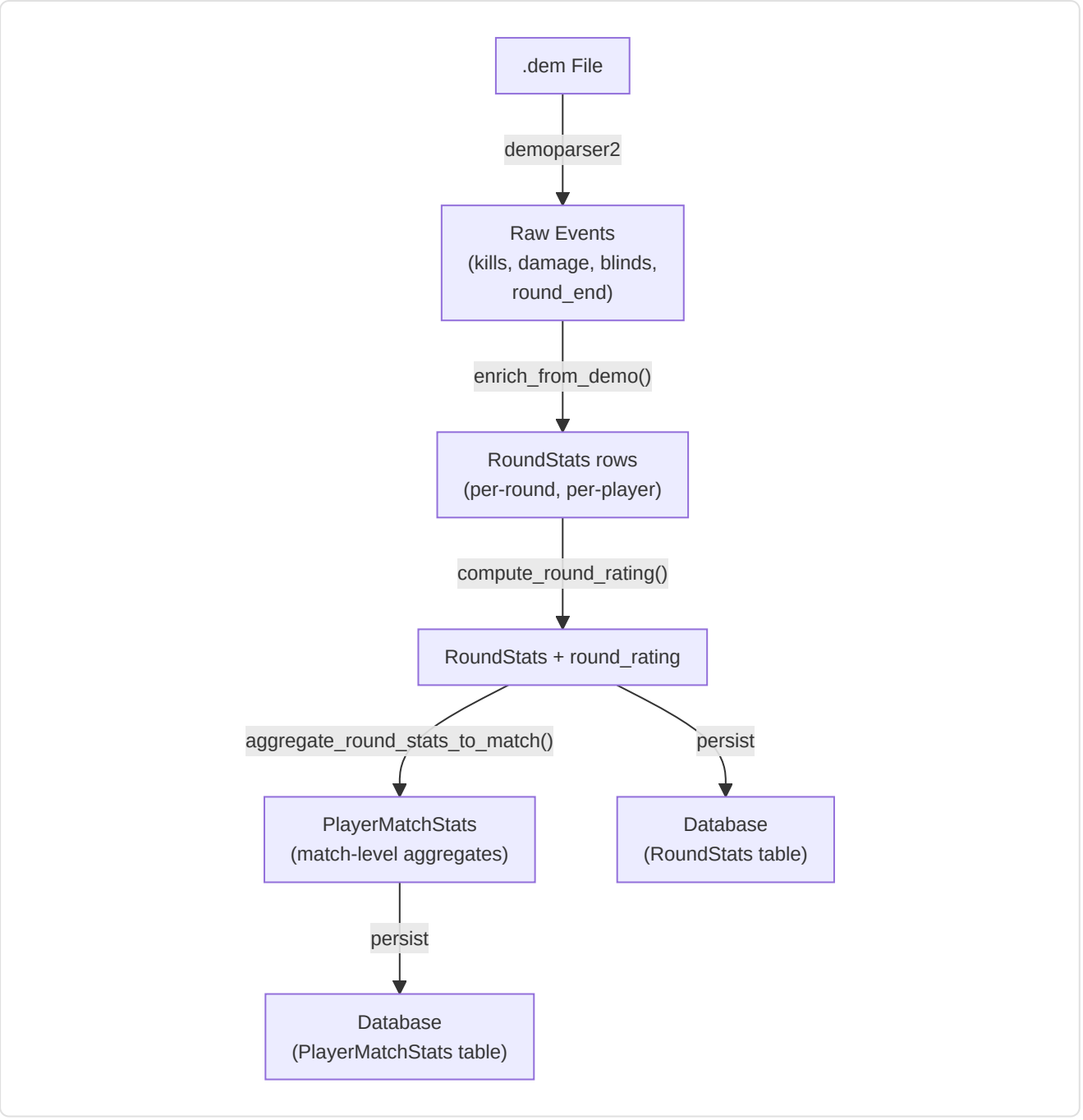
-Per-Round Stats Generator (`round_stats_builder.py`)

Bridges raw demo events to the **per-round isolation layer** (`RoundStats` in `db_models.py`), preventing statistical contamination between rounds and enabling detailed per-round coaching analysis.

Key functions:

Function	Purpose
`build_round_stats(parser, demo_name)`	Parses round_end, player_death, player_hurt, player_blind events and builds per-round stats
`enrich_from_demo(demo_path, demo_name)`	Full enrichment: noscope/blind kills, flash/smoke counts, flash assists, trade kills, utility analysis
`aggregate_round_stats_to_match(round_stats_list)`	Processes round-level stats → match-level PlayerMatchStats
`compute_round_rating(round_stats)`	HLTV 2.0 per-round rating using the unified rating module

Enrichment pipeline:



Kill enrichment fields added by `enrich_from_demo()`:

Field	Detection method
`noscope_kills`	Kills where AWP/Scout/SSG is equipped but `is_scoped=False`
`blind_kills`	Kills where `attacker_blinded=True` at the moment of death
`flash_assists`	Kills within 128 ticks (2s) of an enemy being flashed by the same team
`thrusmoke_kills`	Kills through a smoke grenade (from demo event flags)
`wallbang_kills`	Kills through wall penetration (from demo event flags)
`trade_kills`	Kills within 5s of a teammate's death at the hands of the same enemy

Integration with ingestion pipelines: Both `user_ingest.py` and `pro_ingest.py` now call `enrich_from_demo()` after demo parsing and persist the resulting `RoundStats` objects to the database. This ensures every ingested demo, user or pro, produces granular per-round data.

-Temporal Baseline Decay (`pro_baseline.py`)

The `TemporalBaselineDecay` class complements (it does not replace) the existing `get_pro_baseline()` function with a time-weighted average, ensuring that coach comparisons reflect the **current CS2 meta** rather than stale historical averages.

Decay formula:

```
weight(age_days) = max(MIN_WEIGHT, exp(-ln(2) × age_days / HALF_LIFE))
```

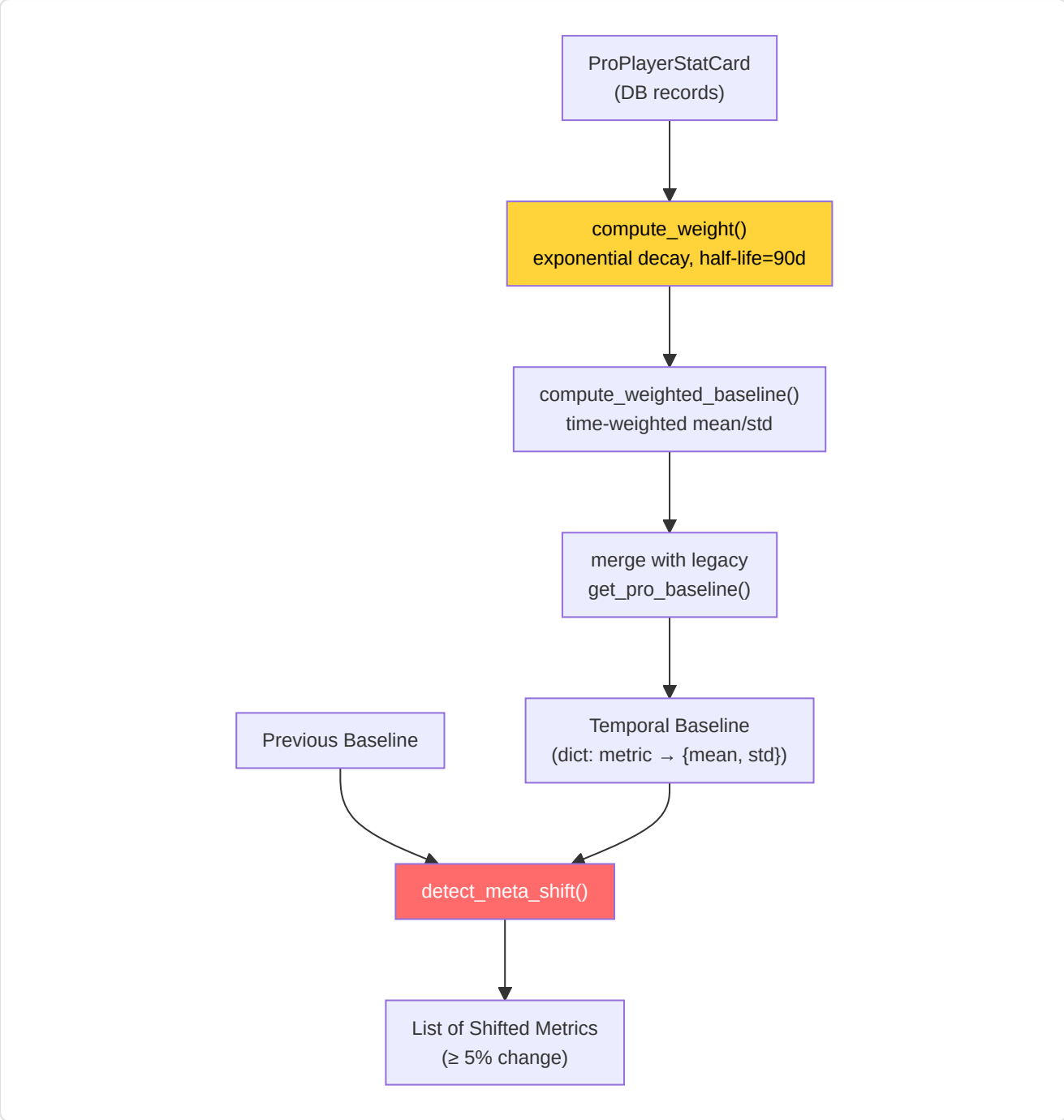
Parameter	Value	Meaning
`HALF_LIFE_DAYS`	90	Weight halves every 90 days
`MIN_WEIGHT`	0.1	Minimum threshold: very old data still contributes 10%
`META_SHIFT_THRESHOLD`	0.05	A 5% variation between epochs signals a meta shift

Key methods:

Method	Purpose
`compute_weight(stat_date)`	Exponential decay weight for a single stat card
`compute_weighted_baseline(stat_cards)`	Time-weighted mean/std across all ProPlayerStatCard records
`get_temporal_baseline(map_name)`	Complete pipeline: fetch cards → weight → merge with legacy defaults
`detect_meta_shift(old, new)`	Flags metrics that have drifted by ≥ 5% between reference epochs

Integration points:

- **CoachingService:** `_get_temporal_baseline()` uses it for COPER enrichment
- **Teacher Daemon:** `_check_meta_shift()` runs after every retraining cycle, logging drifted metrics



-Validation (`validation/`)

- **drift.py:** Data distribution drift detection with DriftReport objects
- **schema.py:** Schema validation for database records
- **sanity.py**** / dem_validator.py:** Data and demo-file integrity checks

Quantitative coverage: The project includes **1,515+ tests** distributed across 94 test files and **319+ headless validator checks** organized across 24+ validation phases. This coverage spans DB schema integrity, embedding vector consistency, training pipeline correctness, and end-to-end validation of coaching flows.

-PlayerKnowledge — NO-WALLHACK Perception System (`player_knowledge.py`)

Player-POV perception model that reconstructs what a player legitimately knows at every tick, without wallhack information. This is the foundation for ethically correct coaching: the coach sees only what the player saw.

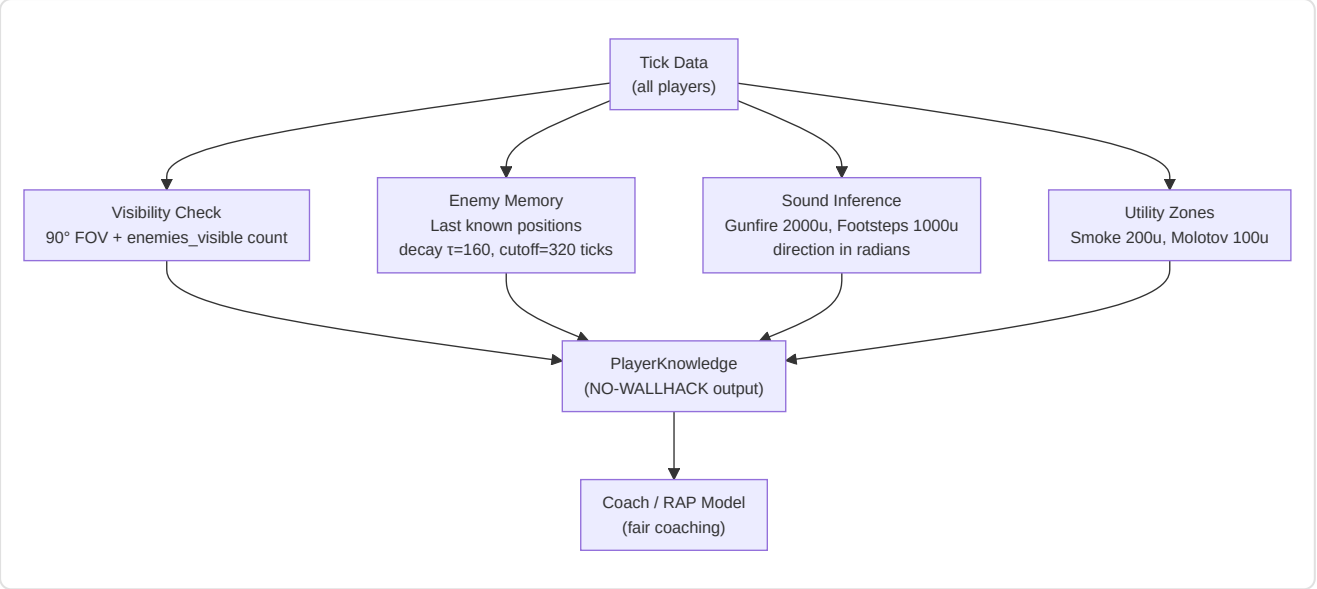
Perception dataclasses:

Dataclass	Fields	Description
`VisibleEntity`	position, distance, is_teammate, health, weapon	Entity visible in the FOV
`LastKnownEnemy`	position, tick_seen, confidence	Enemy position from memory (decay)
`HeardEvent`	position, distance, direction_rad, event_type	Perceived sound event
`UtilityZone`	position, radius, utility_type, team	Active utility zone
`PlayerKnowledge`	own_state, visible_entities, last_known, heard_events, utility_zones	Complete output

Sensory constants:

Constant	Value	CS2 meaning
`FOV_DEGREES`	90.0	CS2 horizontal field of view
`HEARING_RANGE_GUNFIRE`	2000.0	Max distance for gunfire perception (world units)
`HEARING_RANGE_FOOTSTEP`	1000.0	Max distance for footstep perception
`MEMORY_DECAY_TAU`	160	Memory half-life: 2.5s at 64 tick/s
`MEMORY_CUTOFF_TICKS`	320	Memory cutoff: 5 seconds max
`SMOKE_RADIUS`	200.0	Smoke zone radius
`MOLOTOV_RADIUS`	100.0	Molotov zone radius

Perception pipeline:



FOV check: `_is_in_fov()` uses trigonometry (atan2 + angle_diff) to determine whether a target falls within the player's horizontal field of view. Only enemies confirmed visible (from the demo's `enemies_visible` counter) are included.

-RAPStateReconstructor (`state_reconstructor.py`)

Vision Bridge Phase 1: Converts sequences of `PlayerTickState` from the DB into perception tensors for the RAP-Coach model:

- **Metadata:** METADATA_DIM (25) vector from the unified `FeatureExtractor`
- **Perception:** view/map/motion tensors from `TensorFactory`
- **Player-POV:** Optional mode via the `knowledge` parameter (PlayerKnowledge)
- **Windowing:** Temporal windows of `sequence_length=32` ticks with 50% overlap

-ConnectMapContext (`connect_map_context.py`)

Z-aware spatial features for multilevel maps (Task 2.17.1):

Constant	Value	Use
`Z_LEVEL_THRESHOLD`	200	Separation between levels (world units)
`Z_PENALTY_FACTOR`	2.0×	Cross-level distance multiplier

Output: A 6-feature normalized spatial vector [0,1]: distance to bombsite A/B, distance to T/CT spawn, distance to mid, Z penalty. Distances computed with `distance_with_z_penalty()` to penalize cross-level paths on Nuke/Vertigo.

-CVFrameBuffer (`cv_framebuffer.py`)

Thread-safe ring buffer for RGB frames (Task 2.24.2) with HUD region extraction:

Region	Coordinates (1920×1080)	Content
`MINIMAP_REGION`	(0, 0, 320, 320)	Minimap (top-left)
`KILL_FEED_REGION`	(1520, 0, 1920, 300)	Kill feed (top-right)
`SCOREBOARD_REGION`	(760, 0, 1160, 60)	Scoreboard (top-center)

Dynamic resolution scaling: All regions scale automatically relative to `REFERENCE_RESOLUTION = (1920, 1080)` to support different resolutions. BGR→RGB conversion is integrated into the `capture_frame()` method.

-EliteAnalytics (`external_analytics.py`)

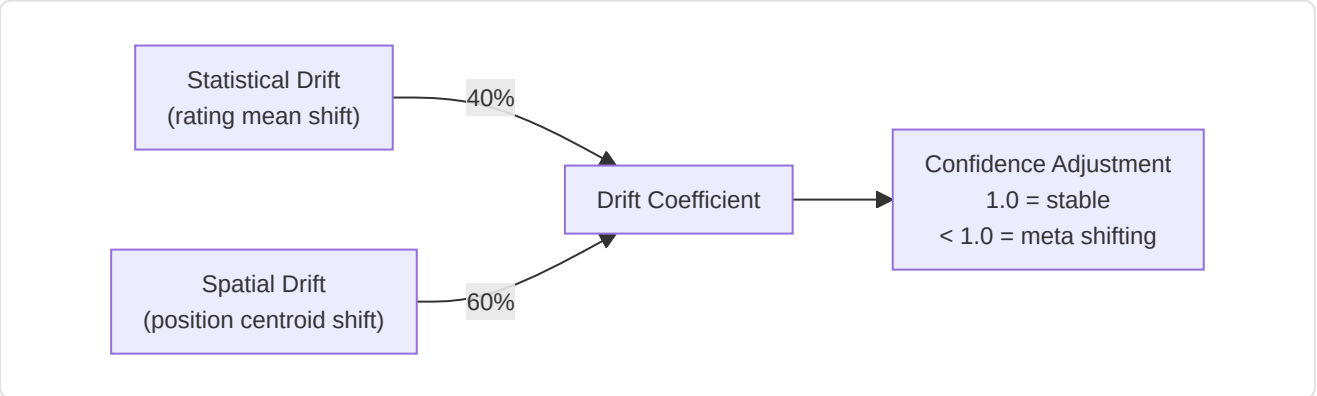
Loads and analyzes CSV reference datasets (top 100 players, match stats, tournament data):

- **7 datasets:** top100, top100_core, match_stats, player_stats, kills_processed, historical_stats, tournament_stats
- **Z-score computation:** `_calc_z_scores()` compares user vs historical mean for ADR, deaths, kills, rating, HS%
- **Tournament Z-score:** `_calc_tournament_z()` compares vs tournament baseline for accuracy, econ_rating, utility_value
- **Graceful degradation:** `is_healthy()` = True if ≥1 dataset was loaded successfully

-MetaDriftEngine (`meta_drift.py`)

Pillar 2 Phase 3 — Meta-Drift Surveillance:

Compares pro positions from the last 30 days vs historical to detect meta shifts:



Integration: `HybridCoachingEngine._calculate_confidence()` calls `MetaDriftEngine.get_meta_confidence_adjustment()` to penalize insight confidence when the meta is unstable.

-NicknameResolver (`nickname_resolver.py`)

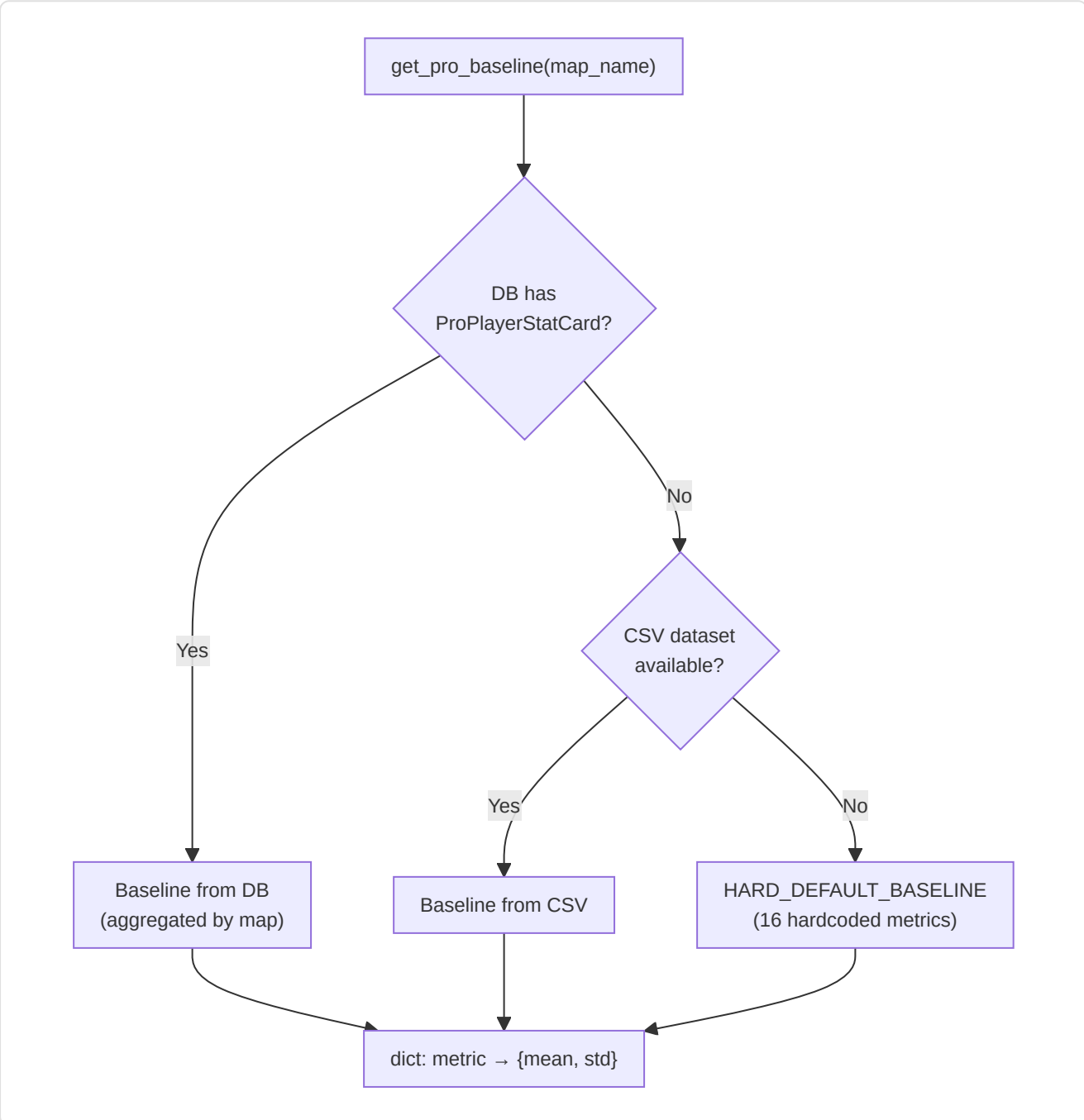
Task 2.18.2: Resolves in-game demo nicknames to HLTV ProPlayer IDs with 3-tier matching:

Tier	Method	Example
1. Exact	Case-insensitive match	"s1mple" → ProPlayer.hltv_id
2. Substring	Nickname contained in demo name	"FaZe_ropz" contains "ropz"
3. Fuzzy	SequenceMatcher ≥ 0.8 threshold	"s1mpl3" ≈ "s1mple"

Handles team tags and clan prefixes. O(n) complexity per query, acceptable for <1,000 pros in the DB.

-ProBaseline (`pro_baseline.py`)

Task 2.18.1: Pro baseline system with map-specific support and a 3-tier fallback chain:



HARD_DEFAULT_BASELINE (16 metrics with mean + std):

Metric	Mean	Std
rating	1.06	0.15
kpr	0.68	0.12
dpr	0.62	0.10
adr	77.8	12.0
kast	0.70	0.06
hs_pct	0.45	0.10
impact	1.05	0.20
opening_kill_ratio	1.05	0.30
clutch_win_pct	0.15	0.08
...	...	...

`calculate_deviations(player_stats, baseline)` — Computes the Z-score for each feature: `z = (player_value - mean) / std`. Handles both flat baselines (legacy) and structured baselines (dict with mean/std).

-RoleThresholdStore (`role_thresholds.py`)

Anti-mock principle: All role thresholds learn exclusively from real pro data (HLTV, demos, ML):

Statistic	Description	Cold-Start Default
`awp_kill_ratio`	% of kills with AWP	—
`entry_rate`	Entry frag rate	—
`assist_rate`	Assist rate	—
`survival_rate`	Survival rate	—
`solo_kill_rate`	Solo kill rate	—
`first_death_rate`	First death rate	—
`utility_damage_rate`	Utility damage per round	—
`clutch_rate`	Clutch win rate	—
`trade_rate`	Trade kill rate	—

Cold-start protection:

Requirement	Value	Meaning
`MIN_SAMPLES_FOR_VALIDITY`	10	Minimum samples for a valid threshold
Minimum valid thresholds	3	Required to exit cold-start
Cold-start output	`(FLEX, 0.0)`	Generic role, 0% confidence

Persistence: `persist_to_db()` / `load_from_db()` — learned thresholds survive restarts. `validate_consistency()` verifies internal consistency (e.g. entry_rate cannot be negative).

---

9. Subsystem 7 — Control Module

Folder in the repo: `backend/control/` **Files:** 4 modules

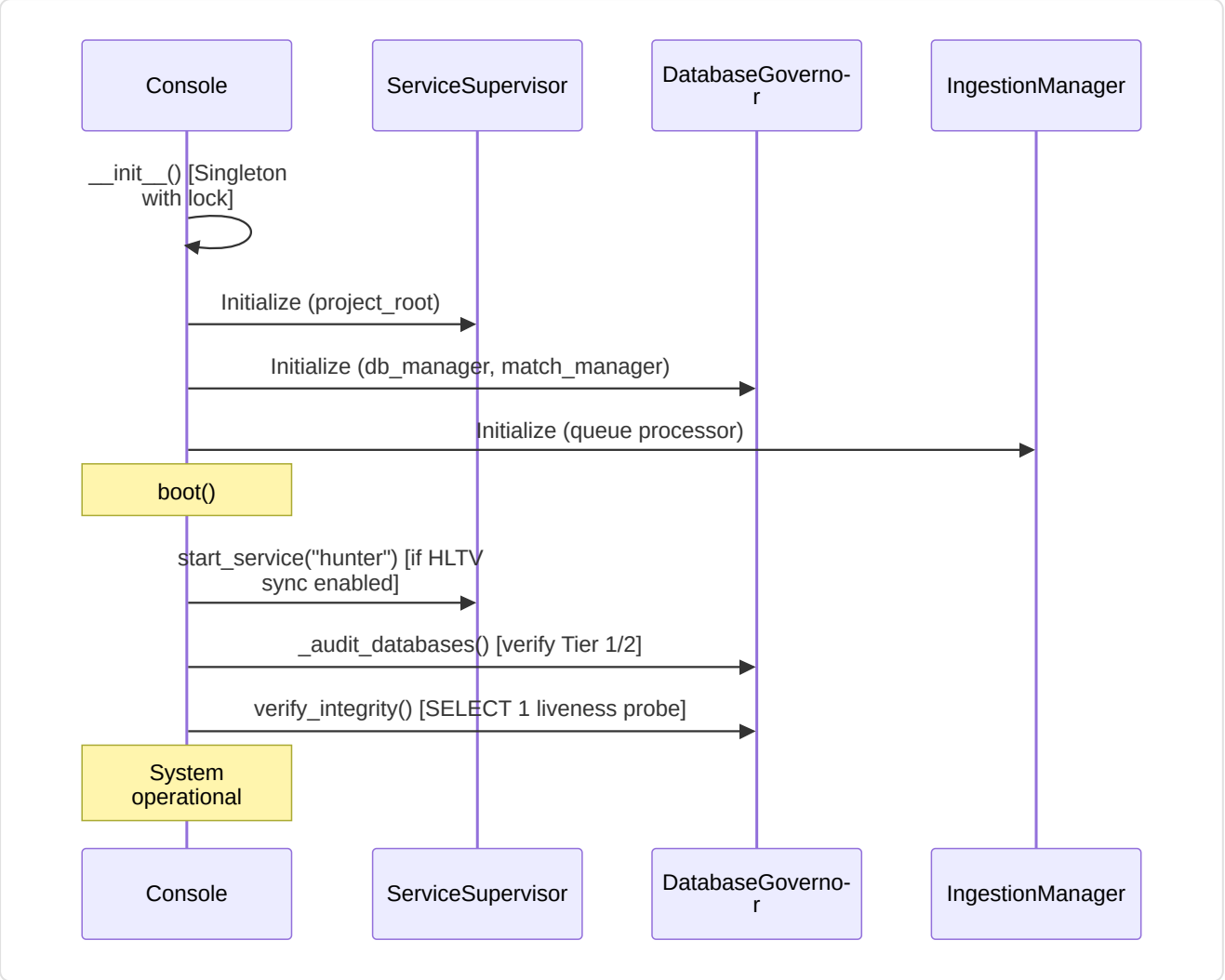
This subsystem is the system's **control tower**: it supervises background services, governs the databases, manages demo ingestion, and controls the ML training lifecycle.

-Console (`console.py`) — Singleton

The **Unified Control Console**: central authority for ML, Ingestion, and System State.

Component	Role
`ServiceSupervisor`	Background-daemon supervisor (PID, liveness, auto-restart)
`IngestionManager`	Operator-governed demo ingestion controller
`DatabaseGovernor`	Tier 1/2/3 database integrity controller
`MLController`	ML training lifecycle supervisor

Boot sequence:



State caches:

Cache	TTL	Content
`_baseline_cache`	60s	Temporal baseline state (stat_cards count, mode)
`_training_data_cache`	120s	Demo progress (pro/user processed, on disk, trained_on)

Graceful shutdown: `shutdown()` → stops hunter → stops ingestion → stops ML training → waits up to 5s for confirmation → logs warning on timeout.

> **Fix M-08 — Settings file validation:** `load_user_settings()` validates the JSON structure on load: checks that the payload is a `dict` (not a list or primitive type), logs a detailed error if the structure is invalid, and applies a graceful fallback to default values. This prevents a corrupted or tampered `settings.json` file from causing cascading crashes in the configuration module.
>
> **Fix C-09 — Early bootstrap:** Bootstrap phases prior to logger initialization use `print()` for diagnostic messages. This avoids the `NameError` that occurred when code tried to call `logger.info()` before the logger was configured.

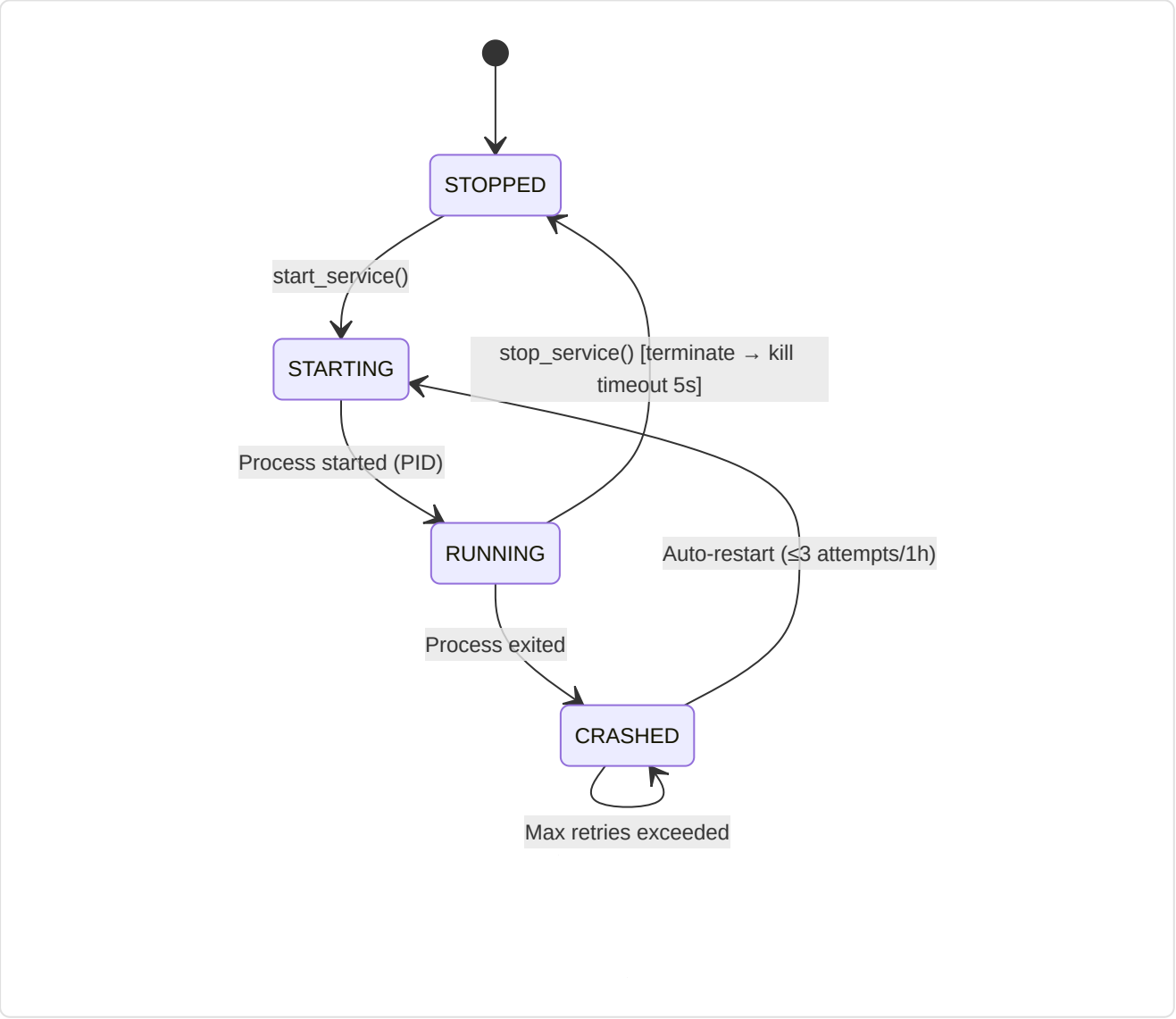
Health report aggregation: `get_system_status()` → timestamp, state, services, teacher, ml_controller, ingestion, storage, baseline, training_data. Each subsystem is isolated with `_safe_call()` — an error in one subsystem does not prevent reporting on the others.

-ServiceSupervisor (`console.py`, internal class)

Authoritative manager for background daemons with auto-restart:

Constant	Value	Description
`_MAX_RETRIES`	3	Max auto-restarts before giving up
`_RETRY_RESET_WINDOW_S`	3600 (1h)	Retry counter reset window
`_RESTART_DELAY_S`	5.0	Delay between restart attempts

Service lifecycle:



Managed services: Currently only `hunter` (HLTV sync daemon: `hltv_sync_service.py`). Architecture extensible to additional daemons.

-DatabaseGovernor (`db_governor.py`)

Authoritative controller for Tier 1, 2, and 3 databases:

Method	Purpose
`audit_storage()`	Computes monolith + WAL + SHM size, counts Tier 3 matches, detects anomalies
`verify_integrity(full=False)`	Lightweight: `SELECT 1`; Full: `PRAGMA quick_check` (slow on large DBs)
`prune_match_data(match_id)`	Privileged deletion of match telemetry data
`rebuild_indexes()`	Maintenance: `REINDEX` via the ORM engine

Anomalies detected:

- `CRITICAL: Monolith database.db not found!`
- `CRITICAL: hltv_metadata.db missing and no backup found.`
- `WARNING: hltv_metadata.db missing, but .bak exists. Restore required.`

Tier Architecture:

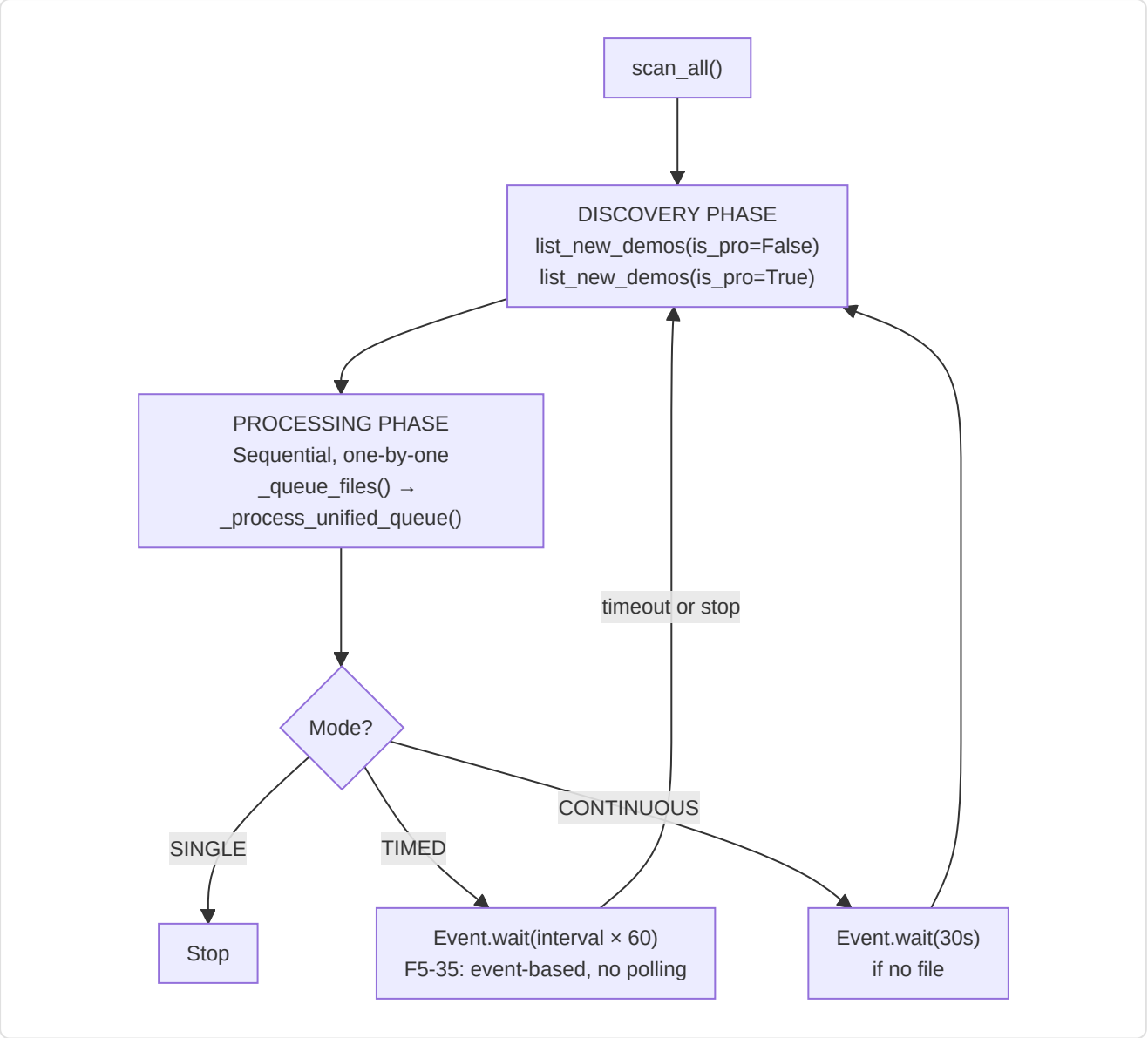
Tier	Database	Content
Tier 1	`database.db` (monolith)	Players, insights, profiles, models, knowledge base
Tier 2	`hltv_metadata.db`	HLTV metadata, ProPlayer, ProPlayerStatCard
Tier 3	`match_*.db` (per-match)	Tick-by-tick telemetry data per match

-IngestionManager (`ingest_manager.py`)

Demo ingestion controller with 3 operating modes:

Mode	Behavior
`SINGLE`	Process one demo and stop
`CONTINUOUS`	Immediate re-scan, 30s pause if no file is found
`TIMED`	Re-scan every N minutes (default 30)

Unified cycle:



Stop signal (F5-35): `threading.Event()` for non-blocking stop — eliminates the 1-second polling in wait loops.

StateManager integration: Real-time parsing progress (0-100%) exposed via `state_manager.update_parsing_progress()`.

-MLController (`ml_controller.py`)

Supervisor of the ML lifecycle with real-time intervention:

Class	Responsibility
`MLControlContext`	Control token passed to ML loops
`MLController`	Thread-safe supervisor for training

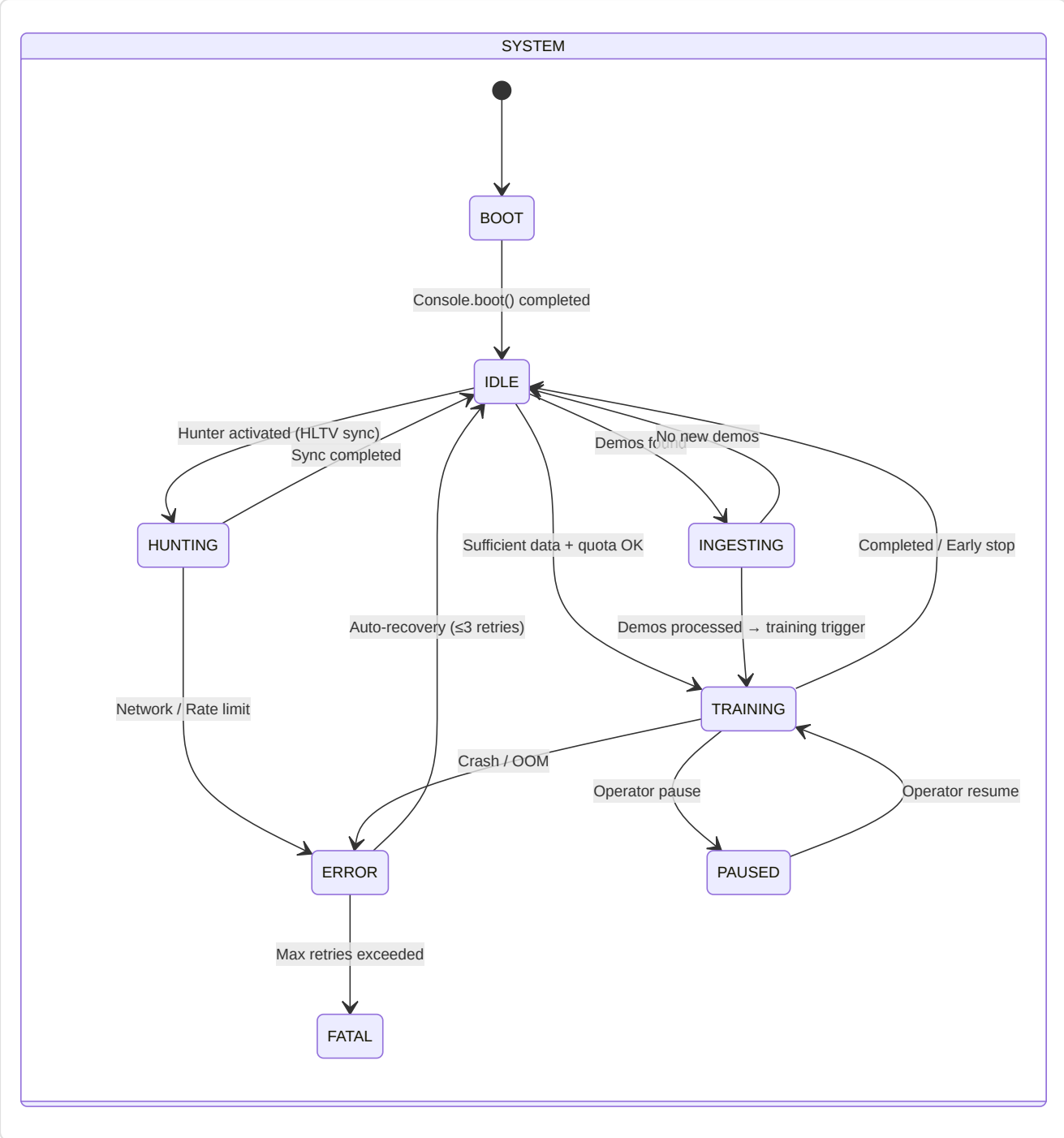
MLControlContext — Operator commands:

Method	Effect
`check_state()`	Blocks if pause is active (F5-15 event-based), raises `TrainingStopRequested` if stop
`request_stop()`	Soft-stop at the next safe checkpoint
`request_pause()`	Blocks `check_state()` via `_resume_event.clear()`
`request_resume()`	Unblocks via `_resume_event.set()`
`set_throttle(value)`	0.0 = full speed, 1.0 = max delay

Pipeline: `start_training()` → daemon thread → `CoachTrainingManager.run_full_cycle(context=self.context)` → `TrainingStopRequested` caught for graceful stop → `set_error()` on crash.

Inter-Daemon Coordination

The Control Module orchestrates the system's 4 daemons (Hunter, Digester, Teacher, Pulse) through communication channels based on shared state (`CoachState`) and event-based signals:



10. Subsystem 8 — Progress and Trends

Folder in the repo: `backend/progress/` **Files:** 2 modules

Lightweight subsystem dedicated to tracking **performance trajectories** over time.

-FeatureTrend (`longitudinal.py`)

Minimal dataclass representing the trend of a single feature:

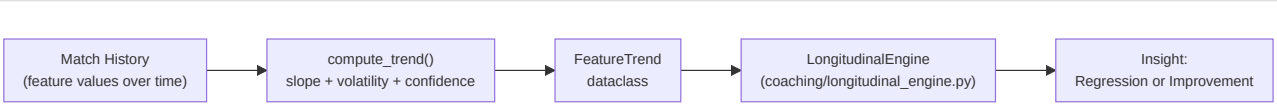
Field	Type	Description
`feature`	str	Feature name (e.g. "avg_adr")
`slope`	float	Linear regression slope (positive = improvement)
`volatility`	float	Standard deviation of values (stability)
`confidence`	float	min(1.0, num_samples / 30) — requires 30+ matches for full confidence

-TrendAnalysis (`trend_analysis.py`)

`compute_trend(values)` — Computes slope, volatility, and confidence from a series of values:

- **Slope:** Linear coefficient via `np.polyfit(x, y, 1)[0]`
- **Volatility:** `np.std(values)`
- **Confidence:** Scales linearly with sample count, full confidence at 30+

Integration: The results of `compute_trend()` feed `FeatureTrend` → `LongitudinalEngine.generate_longitudinal_coaching()` → trend-aware insights in coaching.



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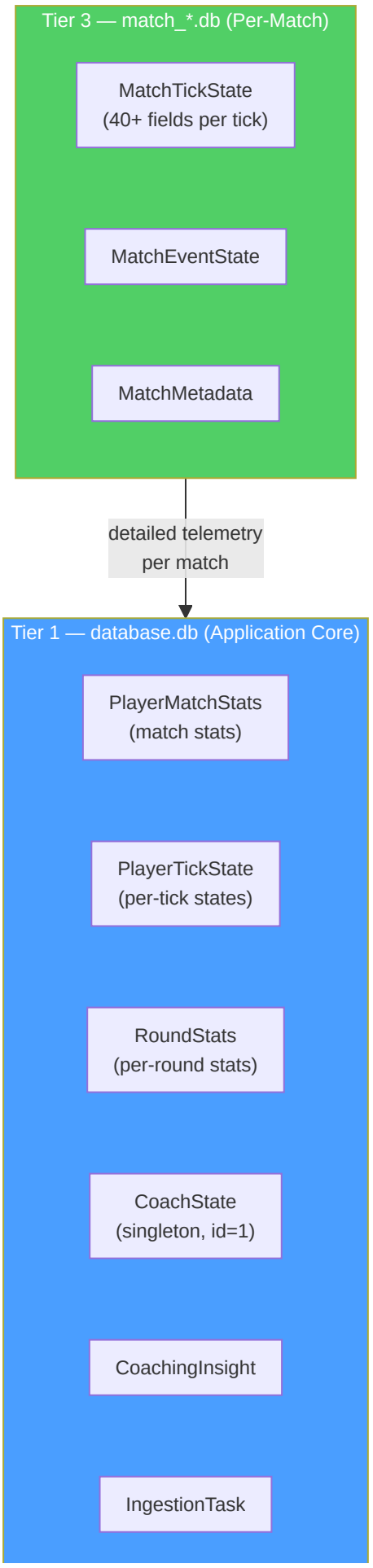
11. Subsystem 9 — Database and Storage

Folder in the repo: `backend/storage/` **Files:** 10 modules

This subsystem is the **permanent storage system** for the entire project: every piece of data — from match statistics to trained models, from coaching experiences to professional player profiles — is saved, protected, versioned, and made available through this infrastructure.

Tri-Database architecture:


Tier 3 — match_*.db (Per-Match)
MatchTickState
(40+ fields per tick)
MatchEventState
MatchMetadata
detailed telemetry
per match
Tier 1 — database.db (Application Core)
PlayerMatchStats
(match stats)
PlayerTickState
(per-tick states)
RoundStats
(per-round stats)
CoachState
(singleton, id=1)
CoachingInsight
IngestionTask

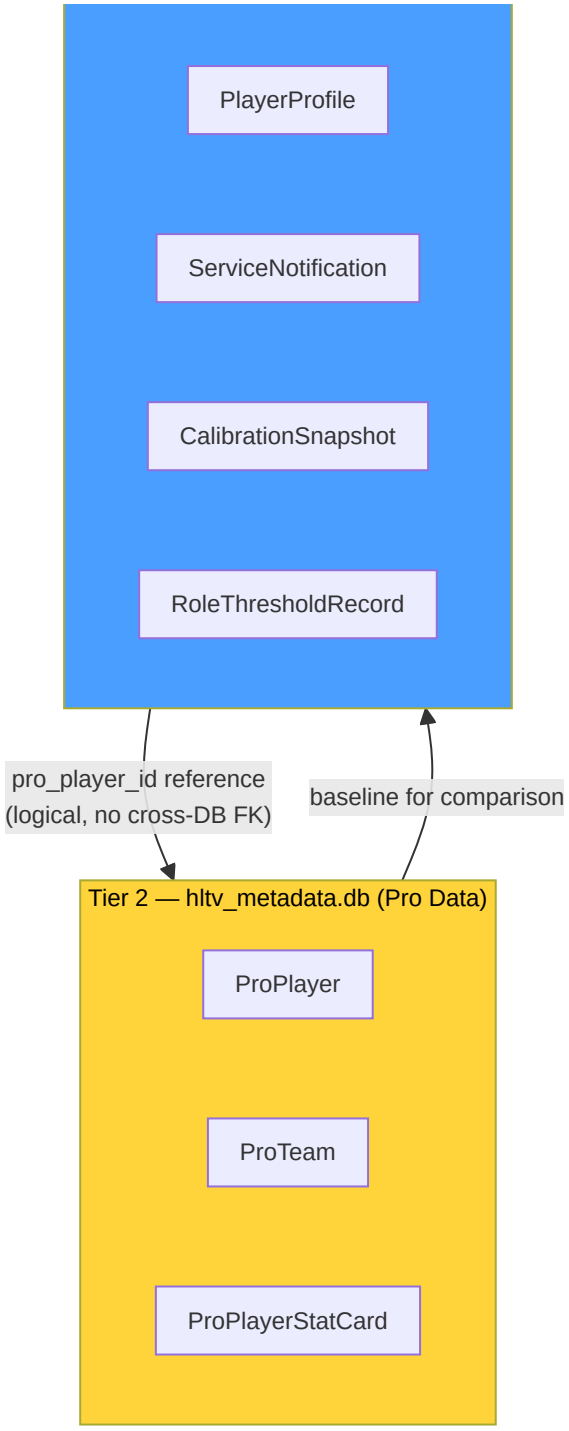


Diagram Explanation: The three databases are physically separated: Tier 1 contains the application's core data (17 tables), Tier 2 the professional data scraped from HLTV (3 tables), and Tier 3 the per-match telemetry data in individual SQLite files. The separation avoids WAL contention: ingestion writes (Tier 1) do not block HLTV reads (Tier 2) or telemetry recording (Tier 3). Cross-database references are **logical** (not real FKs) since SQLite does not support cross-file FKs.

-Data Models (`db_models.py`)

The **canonical schema** of the whole system: defines every table, constraint, index, and validator through SQLModel (Pydantic + SQLAlchemy). Two enums and 20+ models organized per tier.

Integrity enums:

Enum	Values	Use
`DatasetSplit`	`TRAIN`, `VAL`, `TEST`, `UNASSIGNED`	ML split in `PlayerMatchStats`
`CoachStatus`	`Paused`, `Training`, `Idle`, `Error`	ML pipeline state in `CoachState`

Safety constant: `MAX_GAME_STATE_JSON_BYTES = 16,384` (16 KB) — cap for `game_state_json` in `CoachingExperience`, validated by a Pydantic `@field_validator`. Prevents unbounded DB growth from oversized state snapshots.

Model catalog (Tier 1 — `database.db`):

Model	Key Fields	Constraints / Indexes	Purpose
`PlayerMatchStats`	40+ fields: kills, ADR, rating, HLTV 2.0 components, trade metrics, utility breakdown	`UNIQUE(demo_name, player_name)`, `CHECK(avg_kills≥0)`, `CHECK(0≤rating≤5.0)`	Aggregated per-match stats per player
`PlayerTickState`	pos_x/y/z, view_x/y, health, armor, weapon, enemies_visible, round context	`INDEX(demo_name, tick)`, `INDEX(player_name, demo_name)` (P2-05)	Per-tick player state for ML training
`PlayerProfile`	player_name (unique), bio, role, profile_pic_path	`UNIQUE(player_name)`	User profile with role and bio
`Ext_TeamRoundStats`	equipment_value, damage, kills, deaths, accuracy	`INDEX(match_id)`, `INDEX(map_name)`	Round stats from external CSVs (tournaments)
`Ext_PlayerPlaystyle`	6 role probabilities, aggregate metrics, user config (social, specs, cfg)	`UNIQUE(steam_id)`	Playstyle data + user profile (conflated table, future split candidate)
`CoachingInsight`	title, severity, message, focus_area, user_id	`INDEX(player_name)`, `INDEX(demo_name)`	System-generated coaching feedback
`IngestionTask`	demo_path (unique), status, retry_count, error_message	`UNIQUE(demo_path)`, `INDEX(status)`, R2-05: auto-refresh `updated_at`	Demo ingestion queue with retry tracking
`TacticalKnowledge`	title, description, category, situation, embedding (384-dim JSON)	`INDEX(title)`, `INDEX(category)`, `INDEX(map_name)`	RAG knowledge base for tactical coaching
`CoachState`	status, daemon tracking (3 daemons), epoch/loss, heartbeat, resource limits	`CHECK(id=1)` singleton (P2-01)	ML pipeline state for the GUI — exactly 1 row
`ServiceNotification`	daemon, severity, message, is_read	`INDEX(daemon)`, `INDEX(is_read)`	Error/event notification queue for the UI
`RoundStats`	kills, deaths, assists, damage, utility, economy, round_won, round_rating	`INDEX(demo_name, player_name)`, `INDEX(demo_name, round_number)` (P2-05)	Per-round per-player stats isolation
`CalibrationSnapshot`	calibration_type, parameters_json, sample_count, source	`INDEX(calibration_type)`	Bayesian model calibration history
`RoleThresholdRecord`	stat_name (unique), value, sample_count, source	`UNIQUE(stat_name)`	Role thresholds learned from pro data

Match and pro models (cross-tier):

Model	Tier	Key Fields	Purpose
`MatchResult`	Tier 1	match_id (PK), team_a/b, winner, event_name	Match metadata
`MapVeto`	Tier 1	match_id (FK), map_name, action (pick/ban)	Map vetoes
`CoachingExperience`	Tier 1	context_hash, map/phase/side, game_state_json (16KB cap), action, outcome, embedding, effectiveness_score, feedback loop fields	COPER experience bank with semantic search and feedback loop
`ProTeam`	Tier 2	hltv_id (unique), name, world_rank	Professional teams
`ProPlayer`	Tier 2	hltv_id (unique), nickname, team_id (FK→ProTeam)	Professional players
`ProPlayerStatCard`	Tier 2	player_id (FK), rating_2_0, kpr, adr, kast, detailed_stats_json	HLTV stat cards with granular data

> **Architectural note:** `PlayerMatchStats.pro_player_id` is a **logical** reference (not a real FK) to `ProPlayer.hltv_id` because they live in separate databases. SQLite does not support cross-file FKs — the join is performed at the application level.

> **Note (P2-01):** The `CHECK(id=1)` constraint on `CoachState` ensures exactly one row exists in the database, turning it into a persistent singleton. Any attempt to insert a second row fails at the database level, not just at the application level.

-DatabaseManager (`database.py`)

The **archive keeper**: manages SQLite connections with mandatory WAL mode, physically separating the monolith database (Tier 1) from the HLTV database (Tier 2).

Two classes, two databases:

Class	Database	Tables	Engine
`DatabaseManager`	`database.db`	17 tables (`_MONOLITH_TABLES`)	`pool_size=1` , `max_overflow=4`
`HLTVDatabaseManager`	`hltv_metadata.db`	3 tables (`_HLTV_TABLES`)	`pool_size=1` , `max_overflow=4`

SQLite PRAGMAs (applied to every connection via `@event.listens_for`):

PRAGMA	Value	Purpose
`journal_mode`	`WAL`	Concurrent reads + single writer
`synchronous`	`NORMAL`	Balance performance/durability
`busy_timeout`	`30000` (30s)	WAL-lock contention timeout

Public API:

Method	Description
`create_db_and_tables()`	Creates schema in the monolith (only `_MONOLITH_TABLES`, not all SQLModel tables)
`get_session()`	Transactional context manager: auto-commit on success, auto-rollback on exception
`upsert(model_instance)`	Atomic upsert; special handling for `PlayerMatchStats` (lookup by demo_name + player_name)
`get(model_class, pk)`	Read by primary key

Singleton: `get_db_manager()` and `get_hltv_db_manager()` with double-checked locking (identical pattern). Prevents duplicate engine creation during imports from multiple threads.

Table isolation (critical): The explicit `_MONOLITH_TABLES` list prevents per-match models (`MatchTickState`, `MatchEventState`, `MatchMetadata`) from being created in the monolith database if their modules are imported before `create_db_and_tables()`. Without this guard, every globally registered SQLModel would be created in `database.db`.

> **HLTV reconciliation note:** `HLTVDatabaseManager._reconcile_stale_schema()` detects missing columns in existing HLTV tables (updated schema but old DB) and recreates the affected tables via `DROP TABLE` + `CREATE TABLE`, operating as an emergency destructive migration — acceptable because HLTV data is re-downloadable.

-MatchDataManager (`match_data_manager.py`)

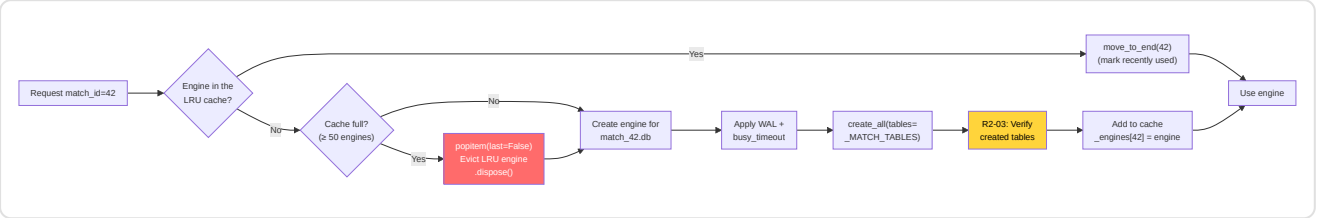
The project's **security-camera system**: every match gets its own dedicated SQLite file (`match_{id}.db`) containing tick-by-tick telemetry — up to **1.7 million rows per match**.

3 per-match models (Tier 3):

Model	Fields	Purpose
`MatchTickState`	40+ fields: position, state, equipment, visibility, round context, match cumulatives	Player state at every tick (128 Hz)
`MatchEventState`	tick, event_type, player/victim info, position, weapon, damage, entity_id	Game events for Player-POV reconstruction
`MatchMetadata`	match_id, demo_name, map_name, tick/round/player count, team names/scores	Metadata for access without the monolith DB

> **Note (C-08):** `MatchEventState.entity_id` uses `-1` as the sentinel value (not `0`) to distinguish "ID not populated by parser" from "real entity ID = 0", preventing incorrect smoke/molotov coupling.

LRU cache (M-18):



> **Note (R2-03):** The `tables=_MATCH_TABLES` filter in `create_all()` is **critical**. Without it, SQLModel would create all ~20 global tables in every per-match file. A defensive post-creation check verifies that only the 3 expected tables exist in the DB, logging a warning if unexpected tables appear.

Public API:

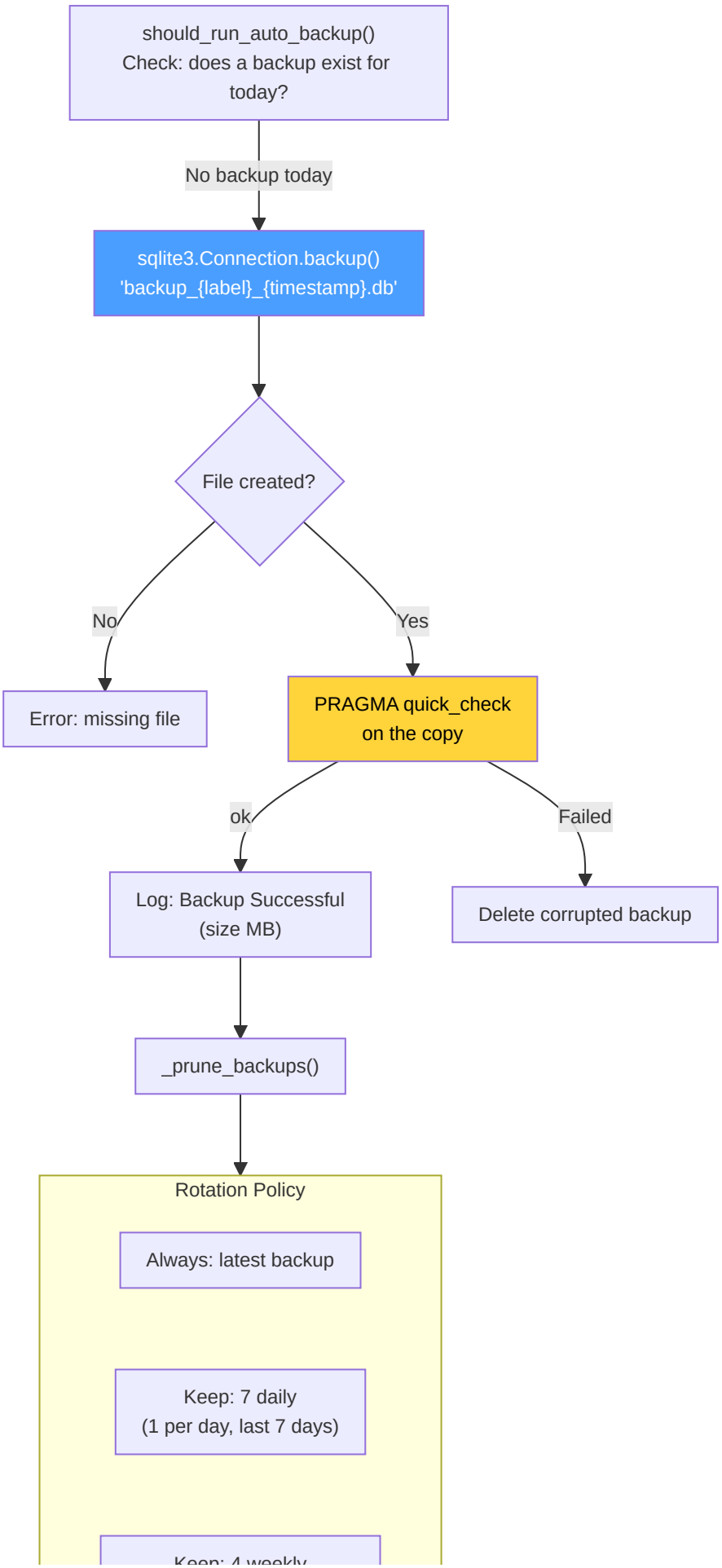
Method	Description
`get_match_session(match_id)`	Transactional context manager for a specific match
`store_tick_batch(match_id, ticks)`	Batch insert of `MatchTickState` — returns count
`store_metadata(match_id, metadata)`	Upsert match metadata
`get_ticks_for_round(match_id, round)`	Query ticks for a specific round
`get_player_ticks(match_id, player, start, end)`	Tick window for player + interval
`get_all_players_at_tick(match_id, tick)`	Snapshot of all players at a tick
`get_all_players_tick_window(match_id, start, end)`	Tick window for all players (RAP training)

-BackupManager (`backup_manager.py`)

The **data security officer**: creates non-blocking backup copies via the **SQLite Online Backup API** (`sqlite3.Connection.backup()`) and manages them with a rotation policy. The online backup API (not `VACUUM INTO`) is the correct choice because it operates safely with WAL-mode and concurrent access — it copies pages incrementally without requiring an exclusive lock on the entire database.

> **Fix H-02 — DB handle leak:** `_verify_integrity()` uses a `try/finally` block to guarantee closing the `sqlite3` connection even on error. The PRAGMA used is `quick_check` (not `integrity_check`) to reduce verification time on large databases — `quick_check` skips index validation, which is sufficient to verify the structural integrity of backup pages.

Backup pipeline:


should_run_auto_backup()
Check: does a backup exist for today?
No backup today
sqlite3.Connection.backup()
'backup_{label}_{timestamp}.db'
File created?
No
Yes
Error: missing file
PRAGMA quick_check on the copy
ok
Failed
Log: Backup Successful (size MB)
Delete corrupted backup
_prune_backups()
Rotation Policy
Always: latest backup
Keep: 7 daily
(1 per day, last 7 days)
Keep: 4 weekly

Path safety: The backup label is pre-validated with regex `_SAFE_BACKUP_LABEL_RE = ^[a-zA-Z0-9_\-]{1,64}$` (BE-01, DB-03). The destination path is verified via `Path.resolve().relative_to()` to ensure it resides within the backup directory — defense in depth (BE-06) against path traversal. The online backup API (`sqlite3.Connection.backup()`) does not involve SQL queries for the path, eliminating injection risk at this level.

-StorageManager (`storage_manager.py`)

Manager for **local storage** of CS2 demos: ingestion folders, post-processing archive, and disk quotas.

Feature	Detail
Ingestion folder	Configurable path (`DEFAULT_DEMO_PATH`), fallback to in-project `data/`
Archive	`datasets/user_archive/` and `datasets/pro_archive/` in the "Brain Root"
Quotas	`MAX_DEMOS_PER_MONTH` and `MAX_TOTAL_DEMOS_PER_USER` from config
Disk quota	`LOCAL_QUOTA_GB` (default 10 GB), verified before archiving
Discovery	`list_new_demos(is_pro)` — scans for `.dem` not yet processed in `PRO_DEMO_PATH` or `DEFAULT_DEMO_PATH`

-StateManager (`state_manager.py`)

Centralized DAO for the `CoachState` singleton: thread-safe interface between daemons and the GUI.

Method	Description
`get_state()`	Retrieves the singleton (creates it if missing, P0-04 lock)
`update_status(daemon, status, detail)`	Updates the status of a specific daemon (hunter/digester/teacher/global)
`update_training_progress(epoch, total, train_loss, val_loss, eta)`	Real-time training progress for the GUI
`update_parsing_progress(progress)`	Demo parsing progress bar (0.0–100.0)
`push_notification(daemon, severity, message)`	Inserts a `ServiceNotification` into the queue for the UI
`get_unread_notifications(limit)`	Retrieves unread notifications for display

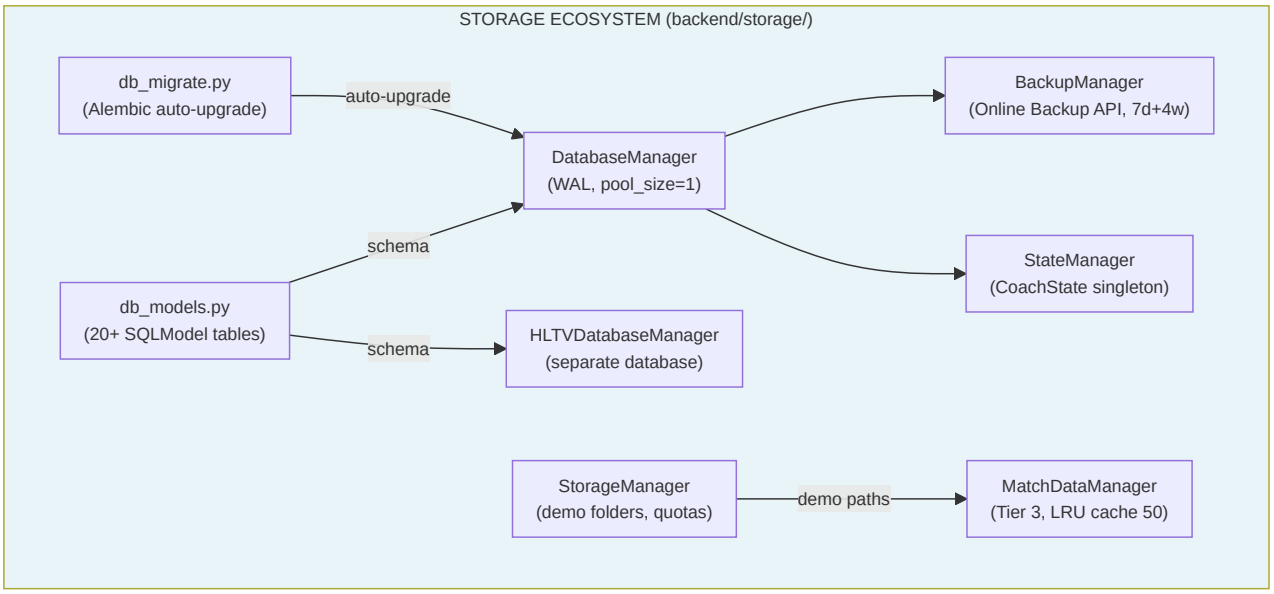
-Additional Storage Services

`stat_aggregator.py` — HLTV spider data persistence: `persist_player_card(card_data, player_hltv_id)` converts data extracted by the spider into a `ProPlayerStatCard` and saves it to the HLTV database. `persist_team(team_data)` saves/updates a `ProTeam`.

`db_migrate.py` — Alembic migration orchestration: `ensure_database_current()` is called at application startup to auto-upgrade the schema. Compares `current_rev` vs `head_rev` and, if different, executes `alembic.command.upgrade(cfg, "head")`. If `alembic.ini` does not exist (development without Alembic), returns `True` silently.

`maintenance.py` — Database maintenance: `prune_old_metadata(days=90)` deletes old records in batches of 500 (`SQLITE_MAX_VARIABLE_NUMBER` safety), avoiding DELETE queries with thousands of parameters that SQLite would reject.

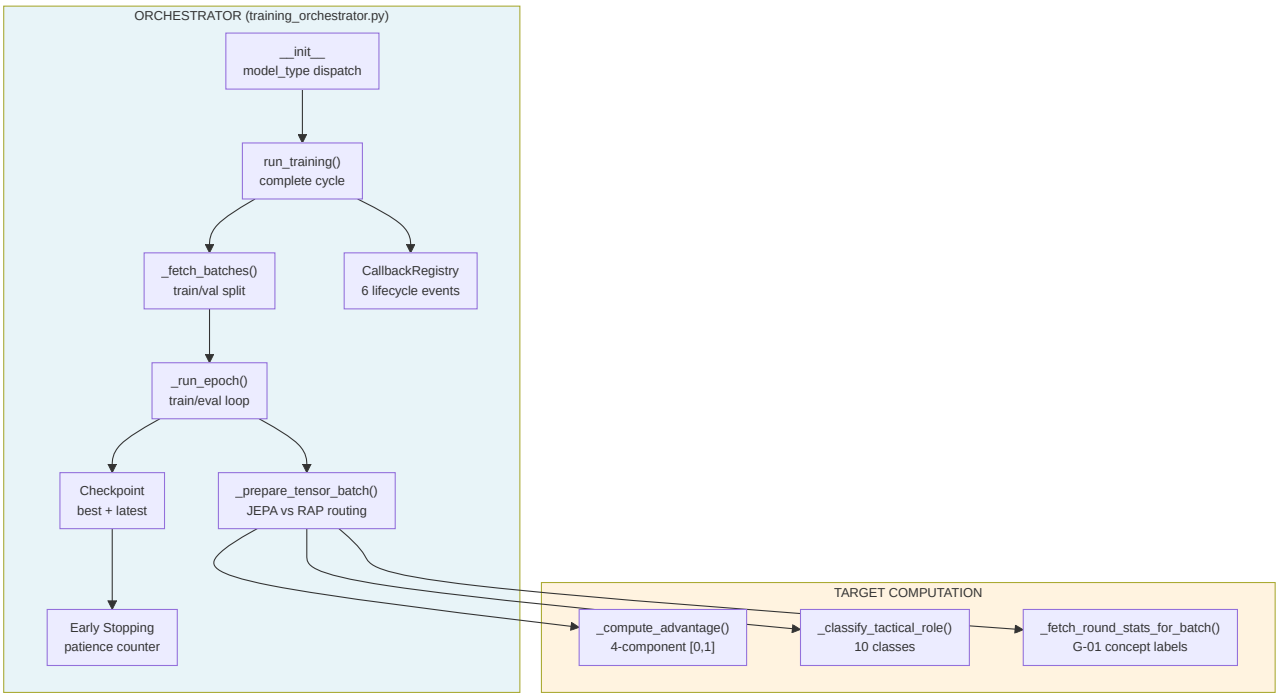
`remote_file_server.py` — Personal cloud storage server built on FastAPI for remote access to application data. Implements authentication via API key with HMAC-safe comparison (`hmac.compare_digest`) to prevent timing attacks, sliding-window rate limiting (10 req/min per IP) via ASGI middleware, and path traversal protection on all file-access endpoints. **Security BE-07:** refuses binding on non-localhost addresses without TLS — if certificate and key are provided, serves over HTTPS via uvicorn; otherwise, restricts to `localhost`. Endpoints: `GET /list` (file listing with auth), `GET /download/{filename}` (download with auth and path traversal protection), `GET /health` (health check without auth).



12. Training and Orchestration Pipeline

The training subsystem has three distinct responsibilities:

1. **Orchestration** — Full lifecycle: data loading → epochs → checkpoint → early stopping
2. **Tensor Preparation** — Conversion of raw ticks from the database into model-ready tensor batches
3. **Target Computation** — Generation of training labels: advantage function (continuous) and tactical role (10 classes)



-TrainingOrchestrator (training_orchestrator.py)

The central class of the training pipeline. Handles JEPA, VL-JEPA, and RAP from a unified entry point, with dispatch based on `model_type`.

Constructor and configuration:

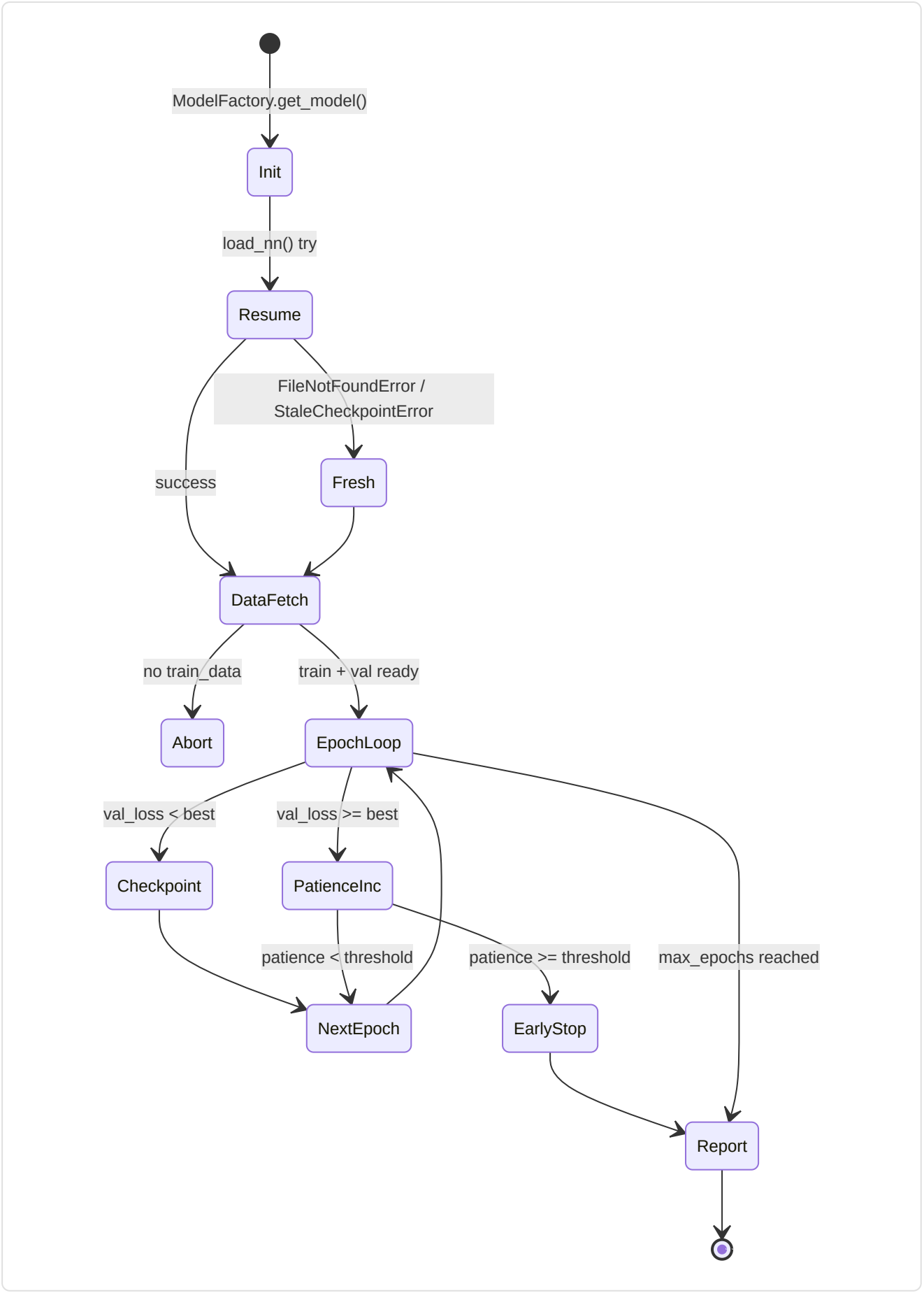
Parameter	Default	Description
`model_type`	`"jepa"`	Routing: `"jepa"` → JEPATrainer, `"vl-jepa"` → JEPATrainer (VL mode), `"rap"` → RAPTrainer
`max_epochs`	100	Upper bound on epochs
`patience`	10	Epochs without improvement before early stopping
`batch_size`	32	Samples per batch
`callbacks`	`CallbackRegistry()`	Plugin system for TensorBoard, logging, etc.

Notable constructor details:

- **Deterministic RNG** (F3-02): `np.random.default_rng(seed=42)` for JEPA negative sampling — guarantees reproducibility
- **Fallback counters** (F3-11): `_total_samples` and `_total_fallbacks` track the aggregate zero-tensor rate over the entire cycle
- **RAP gated**: The RAP model requires `USE_RAP_MODEL=True` in settings — if disabled, `ValueError` is raised immediately
- **Learning rate**: JEPA 1e-4, RAP 5e-5 (more conservative for multi-task)

Lifecycle: `run_training()`

The main method executes 6 sequential phases:


ModelFactory.get_model()
Init
load_nn() try
Resume
FileNotFoundError /
StaleCheckpointError
success
Fresh
DataFetch
no train_data
train + val ready
Abort
EpochLoop
val_loss < best
val_loss >= best
Checkpoint
PatienceInc
patience < threshold
patience >= threshold
max_epochs reached
NextEpoch
EarlyStop
Report

1. **Init**: `ModelFactory.get_model(type)` instantiates the model, then attempts `load_nn()` to resume from an existing checkpoint. Handles 3 exceptions: `FileNotFoundError` (first training), `StaleCheckpointError` (architecture changed), and generic (corruption)
2. **Data Fetch**: `_fetch_batches(is_train=True/False)` with train/val split
3. **Epoch Loop**: For each epoch, `_run_epoch()` in train mode and then eval. The `context.check_state()` call allows cooperative interruption (TrainingStopRequested)
4. **Checkpoint**: Double save — `model_name` (best) and `model_name_latest` (always). Only the best is saved when `val_loss` improves
5. **Early Stopping**: The `patience_counter` is incremented on every epoch without improvement
6. **Report**: `on_train_end` callback with final metrics + log of the F3-11 fallback rate

Data Preparation: `_fetch_batches()`

Retrieves ticks from the database via the manager, respecting the train/val split:

- Calls `manager._fetch_jepa_ticks(is_pro, split)` for all model types (RAP included — RAP-specific data pipeline not yet implemented, with an explicit warning)
- **Temporal order preserved** — no shuffling, because the models are sequential
- Batching into batches of size `self.batch_size`

Tensor Routing: `_prepare_tensor_batch()`

The most critical method: converts `PlayerTickState` objects from the database into PyTorch tensor dictionaries.

JEPA flow (context/target/negatives):

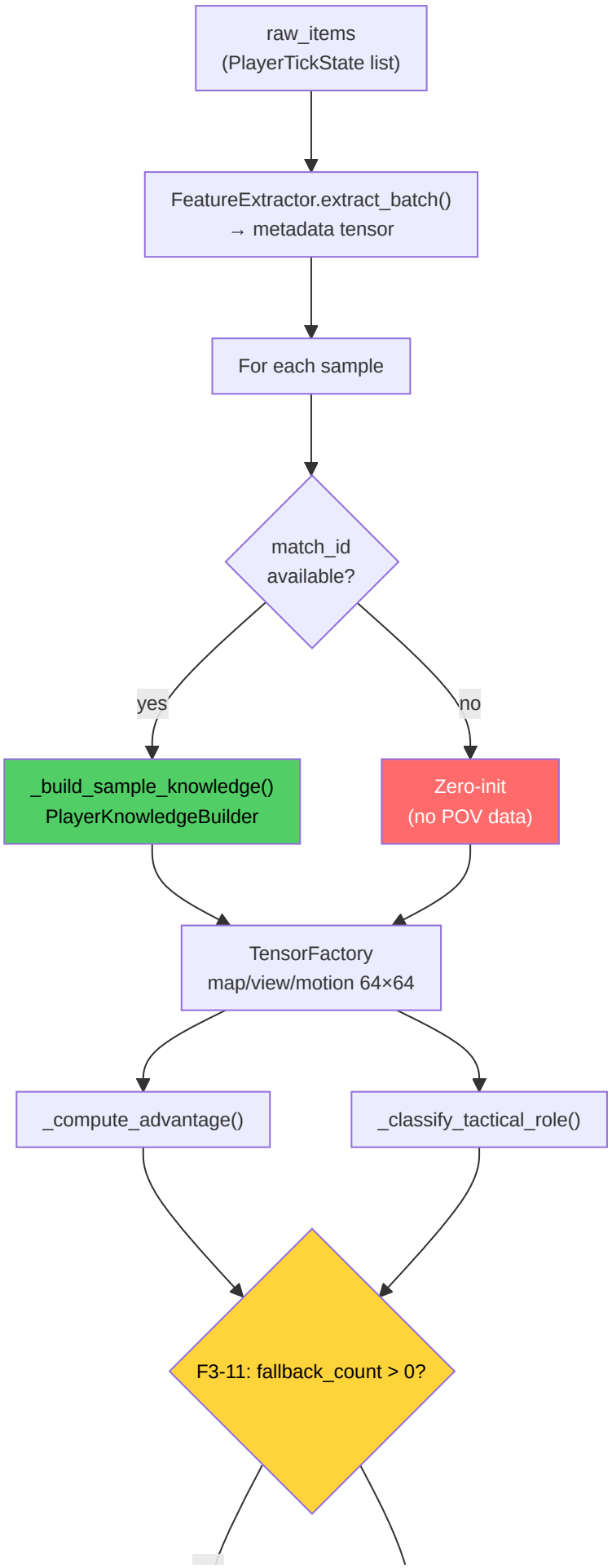
Tensor	Shape	Construction
`context`	`(1, 10, METADATA_DIM)`	First 10 ticks of the batch, padding if < 10
`target`	`(1, METADATA_DIM)`	Mean of the last tick (next-item prediction)
`negatives`	`(1, 5, METADATA_DIM)`	5 random samples from the batch (deterministic RNG)

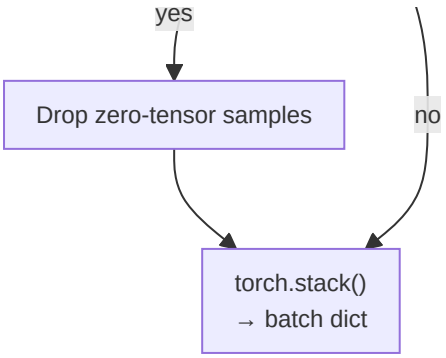
Requires at least 5 real samples for the contrastive negatives — if the batch is too small, it returns `None` (skip).

> **Fix G-01:** For VL-JEPA, `_fetch_round_stats_for_batch()` is also called to provide concept labels based on real round outcomes (eliminates label leakage from heuristic labeling).

RAP flow (full Player-POV):

The RAP path is significantly more complex — delegated to `_prepare_rap_batch()`:


raw_items
(PlayerTickState list)
FeatureExtractor.extract_batch()
→ metadata tensor
For each sample
match_id
available?
yes
no
_build_sample_knowledge()
PlayerKnowledgeBuilder
Zero-init
(no POV data)
TensorFactory
map/view/motion 64×64
_compute_advantage()
_classify_tactical_role()
F3-11: fallback_count > 0?



For each sample in the batch:

1. **match_id resolution** → query the `MatchDataManager` for all players at the tick
2. **PlayerKnowledgeBuilder** → NO-WALLHACK sensory model (visibility, sound, utility)
3. **TensorFactory** → 64×64 map/view/motion tensors (training resolution)
4. **Target computation** → advantage [0,1] + tactical role (10 classes)

Per-batch cache (4 dictionaries) avoids redundant queries on the same match/tick:

Cache	Key	Value
`_all_players_cache`	`(match_id, tick)`	Player list at the tick
`_window_cache`	`(match_id, tick)`	320-tick history for enemy memory
`_event_cache`	`(match_id, tick)`	Events in the interval [-320, +64]
`_metadata_cache`	`match_id`	Match metadata (map, etc.)

> **Fix F3-11:** Samples with zero-tensor fallback are **discarded** (not used for training). The aggregate rate is tracked and, if it exceeds 10%, a warning is emitted. If the entire batch is fallback, it is skipped completely.
>
> **Fix H-03 — Removal of silent batch swallowing:** The RAP training and validation loops no longer contain the `except (KeyError, TypeError): continue` block that silently masked errors in the data. Now `train_step()` must return a `dict` with the `'loss'` key, or an explicit `ValueError` is raised. This ensures structural errors in batches are detected immediately instead of being ignored, improving the training pipeline's diagnosability.

Advantage Function: `_compute_advantage()`

Continuous [0, 1] formula that replaces the binary win/lose target (G-04):

```
advantage = 0.4 × alive_diff + 0.2 × hp_ratio + 0.2 × equip_ratio + 0.2 × bomb_factor
```

Component	Weight	Computation	Range
`alive_diff`	0.4	`(team_alive - enemy_alive + 5) / 10`	[0, 1]
`hp_ratio`	0.2	`team_hp / (team_hp + enemy_hp)`	[0, 1]
`equip_ratio`	0.2	`team_equip / (team_equip + enemy_equip)`	[0, 1]
`bomb_factor`	0.2	Planted: T=0.7, CT=0.3 / Not planted: 0.5	{0.3, 0.5, 0.7}

The `alive_diff` has the largest weight (0.4) because numerical superiority is the single most predictive factor in CS2. The `(diff + 5) / 10` normalization maps the range [-5, +5] (5v0 → 0v5) into [0, 1].

Tactical Role Classification: `_classify_tactical_role()`

Assigns one of 10 tactical roles to each tick based on heuristics informed by `PlayerKnowledge`:

Index	Role	Condition
0	Site Take	T default (no special condition)
1	Rotation	— (reserved)
2	Entry Frag	Enemies visible + distance < 800u
3	Support	T + average team distance < 500u + ≥2 teammates
4	Anchor	CT + crouched
5	Lurk	Average distance from team > 1500u + no enemy visible
6	Retake	CT + bomb planted
7	Save	Equipment < $1500 (maximum priority)
8	Aggressive Push	Enemies visible + distance ≥ 800u
9	Passive Hold	CT default (without knowledge)

The priority chain is: Save → Retake → Lurk → Entry Frag → Aggressive Push → Anchor/Passive Hold → Support → Site Take. Without `PlayerKnowledge`, it falls back to team-based defaults (CT → Passive Hold, T → Site Take).

Map Resolution: `_resolve_map_name()`

3-tier fallback chain to determine the map of every sample:

1. **Match metadata DB** → `MatchDataManager.get_metadata(match_id).map_name`
2. **Regex on demo_name** → searches for known map names (mirage, inferno, dust2, etc.)
3. **Static fallback** → `"de_mirage"` (most played map)

Contrastive Validation (Evaluation)

During validation, the JEPA loss computation requires an extra step for negatives:

1. Raw negatives `[batch, n_neg, feat_dim]` → flatten to `[batch*n_neg, feat_dim]`
2. Encode via `target_encoder` → `[batch*n_neg, latent_dim]`
3. Reshape to `[batch, n_neg, latent_dim]`
4. Compute `jepa_contrastive_loss(pred, target, neg_latent)`

For RAP, validation uses `trainer.criterion_val` (MSE on value_estimate vs target_val).

-JEPATrainer (jepa_trainer.py)

The specialized trainer for the JEPA architecture. Manages self-supervised pre-training, drift monitoring, and the VL-JEPA triple-loss protocol.

Constructor:

Parameter	Default	Detail
`lr`	1e-4	Learning rate for AdamW
`weight_decay`	1e-4	L2 regularization
`drift_threshold`	2.5	z-score threshold for DriftMonitor

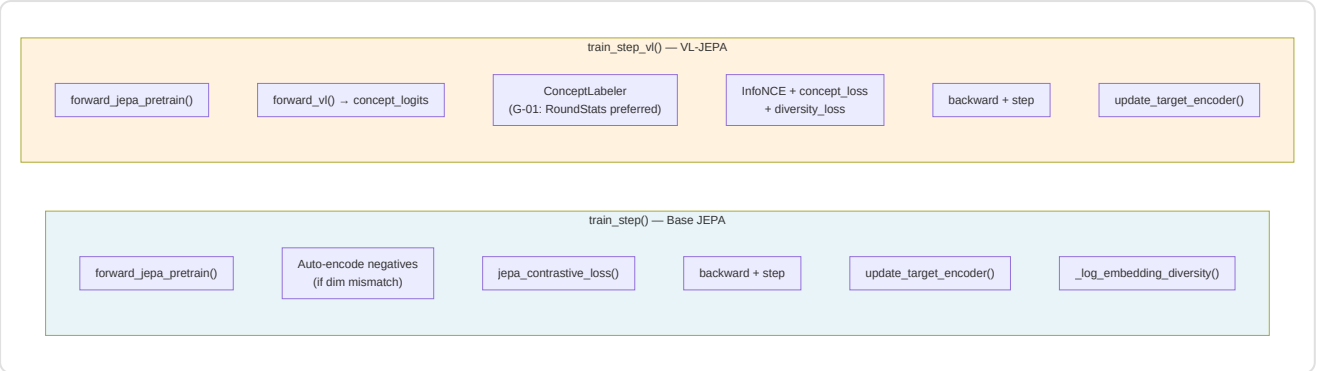
- **NN-36**: `target_encoder` parameters are **excluded** from the optimizer — they are updated only via EMA (Exponential Moving Average), never by direct gradients
- **KT-05**: `concept` layer parameters receive a **0.05× LR multiplier** (5% of the base learning rate) via separate parameter groups in the optimizer. This prevents collapse of concept embeddings during VL-JEPA training (per VL-JEPA paper, Section 4.6, Assran et al.)
- **Scheduler**: `CosineAnnealingLR` with `T_max=100`, stepped once per epoch

EMA Momentum Schedule (J-6):

```
τ(t) = 1 - (1 - τ_base) · (cos(πt/T) + 1) / 2
```

With `τ_base = 0.996`, momentum starts at 0.996 (fast target encoder tracking) and converges to 1.0 (target encoder frozen) during training. This cosine schedule, adapted from Assran et al. (CVPR 2023, Section 3.2), allows the target encoder to quickly track the online encoder in the early phases and then stabilize to provide consistent targets in the advanced phase of training.

Two training-step modes:



`train_step()` (base JEPA):

1. Forward → `pred_embedding`, `target_embedding`
2. Auto-detect raw negatives (dim mismatch with `pred_embedding`) → encode via `target_encoder`
3. `jepa_contrastive_loss(pred, target, negatives)` — InfoNCE
4. Backward + optimizer step
5. EMA update of the target encoder
6. **P9-02**: Embedding variance monitoring — warning if < 0.01 (collapse risk)

`train_step_vl()` (VL-JEPA):

Three loss components:

Loss	Weight	Function
InfoNCE	1.0	Contrastive self-supervised
Concept alignment	α = 0.5	Alignment of concepts to labels
Diversity regularization	β = 0.1	Prevents concept collapse

> **Fix G-01:** Concept labels are preferably generated by `ConceptLabeler.label_from_round_stats(rs)` using real round outcome data. Only as a fallback is heuristic labeling used (with a warning logged only once).

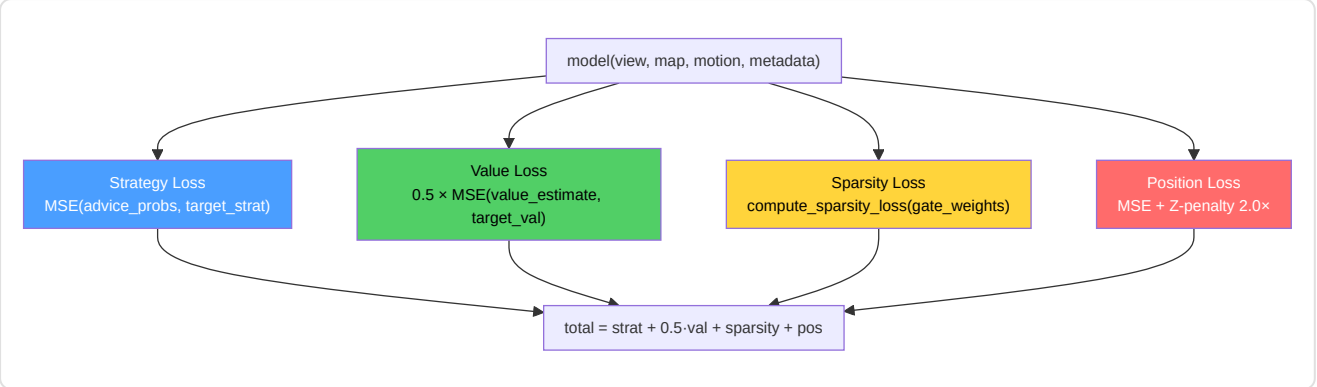
Drift system (Task 2.19.3):

- `check_val_drift(val_df, reference_stats)` → `DriftMonitor.check_drift()` with z-threshold 2.5
- `should_retrain(history, window=5)` → True if drift persists in the last 5 epochs
- `retrain_if_needed()` → full retraining cycle with the scheduler reset

-RAPTrainer (rap_coach/trainer.py)

The multi-task trainer for the RAP Coach model. Manages 4 loss components simultaneously.

Loss composition:



Loss	Weight	Criterion	Target
Strategy	1.0	MSE	`target_strat` (one-hot tactical role, 10 classes)
Value	0.5	MSE	`target_val` (advantage [0,1])
Sparsity	1.0	L1 on gate weights	Interpretability: gate weights → ~0
Position	1.0	MSE + Z×2.0	`target_pos` delta (optional)

Task 2.17.1 — Z-axis penalty: `compute_position_loss()` applies a 2× weight on Z-axis error (verticality), because in CS2 elevation errors (e.g. enemy on a stairway vs ground floor) are tactically critical.

F3-07: `gate_weights` are passed explicitly to `compute_sparsity_loss()` instead of read from the model instance — ensuring thread safety.

-Training Entry Point (train.py)

Entry-point module that provides the `train_nn()` function with automatic dispatch and standalone functions.

`train_nn(X, y, X_val, y_val, model, config_name, context)`:

config_name	Path	Pipeline
`"jepa"`	`_train_jepa_self_supervised()`	Self-supervised, InfoNCE, 5-epoch prototype
other	`_execute_validated_loop()`	Supervised, MSE, standard DataLoader

Helper functions:

Function	Responsibility
`_prepare_splits(X, y)`	Train/val split 80/20 with `random_state=42`. Rejects if < 20 samples (P1-04)
`_train_jepa_self_supervised()`	JEPA prototype: `SelfSupervisedDataset`, context_len=10, prediction_len=5, clip_grad=1.0
`_execute_validated_loop()`	Supervised loop with `EarlyStopping(patience=10)`, configurable throttling
`run_training()`	Standalone entry: uses `CoachTrainingManager` to fetch pro data

> **Note P1-02:** All paths call `set_global_seed()` before training to guarantee reproducibility. The global seed is configured in the `config.py` module.

-WinProbabilityTrainer (win_probability_trainer.py)

Module dedicated to the win-probability model — the simplest network in the ecosystem.

`WinProbabilityTrainerNN` architecture:

```
Input(9) → Linear(32) → ReLU → Linear(16) → ReLU → Linear(1) → Sigmoid
```

9 input features:

#	Feature	Type
1	`ct_alive`	Alive CT players
2	`t_alive`	Alive T players
3	`ct_health`	CT total HP
4	`t_health`	T total HP
5	`ct_armor`	CT total armor
6	`t_armor`	T total armor
7	`ct_eqp`	CT equipment value
8	`t_eqp`	T equipment value
9	`bomb_planted`	Bomb planted (0/1)

Training:

- Loss: **BCELoss** (Binary Cross-Entropy — output is a probability [0,1])
- Optimizer: Adam (lr=0.001) — not AdamW, simpler for this network
- Early stopping: patience=10, min_delta=1e-4
- Target: `did_ct_win` (binary)
- Split: 80/20 with `random_state=42`
- Minimum samples: 20 (AR-6)
- Saving: `torch.save(state_dict, model_path)` — direct persistence, not via `save_nn()`

`predict_win_prob(model, state_dict)` — Single inference: builds the tensor from the 9 features and returns the CT-win probability as a float.

> **Note A-12 — Distinct architectures:** This `WinProbabilityTrainerNN` (9 features, offline training) is a *separate and incompatible* model from the 12-feature `WinProbabilityNN` described in Section 7. The former is trained on aggregated demo data; the latter operates in real time during live analysis. The checkpoints are not interchangeable.

-Training Utilities

The following modules provide the support infrastructure for the training cycle — configuration, control, monitoring, datasets, and early stopping.

`training_config.py` — Centralized dataclasses for hyperparameters:

Dataclass	Key parameters	Use
`TrainingConfig`	`base_lr=1e-4`, `max_epochs=100`, `patience=10`, `batch_size=1`, `gradient_clip=1.0`, `ema_decay=0.999`	Base configuration for all models
`JEPATrainingConfig`	Extends TrainingConfig + `latent_dim=256`, `contrastive_temperature=0.07`, `momentum_target=0.996`, `pretraining_epochs=50`	Specific to JEPA self-supervised

`training_controller.py` — Gatekeeper that decides **whether** a new demo should be used for training:

- **Monthly quota**: `MAX_DEMOS_PER_MONTH = 10` — counts `PlayerMatchStats` processed in the last 30 days
- **Diversity score**: Compares the new demo with the last 5 matches via cosine similarity over 6 normalized features (kills, deaths, ADR, HS%, utility, opening duels). If `diversity < 0.3`, the demo is rejected
- Returns a `TrainingDecision(should_train, reason, diversity_score)` — the caller decides whether to proceed

`training_monitor.py` — Persists metrics in JSON for real-time monitoring and post-training analysis. Tracks: epochs, train_loss, val_loss, learning_rate, best_val_loss, state (in progress / early stopped / completed).

`early_stopping.py` — Callable early-stopping implementation:

```
stopper = EarlyStopping(patience=10, min_delta=1e-4)
if stopper(val_loss):  # True = stop training
    break
```

Logic: if `val_loss < best_loss - min_delta`, reset the counter. Otherwise increment. When `counter >= patience`, it returns `True`. Used by all trainers (JEPA, RAP, Legacy, WinProb).

`dataset.py` — Two PyTorch Datasets:

Dataset	Input	Output	Use
`ProPerformanceDataset`	`(X, y)` tensors	`(x[i], y[i])` tuples	Legacy supervised training
`SelfSupervisedDataset`	`X` sequential	`(context, target)` sliding windows	JEPA self-supervised

The `SelfSupervisedDataset` uses a **sliding window**: `context_len=10` ticks as input, `prediction_len=5` ticks as target. The total number of samples is `len(X) - context_len - prediction_len`. Raises `ValueError` if data is insufficient (F3-34).

-Callback System (training_callbacks.py + tensorboard_callback.py)

Two-layer plugin architecture for training observability, without modifying the main loop.

Layer 1 — `training_callbacks.py`:

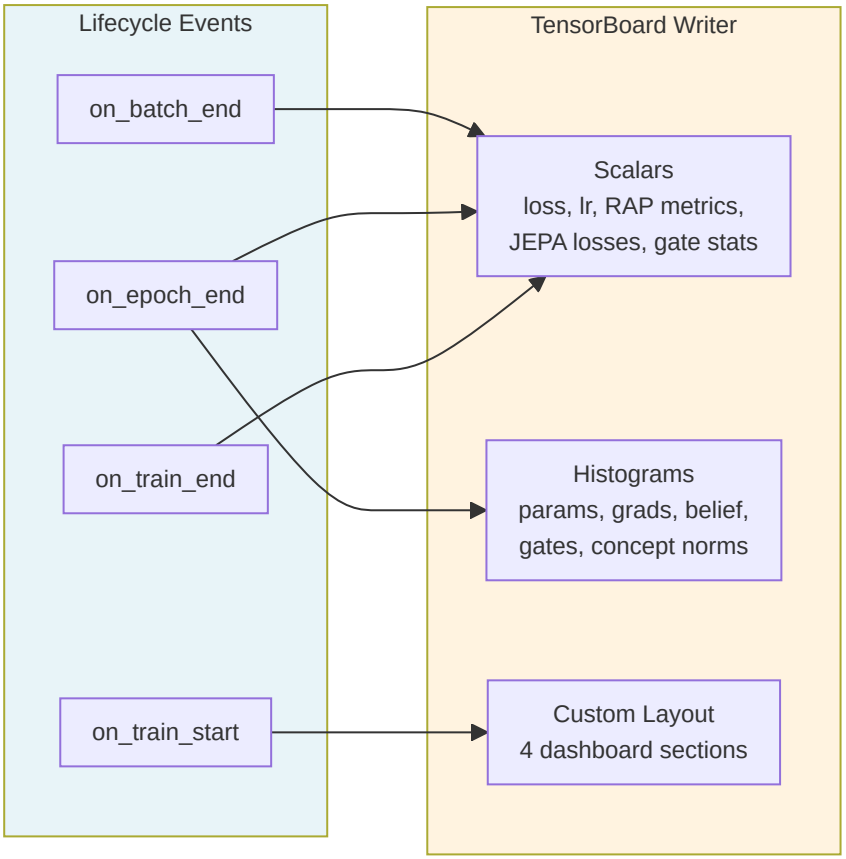
`TrainingCallback` (ABC) defines 7 lifecycle hooks:

Hook	When	Parameters
`on_train_start`	Before the first epoch	model, config dict
`on_epoch_start`	Start of every epoch	epoch
`on_batch_end`	After every training batch	batch_idx, loss, outputs dict
`on_epoch_end`	End of every epoch	epoch, train_loss, val_loss, model, optimizer
`on_validation_end`	After the validation pass	epoch, val_loss, model
`on_train_end`	After completion	model, final_metrics dict
`close`	Release resources	—

> **Note F3-31:** No method is `@abstractmethod` — an opt-in pattern where subclasses implement only the hooks they need. All base methods are no-ops.

`CallbackRegistry` is the dispatcher: `fire(event, **kwargs)` calls the corresponding hook on every registered callback. Errors in callbacks are captured and logged — **they never crash training**.

Layer 2 — `tensorboard_callback.py`:



Signals recorded per model type:

Category	Metrics	Type
Core	`loss/train`, `loss/val`, `loss/gap`, `loss/batch`	Scalars per-epoch/batch
RAP	`rap/sparsity_ratio`, `rap/z_axis_error`, `rap/loss_position`	Scalars per-batch
JEPA	`jepa/infonce_loss`, `jepa/concept_loss`, `jepa/diversity_loss`	Scalars per-batch
Gates	`gates/mean_activation`, `gates/sparsity`, `gates/active_ratio`	Scalars per-batch
Diagnostics	`params/*`, `grads/*`, `belief/vector`, `concepts/embedding_norms`, `gates/activations`	Histograms per-epoch

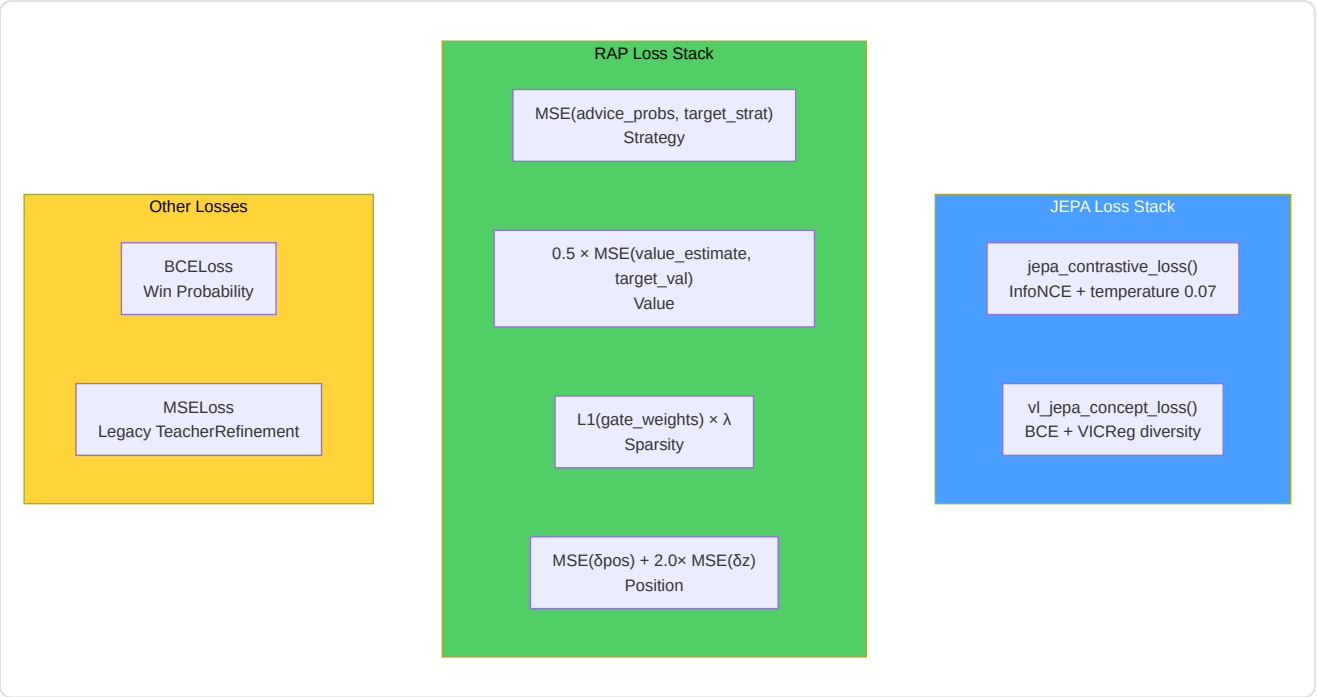
4 custom dashboards: **Coach Vital Signs**, **RAP Coach Internals**, **JEPA Self-Supervised**, **Superposition Gates**.

Graceful import: if `tensorboard` is not installed, the callback becomes a complete no-op (`_TB_AVAILABLE = False`).

---

13. Loss Functions — Implementation Detail

Catalog of Loss Functions



-jepa_contrastive_loss() — InfoNCE

File: `jepa_model.py:326` | **Signature:** `(pred, target, negatives, temperature=0.07) → scalar`

InfoNCE formula (Noise Contrastive Estimation):

```
L = -log( exp(sim(pred, target) / τ) / (exp(sim(pred, target) / τ) + Σ exp(sim(pred, neg_i) / τ
```

Steps:

1. **L2 normalization** of pred, target, negatives (cosine similarity space)
2. **Positive similarity**: `pos_sim = (pred · target) / τ` with `τ = 0.07`
3. **Negative similarities**: `neg_sim = bmm(negatives, pred)` / τ
4. **Cross-entropy**: concat `[pos_sim, neg_sim]` → `F.cross_entropy` with labels=0 (the positive is always index 0)

The temperature `τ = 0.07` is low, which makes the loss **sensitive to small differences** — appropriate for CS2 embeddings where consecutive ticks are very similar.

-vl_jepa_concept_loss() — Concept Alignment

File: `jepa_model.py:913` | **Signature:** `(concept_logits, concept_labels, concept_embeddings, α=0.5, β=0.1) → (total, concept, diversity)`

Two components:

Component	Formula	Weight	Purpose
Concept alignment	`BCE_with_logits(logits, labels)`	α = 0.5	Multi-label: each concept activated independently
Diversity (VICReg)	`-mean(std(emb_norm, dim=0))`	β = 0.1	Prevents collapse: penalizes low variance of concept embeddings

The 16 coaching concepts (`COACHING_CONCEPTS`) cover: positioning (aggressive/passive/exposed), economy (eco/force/full), timing (trade/rotate/patience), utility (flash/smoke/molotov), and communication (info/callout).

-compute_sparsity_loss() — RAP Gate Sparsity

File: `rap_coach/model.py:105` | **Signature:** `(gate_weights) → scalar`

```
L_sparsity = λ × mean(|gate_weights|)
```

L1 norm on `ContextGate` (SuperpositionLayer) weights. The goal is that most gates are close to 0, with a few strong activations — this makes the model **interpretable** (which inputs really influence the decision?).

> **Note F3-07:** `gate_weights` is passed as an explicit parameter instead of read from `self._last_gate_weights` — preventing race conditions in multi-threaded environments.

-RAP Position Loss — Z-axis Penalty

File: `rap_coach/trainer.py:89` | **Signature:** `(pred_delta, target_delta) → (loss, z_error)`

```
L_pos = MSE(Δx) + MSE(Δy) + 2.0 × MSE(Δz)
```

The 2× weight on the Z axis reflects the critical importance of verticality in CS2: confusing the ground floor with the first floor (e.g. Nuke, Vertigo) leads to completely wrong tactical decisions.

-Legacy and Win Probability Losses

- **TeacherRefinementNN**: standard `MSELoss` on supervised predictions (train.py)
- **WinProbabilityNN**: `BCELoss` for the sigmoid output [0,1] → CT-win probability

---